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# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from flask_wtf.csrf import CSRFProtect, CSRFError
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
csrf = CSRFProtect()
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # CSRF 防护：WTF_CSRF_ENABLED 只对 FlaskForm 生效，页面上那些手写 POST
    表单
    # （账号增删改、项目 / 图斑删除等）以前是没校验的；同时模板里的 csrf_token()
    # 也需要它才会存在——少了这一行，保存过一次 AHP 权重后 /ahp/ 直接 500。
    csrf.init_app(app)
    # 模板里展示时间统一走 localdt：库里存 UTC，展示成本地时间
    from app.utils.timefmt import to_local
    app.jinja_env.filters["localdt"] = to_local
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
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app.register_blueprint(constraint.bp)
app.register_blueprint(ahp.bp)
app.register_blueprint(indicator.bp)
app.register_blueprint(evaluate.bp)
app.register_blueprint(report.bp)
app.register_blueprint(search.bp)
# 全局登录保护：除登录页与静态资源外，均需登录
from flask import request, redirect, url_for,
send_from_directory
from flask_login import current_user, login_user
PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
# 浏览器默认会请求根路径下的 /favicon.ico。
# 页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
# 回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    # 截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
# 侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
# 不使用模块编号，避免出现与实际流程不一致的代号。
# 每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
# 与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
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        ("适宜性定量评价", "/evaluate/", "target"),
        ("报告与结果输出", "/report/", "file"),
    ]
    #  「用户与操作日志」已并入「个人设置」，不再是独立菜单，
    #  因此系统管理组只留一个入口。
    ADMIN_LINKS = [
        ("个人设置", "/profile", "user"),
    ]
    @app.context_processor
    def inject_nav():
        path = request.path or "/"
        def hit(prefix):
            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
        badges = {}
        if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
        def build(links):
            return [{"label": label, "href": href, "icon":
    icon,
                     "active": hit(href), "badge":
    badges.get(href, 0)}
                    for label, href, icon in links]
        groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
        ]
        #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
        if path == "/":
            page_title = "工作台"
        elif path.startswith("/profile"):
            page_title = "个人设置"
        elif path.startswith(("/users", "/logs")):
            page_title = "账号与日志"
        elif path.startswith("/search"):
            page_title = "搜索结果"
        else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                if hit(href):
                    page_title = label
                    break
        #  层级引导用：项目是归档单位，图斑才是评价对象。
        #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
        from app.models import Project, MinePatch
        projects = Project.query.order_by(Project.id).all()
        return {
            "nav_groups": groups,
            "home_active": path == "/",
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                "page_title": page_title,
                "today": date.today().strftime("%Y-%m-%d"),
                "search_q": request.args.get("q", ""),
                "role_labels": ROLE_LABELS,
                "project_count": len(projects),
                "patch_count": MinePatch.query.count(),
                "projects_without_patch": [p for p in projects
                                           if p.patches.count()
    == 0],
            }
    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    @app.errorhandler(CSRFError)
    def csrf_error(e):
        """令牌过期或来源不符时给一句人话，并把人送回上一页。
        原来校验一失败就是默认英文 400 页，或者（对手写表单）干脆什么都不发生，
        用户只会觉得「点了没反应」。回跳目标只取本站地址，避免开放重定向。
        """
        from flask import flash, redirect, url_for
        flash("页面停留时间过久或提交来源校验未通过，请重新提交一次", "error")
        referrer = request.referrer or ""
        if referrer.startswith(request.host_url):
            return redirect(referrer)
        return redirect(url_for("auth.index"))
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
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    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 模板改动立即生效，不必重启服务。
    # Flask 在非 debug 下默认把模板缓存到进程里，改完模板刷新浏览器还是旧页面，
    # 容易被误当成「改了没生效」。本系统只在本机运行，这点开销可以忽略。
    TEMPLATES_AUTO_RELOAD = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024            # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
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    default=True)                                                    301
    created_at = db.Column(db.DateTime,                              302
    default=datetime.utcnow)                                         303
    # ---- 个人资料（用户可在「个人设置」里自行维护）----                304
    display_name = db.Column(db.String(64), nullable=True)           305
    # 昵称                                                           306
    email = db.Column(db.String(128), nullable=True)                 307
    phone = db.Column(db.String(32), nullable=True)                  308
    gender = db.Column(db.String(8), nullable=True)                  309
    # 男 / 女 / 保密                                                 310
    avatar = db.Column(db.String(256), nullable=True)                311
    # 头像，相对 static 的路径                                       312
    @property                                                        313
    def is_admin(self):                                              314
        return self.role == "admin"                                  315
    @property                                                        316
    def shown_name(self):                                            317
        """界面上显示的名字：优先用昵称，没填就用账号名。"""          318
        return self.display_name or self.username                    319
    @property                                                        320
    def initial(self):                                               321
        """没有头像时用首字占位。"""                                  322
        return (self.shown_name or "?")[0].upper()                   323
    def to_dict(self):                                               324
        return {"id": self.id, "username": self.username,            325
                "display_name": self.display_name,                   326
    "gender": self.gender,                                           327
                "role": self.role, "active": self.active}            328
class OperationLog(db.Model):                                        329
    """操作日志。"""                                                 330
    __tablename__ = "operation_logs"                                 331
    id = db.Column(db.Integer, primary_key=True)                     332
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"         333
    ), nullable=True)                                                334
    action = db.Column(db.String(128), nullable=False)               335
    # 操作类型                                                       336
    target_type = db.Column(db.String(64), nullable=True)            337
    # 对象类型                                                       338
    target_id = db.Column(db.Integer, nullable=True)                 339
    # 对象主键                                                       340
    detail = db.Column(db.Text, nullable=True)                       341
    # 操作详情                                                       342
    created_at = db.Column(db.DateTime,                              343
    default=datetime.utcnow, index=True)                             344
class Project(db.Model):                                             345
    """评价项目（项目与矿山档案）。"""                               346
    __tablename__ = "projects"                                       347
    id = db.Column(db.Integer, primary_key=True)                     348
    name = db.Column(db.String(128), nullable=False)                 349
    code = db.Column(db.String(64), nullable=True)                   350
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    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
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    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)    # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：>  >=  <  <=  ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")   # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
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    id = db.Column(db.Integer, primary_key=True)                    451
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        452
    ches.id"), nullable=False)                                      453
    policy_pass = db.Column(db.Boolean, nullable=True)              454
    # 政策约束                                                      455
    location_pass = db.Column(db.Boolean, nullable=True)            456
    # 区位约束                                                      457
    pollution_pass = db.Column(db.Boolean, nullable=True)           458
    # 土壤污染                                                      459
    geo_pass = db.Column(db.Boolean, nullable=True)                 460
    # 地质灾害                                                      461
    farmland_pass = db.Column(db.Boolean, nullable=True)            462
    # 耕地属性                                                      463
    passed = db.Column(db.Boolean, nullable=True)                   464
    # 总体是否通过                                                  465
    reason = db.Column(db.Text, nullable=True)                      466
    # 不通过原因                                                    467
    checked_at = db.Column(db.DateTime,                             468
    default=datetime.utcnow)                                        469
class MicroUnit(db.Model):                                          470
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""                  471
    __tablename__ = "micro_units"                                   472
    id = db.Column(db.Integer, primary_key=True)                    473
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        474
    ches.id"), nullable=False)                                      475
    unit_type = db.Column(db.String(32), nullable=False)            476
    # 单元类型                                                      477
    area = db.Column(db.Float, nullable=True)                       478
    # 面积                                                          479
    ratio = db.Column(db.Float, nullable=True)                      480
    # 面积占比（自动计算）                                          481
    description = db.Column(db.Text, nullable=True)                 482
    # 现场描述                                                      483
class Indicator(db.Model):                                          484
    """指标定义：指标树，含角色与启用单元类型。"""                  485
    __tablename__ = "indicators"                                    486
    id = db.Column(db.Integer, primary_key=True)                    487
    code = db.Column(db.String(32), unique=True,                    488
    nullable=False)  # 指标编码                                    489
    name = db.Column(db.String(64), nullable=False)                 490
    # 指标名称                                                      491
    level = db.Column(db.Integer, nullable=False)                   492
    # 1/2/3 级                                                      493
    parent_code = db.Column(db.String(32), nullable=True)           494
    # 父指标编码                                                    495
    # 角色：土壤污染指标同时作为限制性指标与评价指标               496
    is_constraint = db.Column(db.Boolean, default=False)            497
    is_evaluative = db.Column(db.Boolean, default=True)             498
    unit = db.Column(db.String(32), nullable=True)                  499
    # 单位                                                          500
```

```
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 矩阵对应指标层级
```

```
    matrix_json = db.Column(db.Text, nullable=False)
    # 判断矩阵（JSON 二维数组）
    cr = db.Column(db.Float, nullable=True)
    # 该专家矩阵的一致性比例
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class IndicatorData(db.Model):
    """指标实测数据：按微地质单元组织。"""
    __tablename__ = "indicator_data"
    id = db.Column(db.Integer, primary_key=True)
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni
    ts.id"), nullable=False, unique=True)
    soil_ratio = db.Column(db.Float, nullable=True)
    # 土石比
    thickness = db.Column(db.Float, nullable=True)
    # 土层厚度 cm
    ph = db.Column(db.Float, nullable=True)
    organic_matter = db.Column(db.Float, nullable=True)
    # 有机质 g/kg
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡/阴阳坡/阴坡
    def heavy_metals_dict(self):
        """把重金属 JSON 解析成字典。
        字段是手工可改的文本，被脏写或导入半个 JSON 时不能让整页 500：
        解析失败按「未检测」处理（空字典），并把原文留在库里供排查。
        """
        if not self.heavy_metals:
            return {}
        try:
            data = json.loads(self.heavy_metals)
        except (ValueError, TypeError):
            return {}
        return data if isinstance(data, dict) else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
```

```
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator,
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
```

```
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
                                       Length(max=64,

    message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format(),
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
```

```
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称"),
                                       Length(min=2, max=64,
                                              message="账号名称2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码"),
                                         Length(min=6,
                                                max=64,
                                                message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",
                                                message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,
                                             message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),
    EqualTo("new_password",
                                            message="两次输入的新密码不一致")])
    submit_pwd = SubmitField("修改密码")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
```

```
全部数值取自论文，作为系统计算的唯一基准：
- 论文 ４．２．２ 节：指标标准化理想极值与取值依据
- 论文 ４．２．３ 节 / 表 ４：适宜性等级划分
- 论文表 ３：按微地质单元分别确定的一级、二级指标权重
- 论文 ４．２．１ 节：平均随机一致性指标 ＲＩ
"""
# ----------------------------------------------------------------------
    ----------
# 一、微地质单元类型
# ----------------------------------------------------------------------
    ----------
# 论文 ４．１ 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 ３ 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                    # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                        # 不设地形指标
}
# ----------------------------------------------------------------------
    ----------
# 二、指标标准化理想极值（论文 ４．２．２ 节）
# ----------------------------------------------------------------------
    ----------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （ｃｍ）
IDEAL_PH_RANGE = (5.0, 9.0)     # ｐＨ 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （ｇ/ｋｇ），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （％）
IDEAL_SHANNON = 0.7             # Ｓｈａｎｎｏｎ-Ｗｉｅｎｅｒ 多样性指数参照生态系统
# 重金属极值 （ｍｇ/ｋｇ），《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 ＩＶ 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 ４．２．１ 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 ４．２．２ 节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２ 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 ４．２．２ 节："阳坡、阴阳坡、阴坡分别赋值 0．2、0．6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# ｐＨ 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
```

```
PH_TOL_HIGH = 10.0
# ---------------------------------------------------------------------
    ----------------
# 三、适宜性等级划分（论文表 4）
# ---------------------------------------------------------------------
    ----------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ---------------------------------------------------------------------
    ----------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ---------------------------------------------------------------------
    ----------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:
    1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ---------------------------------------------------------------------
    ----------------
# 五、默认权重方案（论文表 3）
# ---------------------------------------------------------------------
    ----------------
# 论文 4.2.2 节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
# 分别计算评价指标权重"（论文摘要、图 2 流程与表 5 均作"碎石边坡"，
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。
# 下表为论文表 3 所载的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS = {
```

```
        "岩质边坡"：  {
            "ｌ１"：  {"植被指标"：  ０．４５，  "地形指标"：  ０．５５},
            "ｌ２"：  {
                "植被指标"：  {"植被覆盖度"：  ０．６７，  "生物多样性"：  ０．３３},
                "地形指标"：  {"坡度"：  ０．６４，  "坡向"：  ０．３６},
            },
        },
        "碎石边坡"：  {
            "ｌ１"：  {"土壤指标"：  ０．４４，  "植被指标"：  ０．３１，  "地形指标"：  ０．２５},
            "ｌ２"：  {
                "土壤指标"：  {"土石比"：  ０．２５，  "土层厚度"：  ０．２５，  "ｐＨ"：  ０．１７，
                              "有机质"：  ０．１７，  "污染指标"：  ０．１７},
                "植被指标"：  {"植被覆盖度"：  ０．６４，  "生物多样性"：  ０．３６},
                "地形指标"：  {"坡度"：  ０．６７，  "坡向"：  ０．３３},
            },
        },
        "平台"：  {
            "ｌ１"：  {"土壤指标"：  ０．５４，  "植被指标"：  ０．４６},
            "ｌ２"：  {
                "土壤指标"：  {"土石比"：  ０．２５，  "土层厚度"：  ０．２５，  "ｐＨ"：  ０．１７，
                              "有机质"：  ０．１７，  "污染指标"：  ０．１７},
                "植被指标"：  {"植被覆盖度"：  ０．６１，  "生物多样性"：  ０．３９},
            },
        },
}
#  - - -
```

```
"""
from sqlalchemy import text
#  （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    #  遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    #  指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    #  多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    #  用户个人资料（账户页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "gender", "VARCHAR(8)"),
    ("users", "avatar", "VARCHAR(256)"),
]
#  职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种，
#  旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
def _add_columns(db):
    added = []
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                            #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
    return added
def _migrate_roles(db):
    """把旧角色名统一为新的两种，返回改动的行数。"""
    changed = 0
    with db.engine.connect() as conn:
        rows = conn.execute(text("PRAGMA
    table_info(users)")).fetchall()
```

```
        if not rows:                                                    951
            return 0                                                    952
        for old, new in ROLE_MIGRATION.items():                         953
            r = conn.execute(text("UPDATE users SET role =              954
    :new "                                                              955
                                  "WHERE role = :old"),                 956
                                 {"new": new, "old": old})              957
            changed += r.rowcount or 0                                  958
        conn.commit()                                                   959
    return changed                                                      960
def ensure_schema(db):                                                  961
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""                    962
    try:                                                                963
        changed = _add_columns(db)                                      964
        n = _migrate_roles(db)                                          965
        if n:                                                           966
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)              967
        return changed                                                  968
    except Exception as e:                                  #           969
    迁移失败不应阻断启动                                                  970
        print("[schema] 结构检查跳过：%s" % e)                             971
        return []                                                       972
# ==== app/utils/uploads.py ====                                        973
# -*- coding: utf-8 -*-                                                 974
"""上传文件处理（遥感影像、批量导入）。"""                                    975
import os                                                               976
import re                                                               977
import uuid                                                             978
from datetime import datetime                                           979
from werkzeug.utils import secure_filename                              980
def _safe_ext(filename):                                                981
    return os.path.splitext(filename)[1].lower().lstrip(".")            982
def ext_allowed(filename, allowed):                                     983
    return _safe_ext(filename) in allowed                               984
def save_upload(file_storage, subdir, allowed_ext,                      985
    upload_root):                                                       986
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。    987
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。                 988
    """                                                                 989
    if file_storage is None or not file_storage.filename:               990
        return None, "未选择文件"                                        991
    ext = _safe_ext(file_storage.filename)                              992
    if ext not in allowed_ext:                                          993
        return None, "不支持的文件类型：.%s" % ext                        994
    target_dir = os.path.join(upload_root, subdir)                      995
    os.makedirs(target_dir, exist_ok=True)                              996
    # 仅保留安全字符，并加时间戳与随机串防重名                                997
    base = secure_filename(os.path.splitext(file_storage.fil            998
    ename)[0]) or "file"                                                999
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or                1000
```

```
    "file"                                                                1001
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M1002
    %S"),                                                                 1003
                                uuid.uuid4().hex[:6], ext)                1004
    abs_path = os.path.join(target_dir, name)                             1005
    file_storage.save(abs_path)                                           1006
    rel = os.path.join("uploads", subdir,                                 1007
    name).replace("\\", "/")                                              1008
    return rel, None                                                      1009
def remove_upload(rel_path, upload_root):                                 1010
    """删除已保存的上传文件（相对 static 的路径）。"""                    1011
    if not rel_path:                                                      1012
        return                                                            1013
    abs_path = os.path.join(upload_root,                                  1014
    rel_path.replace("uploads/", "", 1))                                  1015
    try:                                                                  1016
        if os.path.isfile(abs_path):                                      1017
            os.remove(abs_path)                                           1018
    except OSError:                                                       1019
        pass                                                              1020
# ==== app/utils/charts.py ====                                           1021
# -*- coding: utf-8 -*-                                                   1022
"""轻量图表。                                                            1023
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                              1024
- 服务端渲染，页面与打印（PDF）中都能稳定显示                            1025
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）              1026
- 单色方案，符合界面「一个视图一个强调色」的约定                        1027
"""                                                                       1028
from html import escape                                                   1029
ACCENT = "#1f3a5f"                                                        1030
ACCENT_SOFT = "#7f93ad"                                                   1031
INK = "#23262b"                                                           1032
MUTED = "#6b7280"                                                         1033
LINE = "#dfe3e8"                                                          1034
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                            1035
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                      1036
    "#bcc9da", "#dbe3ec"]                                                 1037
def _fmt(v, digits=4):                                                    1038
    try:                                                                  1039
        return ("%%.%df" % digits) % float(v)                             1040
    except (TypeError, ValueError):                                       1041
        return str(v)                                                     1042
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,                1043
    value_w=76):                                                          1044
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                     1045
    if not items:                                                         1046
        return '<p class="muted">暂无数据</p>'                            1047
    n = len(items)                                                        1048
    height = n * (bar_h + gap) + 10                                       1049
    max_v = max((abs(float(v)) for _, v, _ in items),                     1050
```

```
    default=1) or 1                                                    1051
    body_w = width - label_w - value_w                                 1052
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                   1053
    height="%d" '                                                      1054
              'role="img" aria-label="条形图" '                        1055
              'style="max-width:%dpx;font-family:inherit;">'           1056
    % (width, height, height, width)]                                  1057
    for i, (label, value, disp) in enumerate(items):                   1058
        y = i * (bar_h + gap) + 5                                      1059
        frac = (abs(float(value)) / max_v) if max_v else 0             1060
        w = max(2.0, body_w * frac)                                    1061
        fill = GRAYS[i % len(GRAYS)]                                   1062
        parts.append(                                                  1063
            '<text x="%d" y="%d" text-anchor="end"                     1064
    font-size="12" fill="%s">%s</text>'                                1065
            % (label_w - 10, y + bar_h * 0.72, INK,                    1066
    escape(str(label))))                                               1067
        parts.append(                                                  1068
            '<rect x="%d" y="%d" width="%.1f" height="%d"              1069
    rx="3" fill="%s"/>'                                                1070
            % (label_w, y, w, bar_h, fill))                            1071
        parts.append(                                                  1072
            '<text x="%d" y="%d" font-size="12" fill="%s" '            1073
            'style="font-variant-numeric:tabular-nums;">%s</           1074
    text>'                                                             1075
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,               1076
    escape(str(disp))))                                                1077
    parts.append('</svg>')                                             1078
    return "".join(parts)                                              1079
def donut(items, size=210, thickness=34):                              1080
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]                1081
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                  1082
    """                                                                1083
    vals = [(l, float(v), d) for l, v, d in items if                   1084
    float(v) > 0]                                                      1085
    if not vals:                                                       1086
        return '<p class="muted">暂无数据</p>'                          1087
    total = sum(v for _, v, _ in vals) or 1                            1088
    r = (size - thickness) / 2                                         1089
    cx = cy = size / 2                                                 1090
    import math                                                        1091
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                      1092
    height="%d" role="img" '                                           1093
             'aria-label="权重分布环形图">' % (size, size, size,       1094
    size)]                                                             1095
    angle = -math.pi / 2                                               1096
    for i, (label, v, disp) in enumerate(vals):                        1097
        sweep = 2 * math.pi * (v / total)                              1098
        a0, a1 = angle, angle + sweep                                  1099
        x0, y0 = cx + r * math.cos(a0), cy + r *                       1100
```

```
    math.sin(a0)                                                    1101
        x1, y1 = cx + r * math.cos(a1), cy + r *                    1102
    math.sin(a1)                                                    1103
        large = 1 if sweep > math.pi else 0                         1104
        parts.append(                                               1105
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f           1106
    %.2f" '                                                         1107
            'fill="none" stroke="%s" stroke-width="%d"/>'           1108
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %                1109
    len(GRAYS)], thickness))                                        1110
        angle = a1                                                  1111
    parts.append('<text x="%d" y="%d" text-anchor="middle"          1112
    font-size="12" '                                                1113
                 'fill="%s">合计</text>' % (cx, cy - 2,             1114
    MUTED))                                                         1115
    parts.append('<text x="%d" y="%d" text-anchor="middle"          1116
    font-size="15" '                                                1117
                 'fill="%s" style="font-variant-numeric:tabu        1118
    lar-nums;">%s</text>'                                           1119
                 % (cx, cy + 18, INK, _fmt(total, 4)))              1120
    parts.append('</svg>')                                          1121
    legend = ['<table style="width:auto;margin-top:10px;"><t        1122
    head><tr>'                                                      1123
              '<th scope="col">项目</th><th                         1124
    scope="col">数值</th>'                                          1125
              '<th scope="col">占比</th></tr></thead><tbody>']      1126
    for i, (label, v, disp) in enumerate(vals):                     1127
        legend.append(                                              1128
            '<tr><td style="text-align:left;">'                     1129
            '<span style="display:inline-block;width:10px;he        1130
    ight:10px;'                                                     1131
            'border-radius:2px;background:%s;margin-right:6p        1132
    x;"></span>%s</td>'                                             1133
            '<td class="mono">%s</td><td                            1134
    class="mono">%.1f%%</td></tr>'                                  1135
           % (GRAYS[i % len(GRAYS)], escape(str(label)),            1136
    escape(str(disp)),                                              1137
               100.0 * v / total))                                  1138
    legend.append('</tbody></table>')                               1139
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p        1140
    x;align-items:center;">'                                        1141
            + "".join(parts) + "".join(legend) + '</div>')          1142
# ==== app/utils/browser.py ====                                    1143
# -*- coding: utf-8 -*-                                             1144
"""HTML 转 PDF（报告导出）。                                        1145
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、     1146
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，  1147
避免中文路径在 file:// URL 下出问题。                               1148
"""                                                                 1149
import os                                                           1150
```

```
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
```

```
            ]
            p = subprocess.run(cmd, capture_output=True,
      text=True,
                               timeout=timeout,
      encoding="utf-8", errors="replace")
            if not os.path.isfile(pdf_path) or
      os.path.getsize(pdf_path) < 800:
                return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
      "").strip()[:200]
            with open(pdf_path, "rb") as f:
                return f.read(), None
      except subprocess.TimeoutExpired:
            return None, "生成 PDF 超时，请稍后重试"
      except Exception as e:
            return None, "生成 PDF 失败：%s" % e
      finally:
            shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
      """由判断矩阵上三角构造完整互反矩阵。
      参数
      ----
      upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
            例：3 阶时传入 [[a12, a13], [a23]]。
            对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
      """
      upper = np.asarray(upper, dtype=float)
      k = upper.shape[0]
      n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
      A = np.ones((n, n), dtype=float)
      idx = 0
      for i in range(n):
            for j in range(i + 1, n):
                A[i, j] = upper[idx]
                A[j, i] = 1.0 / upper[idx]
                idx += 1
      return A
def weights_and_consistency(A):
      """由判断矩阵 A 计算权重与一致性指标。
      返回
      ----
```

```
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
```

```
                                "expected": expected, "deviation":  1301
    deviation}                                                      1302
    return best                                                     1303
def aggregate_matrices(matrices):                                   1304
    """多专家判断矩阵聚合：几何平均（保持互反性）。                  1305
    参数                                                            1306
    ----                                                            1307
    matrices : list[np.ndarray]，多个同阶判断矩阵。                 1308
    """                                                             1309
    stack = np.stack([np.asarray(m, dtype=float) for m in           1310
    matrices])                                                      1311
    return np.prod(stack, axis=0) ** (1.0 / len(stack))             1312
def judge_from_weights(weights):                                    1313
    """由权重向量构造完全一致判断矩阵并检验。                        1314
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量  1315
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。                       1316
    """                                                             1317
    w = np.asarray(weights, dtype=float)                            1318
    n = w.shape[0]                                                  1319
    A = np.outer(w, 1.0 / w)                                        1320
    r = weights_and_consistency(A)                                  1321
    r["matrix"] = A                                                 1322
    return r                                                        1323
# ==== app/services/standardization.py ====                       1324
# -*- coding: utf-8 -*-                                           1325
"""指标标准化（的一部分）。                                        1326
按论文 4.2.2 节式(8) 实现：                                        1327
    x' = (x - x_min) / (x_max - x_min)                              1328
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。        1329
所有理想极值取自 app.utils.constants，可配置。                     1330
"""                                                                 1331
from app.utils.constants import (                                   1332
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,              1333
    IDEAL_ORGANIC,                                                  1334
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,               1335
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,           1336
    PH_TOL_LOW, PH_TOL_HIGH,                                        1337
)                                                                   1338
def std_benefit(value, ideal):                                      1339
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""  1340
    if value is None:                                               1341
        return None                                                 1342
    return max(0.0, min(1.0, value / ideal))                        1343
def std_cost(value, extreme):                                       1344
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""      1345
    if value is None:                                               1346
        return None                                                 1347
    return max(0.0, min(1.0, 1.0 - value / extreme))                1348
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,                    1349
    tol_hi=PH_TOL_HIGH):                                            1350
```

```
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals, limits=None):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    limits : 可选，金属 → 极值；不传则用代码常量（界面改过就传库里的值）。
    """
    table = limits or HEAVY_METAL_LIMIT
    total = 0.0
    for metal, extreme in table.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(table)
def standardize_unit(unit_type, data, weights=None,
    settings=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    settings : app.services.unit_config.UnitSettings，图斑级的基准值
```

```
与理想取值。
        不传则全部按代码常量 —— 历史数据与验证算例走这条路径，结果不变。
返回
----
dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
标准化值可能为 None，表示该指标不适用或方法待定。
"""
from app.utils.constants import DEFAULT_WEIGHTS
from app.services import unit_config as uc
if weights is None:
    weights = DEFAULT_WEIGHTS[unit_type]
if settings is None:
    settings = uc.UnitSettings()
# 理想取值：库里有数字就用库里的，否则用常量
def ideal(code, fallback):
    return settings.ideal(code, fallback)
l1_enabled = weights["l1"].keys()
l2w = weights["l2"]
result = {}
# 一级指标是否启用：库里配过就按库里的（界面上的「启用单元」），没配按常量
def group_on(l1_name, l1_code):
    return settings.group_enabled(unit_type, l1_name,
l1_code) is not False
# ---- 土壤指标 ----
if "土壤指标" in l1_enabled and group_on("土壤指标", "soil"):
    result["土壤指标"] = {
        "土石比": (data.get("soil_ratio"),
                   std_benefit(data.get("soil_ratio"),
                               ideal("soil_ratio",
IDEAL_SOIL_RATIO)),
                   l2w["土壤指标"]["土石比"]),
        "土层厚度": (data.get("thickness"),
                     std_benefit(data.get("thickness"),
                                 ideal("thickness",
IDEAL_THICKNESS)),
                     l2w["土壤指标"]["土层厚度"]),
        "pH": (data.get("ph"),
               std_mid_ph(data.get("ph"),
*IDEAL_PH_RANGE),
               l2w["土壤指标"]["pH"]),
        "有机质": (data.get("organic_matter"),
                   std_benefit(data.get("organic_matter")
,
                               ideal("organic_matter",
IDEAL_ORGANIC)),
                   l2w["土壤指标"]["有机质"]),
        "污染指标": (data.get("heavy_metals") or {},
                     pollution_score(data.get("heavy_meta
ls") or {},
```

```
        uc.metal_limits(settings)),                                          1451
                                    l2w["土壤指标"]["污染指标"]),              1452
            }                                                                1453
        #  ----  植被指标  ----                                               1454
        if  "植被指标"  in  l1_enabled  and  group_on("植被指标",  "veg"):     1455
            result["植被指标"]  =  {                                         1456
                "植被覆盖度":  (data.get("coverage"),                        1457
                                    std_benefit(data.get("coverage"),        1458
                                                ideal("coverage",            1459
        IDEAL_COVERAGE)),                                                    1460
                                    l2w["植被指标"]["植被覆盖度"]),           1461
                "生物多样性":  (data.get("shannon"),                         1462
                                    std_benefit(data.get("shannon"),         1463
                                                ideal("shannon",             1464
        IDEAL_SHANNON)),                                                     1465
                                    l2w["植被指标"]["生物多样性"]),           1466
            }                                                                1467
        #  ----  地形指标  ----                                               1468
        if  "地形指标"  in  l1_enabled  and  group_on("地形指标",  "terrain"):1469
            result["地形指标"]  =  {                                         1470
                "坡度":  (data.get("slope"),                                 1471
                          std_slope(data.get("slope")),                      1472
                          l2w["地形指标"]["坡度"]),                          1473
                "坡向":  (data.get("aspect"),                                1474
                          aspect_score(data.get("aspect")),                  1475
                          l2w["地形指标"]["坡向"]),                          1476
            }                                                                1477
        return  result                                                       1478
#  ====  app/services/weight_service.py  ====                                1479
#  -*-  coding:  utf-8  -*-                                                  1480
"""权重方案的存取与生效权重解析。                                            1481
系统有两套权重来源：                                                        1482
1．  论文表  3  给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），     1483
    是**权威取值来源**，案例复算与报告结论以此为准；                        1484
2．  通过层次分析法在本系统中实际算出的权重，保存在                         1485
    weight_schemes  /  weight_items  两张表中，可标记为「生效」。           1486
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                    1487
以论文表  3  为底，用生效方案中已保存的条目覆盖。                          1488
没有任何生效方案时，即完全使用论文表  3  的权重。                          1489
这样既保证默认行为与论文一致，又使  AHP  计算结果真正参与评价，            1490
而不是算完即弃。                                                            1491
"""                                                                         1492
import  copy                                                                1493
from  app.models  import  db,  WeightScheme,  WeightItem                    1494
from  app.utils.constants  import  DEFAULT_WEIGHTS                          1495
from  app.services.indicator_service  import  TREE                          1496
#  方案名称：每个微地质单元类型固定一个可生效的方案                          1497
ACTIVE_NAME  =  "自定义权重方案（AHP  计算结果）"                           1498
def  _code_maps():                                                          1499
    """由指标树导出  编码  →  名称  的映射，避免名称在两处各写一遍。"""      1500
```

```
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights       嵌套权重结构，可直接交给标准化与评价引擎
        source_label  权重来源说明，用于界面与报告标注
        scheme        生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme
def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
```

```
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,
    name=ACTIVE_NAME).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
    name=ACTIVE_NAME,
                              level=1, is_default=True)
        db.session.add(scheme)
        db.session.flush()
    if note:
        scheme.level = scheme.level or 1
    if not scheme.is_default:
        scheme.is_default = True
    written = 0
    for code, w in code_weight_pairs:
        if w is None or w <= 0:
            continue
        item = WeightItem.query.filter_by(scheme_id=scheme.i
    d,
    indicator_code=code).first()
        if item is None:
            item = WeightItem(scheme_id=scheme.id,
    indicator_code=code)
            db.session.add(item)
        item.weight = float(w)
        written += 1
    db.session.commit()
    return scheme, written
def reset_weights(unit_type):
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty
    pe,
    name=ACTIVE_NAME).all()
    n = 0
    for s in schemes:
        WeightItem.query.filter_by(scheme_id=s.id).delete(
            synchronize_session=False)
        db.session.delete(s)
        n += 1
    db.session.commit()
    return n
def scheme_items(unit_type):
    """返回该单元类型已保存的自定义权重条目 {指标编码: 权重}。"""
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,
    name=ACTIVE_NAME).first()
```

```
    if scheme is None:                                                    1601
        return {}, None                                                   1602
    items = {i.indicator_code: i.weight                                   1603
             for i in WeightItem.query.filter_by(scheme_id=s              1604
    cheme.id).all()}                                                      1605
    return items, scheme                                                  1606
def summary():                                                            1607
    """各单元类型的权重来源概览，供权重首页展示。"""                                  1608
    rows = []                                                             1609
    for unit_type in DEFAULT_WEIGHTS:                                     1610
        weights, label, scheme = effective_weights(unit_type              1611
    )                                                                     1612
        rows.append({                                                     1613
            "unit_type": unit_type,                                       1614
            "label": label,                                               1615
            "custom": scheme is not None,                                 1616
        })                                                                1617
    return rows                                                           1618
# ==== app/services/evaluation_service.py ====                            1619
# -*- coding: utf-8 -*-                                                   1620
"""适宜性评价引擎。                                                            1621
按论文式（9） 实现：                                                           1622
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）                          1623
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比                             1624
- 等级判定按论文表 4。                                                        1625
"""                                                                       1626
from app.utils.constants import DEFAULT_WEIGHTS, grade_of                 1627
from app.services import standardization as std                           1628
def score_unit(unit_type, data, weights=None,                             1629
    settings=None):                                                       1630
    """计算单个微地质单元的评价因子 F'。                                             1631
    参数                                                                  1632
    ----                                                                  1633
    weights  : 该单元类型的权重；不传用论文默认权重                                   1634
    settings : 图斑级标准化参数（基准值与理想取值）；不传全部按代码常量                        1635
    返回                                                                  1636
    ----                                                                  1637
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:          1638
    [...] }                                                               1639
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，                              1640
    则按其余项权重归一化后计算，保证单元得分始终有定义。                                   1641
    """                                                                   1642
    if weights is None:                                                   1643
        weights = DEFAULT_WEIGHTS[unit_type]                              1644
    std_result = std.standardize_unit(unit_type, data,                    1645
    weights, settings)                                                    1646
    l1_weights = weights["l1"]                                            1647
    l1_scores = {}                                                        1648
    details = []                                                          1649
    for l1_name, items in std_result.items():                             1650
```

```
            wsum = 0.0
            acc = 0.0
            for l2_name, (raw, std_value, w2) in items.items():
                if std_value is None:
                    # 不适用项：不计入
                    details.append({
                        "l1": l1_name, "l2": l2_name, "raw":
raw,
                        "std_value": None, "weight": w2,
"contribution": None,
                    })
                    continue
                contribution = std_value * w2
                acc += contribution
                wsum += w2
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw": raw,
                    "std_value": std_value, "weight": w2,
                    "contribution": contribution,
                })
            l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
        # 一级指标权重按实际参与计算的指标归一化。
        # 论文表 3 的权重本身已归一（和为 1），此处为恒等变换；
        # 若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
        lw_sum = sum(l1_weights[k] for k in l1_scores if k in
l1_weights)
        if lw_sum > 0:
            unit_score = sum(l1_weights[k] * l1_scores[k]
                             for k in l1_scores if k in
l1_weights) / lw_sum
        else:
            unit_score = 0.0
        return {"l1_scores": l1_scores, "unit_score":
unit_score, "details": details}
def score_mine(units, weights_map=None, settings=None):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型: 权重结构}。未提供的单元类型
        使用论文表 3 的默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
    settings : 图斑级标准化参数（参照生态系统基准、指标理想取值），
        由 app.services.unit_config.for_patch() 生成；不传按代码常量。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
```

```
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]),
    settings)
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
            "l1_scores": r["l1_scores"],
            "details": r["details"],
        })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
from app.utils.timefmt import to_local, now_utc
from app.services import workflow as wf
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
```

```
        p.alignment = align                                                    1751
    p.paragraph_format.space_after = Pt(space_after)                           1752
    _set_run(p.add_run(), text, size, bold)                                    1753
    return p                                                                   1754
def _heading(doc, text, level=1):                                              1755
    h = doc.add_heading(text, level=level)                                     1756
    for run in h.runs:                                                         1757
        run.font.name = "Times New Roman"                                      1758
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")                  1759
    return h                                                                   1760
def _table(doc, headers, rows, widths=None):                                   1761
    t = doc.add_table(rows=1, cols=len(headers))                               1762
    t.style = "Table Grid"                                                     1763
    hdr = t.rows[0].cells                                                      1764
    for i, htext in enumerate(headers):                                        1765
        _set_run(hdr[i].paragraphs[0].add_run(), htext,                        1766
    10.5, bold=True)                                                           1767
    for row in rows:                                                           1768
        cells = t.add_row().cells                                              1769
        for i, val in enumerate(row):                                          1770
            _set_run(cells[i].paragraphs[0].add_run(),                         1771
    str(val), 10.5)                                                            1772
    return t                                                                   1773
def build_report(patch, mine_result):                                          1774
    """生成评价报告 Word 文档，返回 BytesIO。                                  1775
    参数                                                                       1776
    ----                                                                       1777
    patch : MinePatch，图斑对象                                                1778
    mine_result : evaluation_service.score_mine 的返回 dict                    1779
    """                                                                        1780
    doc = Document()                                                           1781
    # 页边距                                                                   1782
    for section in doc.sections:                                               1783
        section.top_margin = Cm(2.5)                                           1784
        section.bottom_margin = Cm(2.5)                                        1785
        section.left_margin = Cm(2.5)                                          1786
        section.right_margin = Cm(2.5)                                         1787
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)                 1788
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0",           1789
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)                          1790
    _para(doc, "生成时间：%s" % to_local(now_utc(), "%Y-%m-%d                  1791
    %H:%M:%S"),                                                                1792
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,                          1793
    space_after=12)                                                            1794
    # 图斑信息                                                                 1795
    _heading(doc, "一、图斑基本信息", 1)                                       1796
    info = [                                                                   1797
        ("图斑名称", patch.name),                                              1798
        ("所在行政区", patch.admin_region or "—"),                             1799
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is            1800
```

```
not None else "—"),
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
]
_table(doc, ["项目", "内容"], info)
#  评价结论
_heading(doc, "二、评价结论", 1)
gi = mine_result["grade_info"]
concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
]
_table(doc, ["项目", "结果"], concl)
#  各单元得分
_heading(doc, "三、各微地质单元得分", 1)
rows = []
for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]
* 100),
                     "%.4f" % u["unit_score"], "%.4f" %
u["weighted"]))
_table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],
rows)
#  完整得分明细
_heading(doc, "四、得分明细", 1)
for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],
u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
                raw = "—" if d["raw"] is None else d["raw"]
                std = "—" if d["std_value"] is None else "%.4f"
% d["std_value"]
                w = "%.4f" % d["weight"]
                c = "—" if d["contribution"] is None else
"%.4f" % d["contribution"]
                detail_rows.append((d["l1"], d["l2"], raw, std,
w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
"贡献"],
               detail_rows)
        #  一级指标得分
        l1_row = []
        for k, v in u["l1_scores"].items():
                l1_row.append("%s=%.4f" % (k, v))
        _para(doc, "一级指标得分：" + "，".join(l1_row), size=10.5,
space_after=10)
```

```
    #  数据完整度：未检测项必须写明，否则读报告的人会以为这些项是实测达标
    undetected = wf.undetected_metals(patch.id)
    if undetected:
        _heading(doc, "五、数据完整度说明", 1)
        parts = ["%s：%s" % (t, "、".join(m)) for t, m in
undetected]
        _para(doc, "以下重金属项未检测，按论文注记以「未受污染」计入污染指标"
                   "（并非实测达标）：%s。" % "；".join(parts),
size=10.5)
    buf = io.BytesIO()
    doc.save(buf)
    buf.seek(0)
    return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    #  ----  一级指标  ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    #  ----  二级指标：土壤  ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
```

```
        IDEAL_PH_RANGE, False),                                                 1901
        ("organic_matter", "有机质", 2, "soil", "g/kg",                          1902
        str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",                          1903
         "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),  1904
        ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",               1905
        "cost", "碎石边坡,平台",                                                 1906
         "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),        1907
        # ---- 二级指标：植被 ----                                               1908
        ("coverage", "植被覆盖度", 2, "veg", "%",                                1909
        str(IDEAL_COVERAGE), "benefit", ALL_UNITS,                              1910
         "论文 4.2.2 节：参照生态系统取值 75%", False),                           1911
        ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),             1912
        "benefit", ALL_UNITS,                                                   1913
         "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),              1914
        # ---- 二级指标：地形 ----                                               1915
        ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %                        1916
        SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",                                1917
         "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°  1918
        为陡坡重要分界", False),                                                 1919
        ("aspect", "坡向", 2, "terrain", "—",                                   1920
        "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",           1921
         "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),          1922
]                                                                               1923
# 污染指标下的三级指标（8 种重金属）                                             1924
for _m, _limit in HEAVY_METAL_LIMIT.items():                                    1925
    TREE.append((                                                               1926
        "hm_" + _m, _m, 3, "pollution", "mg/kg",                                1927
    str(_limit), "cost",                                                        1928
        "碎石边坡,平台",                                                         1929
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %    1930
    _limit,                                                                     1931
        False,                                                                  1932
    ))                                                                          1933
def seed_indicators():                                                          1934
    """写入/更新指标定义（幂等）。返回写入条数。"""                                1935
    code_map = {                                                                1936
        "benefit": "越大越优", "cost": "越小越优",                               1937
        "mid": "适中型", "discrete": "离散赋值",                                 1938
    }                                                                           1939
    n = 0                                                                       1940
    for (code, name, level, parent, unit, ideal, itype,                         1941
    units, desc, is_con) in TREE:                                               1942
        ind = Indicator.query.filter_by(code=code).first()                      1943
        if ind is None:                                                         1944
            ind = Indicator(code=code)                                          1945
            db.session.add(ind)                                                 1946
        ind.name = name                                                         1947
        ind.level = level                                                       1948
        ind.parent_code = parent                                                1949
        ind.unit = unit                                                         1950
```

```
            ind.ideal_value = ideal
            ind.ideal_type = itype
            ind.enabled_units = units
            ind.description = desc
            ind.is_constraint = is_con
            ind.is_evaluative = (level in (2, 3))
            n += 1
        db.session.commit()
        return n
def ideal_type_label(code):
        return {"benefit": "越大越优", "cost": "越小越优",
                "mid": "适中型", "discrete": "离散赋值"}.get(code,
        code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
        单元类型， 土石比， 土层厚度， pH， 有机质，
        Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
        植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
        UNIT_TYPES
#  表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
                + list(HEAVY_METALS)
                + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
        "单元类型": "unit_type",
        "土石比": "soil_ratio",
        "土层厚度": "thickness",
        "pH": "ph", "PH": "ph", "ph": "ph",
        "有机质": "organic_matter",
        "植被覆盖度": "coverage",
        "Shannon": "shannon", "shannon": "shannon", "生物多样性":
        "shannon",
        "坡度": "slope",
        "坡向": "aspect",
}
for _m in HEAVY_METALS:
        FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
        """生成导入模板（含表头与示例行）。"""
        buf = io.StringIO()
        w = csv.writer(buf)
```

```
    w.writerow(HEADERS)
    #  示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
    None)
                        for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
```

```
            return None, "「%s」不是有效数值" % s                                    2051
def parse_upload(filename, stream, unit_type_map):                                  2052
    """解析上传文件。                                                               2053
    参数                                                                            2054
    ----                                                                            2055
    unit_type_map : dict，{单元类型: MicroUnit 对象}                                2056
    返回                                                                            2057
    ----                                                                            2058
    (parsed, errors)                                                                2059
      parsed : list[dict]，每项 {unit_type, fields{...},                            2060
    heavy_metals{}}                                                                 2061
      errors : list[str]，逐行问题说明                                              2062
    """                                                                             2063
    rows = _read_rows(filename, stream)                                             2064
    parsed, errors = [], []                                                         2065
    if not rows:                                                                    2066
        return [], ["文件为空或没有数据行"]                                         2067
    # 表头校验                                                                      2068
    unknown = [h for h in (rows[0].keys() if rows else [])                          2069
    if h not in FIELD_ALIASES]                                                      2070
    if unknown:                                                                     2071
        errors.append("存在无法识别的列，将被忽略：%s" %                            2072
"、".join(map(str, unknown)))                                                       2073
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头                2074
        rec = {}                                                                    2075
        for k, v in raw.items():                                                    2076
            key = FIELD_ALIASES.get(str(k).strip())                                 2077
            if key:                                                                 2078
                rec[key] = v                                                        2079
        unit_type = str(rec.get("unit_type") or "").strip()                         2080
        if not unit_type:                                                           2081
            errors.append("第 %d 行：缺少「单元类型」" % idx)                       2082
            continue                                                                2083
        if unit_type not in unit_type_map:                                          2084
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"        2085
                          % (idx, unit_type,                                        2086
"、".join(unit_type_map) or "无"))                                                  2087
            continue                                                                2088
        if unit_type not in UNIT_TYPES:                                             2089
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"          2090
                          % (idx, unit_type))                                       2091
            continue                                                                2092
        fields, heavy = {}, {}                                                      2093
        ok = True                                                                   2094
        for key, val in rec.items():                                                2095
            if key in ("unit_type",):                                               2096
                continue                                                            2097
            if key.startswith("hm_"):                                               2098
                n, err = _num(val)                                                  2099
                if err:                                                             2100
```

```
                    errors.append("第 %d 行:%s %s" % (idx,
    key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
                s = str(val).strip() if val is not None
    else ""
                if s == "":
                    fields["aspect"] = None
                elif s in ASPECT_TYPES:
                    fields["aspect"] = s
                else:
                    errors.append("第 %d 行:坡向「%s」无效（应为 %s）"
                                  % (idx, s,
    "、".join(ASPECT_TYPES)))
                    ok = False
                continue
            n, err = _num(val)
            if err:
                errors.append("第 %d 行:%s %s" % (idx, key,
    err))
                ok = False
            else:
                fields[key] = n
        if ok:
            parsed.append({"unit_type": unit_type,
    "fields": fields,
                           "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
论文 4.1 节确定的评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
from app.services import unit_config
#  指标数据非空字段（用于判断某单元是否已录入数据）
```

```
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
                "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def _data_of(ind):
    """把 IndicatorData 行转成标准化接口要的 dict。"""
    if ind is None:
        return {}
    data = {f: getattr(ind, f, None) for f in DATA_FIELDS}
    data["heavy_metals"] = ind.heavy_metals_dict()
    return data
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units          {done, count, ratio_sum, detail}
        constraint     {done, passed, detail}
        data           {done, ready, missing, detail}
        blocking       未满足的前置步骤列表（按流程顺序）
        ready          是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    # ---- 步骤一：微地质单元 ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    # ---- 步骤二：限制性条件判定 ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)
    \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
```

```
                                "detail": "限制性条件判定未通过（%s）" %
(cc.reason or "")}
    # ---- 步骤三：指标数据 ----
    # 判断标准是「该单元类型启用的必填项是否齐全」，而不是「有没有填过任意一个字段」。
    # 早先只要 8 个字段里有一个非空就算齐，于是碎石边坡只填一个坡度也能跑出
    # F=0.69「基本适宜」；而 score_unit 又会把缺的项剔掉、把权重摊给其余指标，
    # 结论方向还偏乐观。污染指标按论文注记允许「未检测」，不算缺。
    settings = unit_config.for_patch(patch_id)
    missing = []                    # 完全没录数据的单元类型
    incomplete = []                 # 录了但不齐的单元类型 → [(单元类型, [缺的字段])]
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
rst()
        data = _data_of(ind)
        gaps = settings.missing_fields(u.unit_type, data)
        if not _has_data(ind):
            missing.append(u.unit_type)
        elif gaps:
            incomplete.append((u.unit_type, gaps))
    if not units:
        data_step = {"done": False, "ready": False,
"missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
"missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    elif incomplete:
        data_step = {"done": False, "ready": False,
                     "missing": [t for t, _ in incomplete],
                     "detail": "以下微地质单元的必填指标未填齐：%s"
                               % "；".join("%s缺 %s" % (t,
"、".join(g))
                                           for t, g in
incomplete)}
    else:
        data_step = {"done": True, "ready": True,
"missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
len(units)}
    # 占比合计必须落在 100% 附近才算通过。只挡「偏小」是不够的：
    # 库里若出现合计 1.5 的记录（手工改库、导入异常），F = Σ(Si×F'i)
    # 会被放大到 1.5 倍，所以上下都要挡。
    ratio_ok = 0.999 <= ratio_sum <= 1.001 if units else
False
    if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 100%，请检查面积录入）"
    # ---- 汇总未满足的前置条件（按论文 4.1 节的流程顺序：限制性判定优先） ----
    # reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
```

```
    # 缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
    blocking = []
    if not constraint_step["done"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未执行",
            "reason": "五项限制性条件尚未逐项判定。",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去执行判定",
        })
    elif not constraint_step["passed"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未通过",
            "reason": cc.reason or "",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去复核判定结果",
        })
    if not units_step["done"]:
        blocking.append({
            "step": "units", "label": "未划分微地质单元",
            "reason": "该图斑下没有任何微地质单元。",
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去划分",
        })
    elif not ratio_ok:
        blocking.append({
            "step": "units", "label": "面积占比合计不为 100%",
            "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去核对面积",
        })
    if units_step["done"] and not data_step["ready"]:
        blocking.append({
            "step": "data", "label": "指标数据未录入完整",
            "reason": data_step["detail"],
            "fix_url": "/indicator/", "fix_label": "去录入",
        })
    return {
        "units": units_step,
        "constraint": constraint_step,
        "data": data_step,
        "blocking": blocking,
        "ready": not blocking,
    }
def undetected_metals(patch_id):
    """列出该图斑各单元「未检测」的重金属，供结果页与报告如实标注。
    论文注记允许未检测的重金属按未受污染计入，但报告里必须写明是哪些 ——
    否则读报告的人会以为这些项是实测达标。
    只列启用了土壤指标的单元：岩质边坡本来就不评价土壤，列 8 项未检测是噪音。
    """
    settings = unit_config.for_patch(patch_id)
```

```
    out = []
    for u in MicroUnit.query.filter_by(patch_id=patch_id).all():
        if settings.group_enabled(u.unit_type, "土壤指标", "soil") is False:
            continue
        ind = IndicatorData.query.filter_by(unit_id=u.id).first()
        if ind is None:
            continue
        metals = settings.undetected_metals(_data_of(ind))
        if metals:
            out.append((u.unit_type, metals))
    return out
def latest_result(patch_id):
    """取该图斑最近一次的评价记录。"""
    return EvaluationResult.query.filter_by(patch_id=patch_id) \
        .order_by(EvaluationResult.id.desc()).first()
def report_progress(patch_id):
    """报告输出的前置条件：评价流程齐备，且已有评价记录。
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
    导出时读取已保存的评价结果，而不是在导出环节重新计算。
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
    同一份评价出现两个数字，结论无法追溯。
    返回 dict：{ready, blocking, result, flow}
        result 为最近一次评价记录（EvaluationResult），无记录时为空。
    """
    flow = patch_progress(patch_id)
    blocking = list(flow["blocking"])
    record = latest_result(patch_id)
    if not blocking and record is None:
        blocking.append({
            "step": "evaluate", "label": "尚未执行适宜性评价",
            "reason": "该图斑还没有评价记录。",
            "fix_url": "/evaluate/%d" % patch_id,
            "fix_label": "去执行评价",
        })
    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
```

```
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
          href="/project/",
          desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
          href="/standard/",
          desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
    dict(key="constraint", no="03", title="限制性指标判定",
    icon="shield",
          href="/constraint/",
          desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),
    dict(key="unit", no="04", title="微地质单元", icon="layers",
          href="/unit/",
          desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
    dict(key="ahp", no="05", title="指标体系与权重",
    icon="sliders",
          href="/ahp/",
          desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
    dict(key="indicator", no="06", title="指标数据采集",
    icon="database",
          href="/indicator/",
          desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
    dict(key="evaluate", no="07", title="适宜性定量评价",
    icon="target",
          href="/evaluate/",
          desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
    dict(key="report", no="08", title="报告与结果输出",
    icon="file",
          href="/report/",
          desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
]
def _has_data(ind):
    return wf._has_data(ind)
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）
MODULES = [
```

```
        ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),
        ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),
        ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),
        ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),
        ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),
        ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),
        ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),
        ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),
        ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
    ]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
        MicroUnit.patch_id.in_(patch_ids)).all() if
    patch_ids else []
    unit_by_patch = {}
    for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
    unit_ids = [u.id for u in units]
    data_unit_ids = set()
    if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
    ():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
```

```
    for r in EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).all():
        evaluated.add(r.patch_id)
# ---- 各步骤的待办数量 ----
pending = {
    "constraint": sum(1 for p in patches if p.id not in passed_constraint),
    "unit": sum(1 for p in patches if not unit_by_patch.get(p.id)),
    "indicator": 0,
    "evaluate": 0,
}
ready_patches = 0
for p in patches:
    us = unit_by_patch.get(p.id, [])
    if us and any(u.id not in data_unit_ids for u in us):
        pending["indicator"] += 1
    pr = wf.patch_progress(p.id)
    if pr["ready"]:
        ready_patches += 1
        if p.id not in evaluated:
            pending["evaluate"] += 1
# ---- 步骤可用性：前置数据不具备时置灰 ----
avail = {
    "project": True,
    "standard": True,
    "constraint": bool(patches),
    "unit": bool(patches),
    "ahp": True,
    "indicator": bool(units),
    "evaluate": ready_patches > 0,
    "report": bool(evaluated),
}
notes = {
    "constraint": "需先建立矿山图斑",
    "unit": "需先建立矿山图斑",
    "indicator": "需先划分微地质单元",
    "evaluate": "需前置步骤齐备的图斑",
    "report": "需先执行适宜性评价",
}
steps = []
for s in STEPS:
    steps.append(dict(
        s,
        available=avail[s["key"]],
        pending=pending.get(s["key"], 0),
        note=None if avail[s["key"]] else
notes.get(s["key"], ""),
```

```
        ))                                                                  2501
    unit_types = len({u.unit_type for u in units})                          2502
    stats = {                                                               2503
        "projects": projects,                                               2504
        "patches": len(patches),                                            2505
        "unit_types": unit_types,                                           2506
        "standards": Standard.query.count(),                                2507
    }                                                                       2508
    return {                                                                2509
        "stats": stats,                                                     2510
        "steps": steps,                                                     2511
        "badges": {k: v for k, v in pending.items() if v},                  2512
        "progress": {                                                       2513
            "patches": len(patches),                                        2514
            "units": len(units),                                            2515
            "data_units": len(data_unit_ids),                               2516
            "ready": ready_patches,                                         2517
            "evaluated": len(evaluated),                                    2518
        },                                                                  2519
    }                                                                       2520
def nav_badges():                                                           2521
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""                2522
    snap = snapshot()                                                       2523
    return {                                                                2524
        "/constraint/": snap["badges"].get("constraint", 0),               2525
        "/unit/": snap["badges"].get("unit", 0),                            2526
        "/indicator/": snap["badges"].get("indicator", 0),                  2527
        "/evaluate/": snap["badges"].get("evaluate", 0),                    2528
    }                                                                       2529
# ==== app/services/audit.py ====                                           2530
# -*- coding: utf-8 -*-                                                     2531
"""操作审计。                                                               2532
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）  2533
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。               2534
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。       2535
"""                                                                         2536
from flask import request                                                   2537
from flask_login import current_user                                        2538
from app.models import db, OperationLog                                     2539
# 端点 → 可读动作名                                                         2540
ACTION_NAMES = {                                                            2541
    "auth.login": "登录系统",                                               2542
    "auth.register": "注册账号",                                            2543
    "auth.logout": "退出登录",                                              2544
    "auth.add_user": "新增账号",                                            2545
    "auth.edit_user": "编辑账号",                                           2546
    "auth.delete_user": "删除账号",                                         2547
    "auth.reset_password": "重置用户密码",                                  2548
    "auth.profile": "修改个人资料",                                         2549
    "auth.upload_avatar": "更新头像",                                       2550
```

```
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    # 个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
```

```
            return                                                    2601
        try:                                                          2602
            detail = "路径 %s" % request.path                         2603
            if status_code >= 400:                                    2604
                detail += "；结果 HTTP %d" % status_code              2605
            db.session.add(OperationLog(                              2606
                user_id=current_user.id if                            2607
        current_user.is_authenticated else None,                      2608
                action=action_name(endpoint),                         2609
                target_type=(endpoint or "").split(".")[0] or         2610
        None,                                                         2611
                target_id=extract_target_id(),                        2612
                detail=detail,                                        2613
            ))                                                        2614
            db.session.commit()                                       2615
        except Exception:                                             2616
            #  审计失败不应影响业务响应                               2617
            db.session.rollback()                                     2618
#  ==== app/views/auth.py ====                                       2619
#  -*- coding: utf-8 -*-                                             2620
"""账号与权限 —— 登录、注册、个人资料、用户维护、操作日志。          2621
页面组织：                                                          2622
   /login          登录页（含「点我注册」入口）                     2623
   /register       自助注册（默认「操作员」）                       2624
   /profile        个人设置，分类切换：info / security / users / logs 2625
                   后两个分类仅管理员可见，由本模块统一管控。       2626
"""                                                                 2627
from flask import (Blueprint, render_template, redirect,            2628
        url_for, request,                                           2629
                           flash, abort, current_app)               2630
from flask_login import login_user, logout_user,                    2631
        login_required, current_user                                2632
from werkzeug.security import check_password_hash,                  2633
        generate_password_hash                                      2634
from app.forms import (LoginForm, EmptyForm, ProfileForm,           2635
        PasswordForm,                                               2636
                                RegisterForm, email_ok)             2637
from app.models import db, User, OperationLog                       2638
from app.utils import uploads as up                                 2639
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE,          2640
        GENDERS                                                     2641
bp = Blueprint("auth", __name__)                                    2642
#  「个人设置」的四个分类；后两个是管理员专属                       2643
PROFILE_TABS = ("info", "security", "users", "logs")                2644
ADMIN_TABS = ("users", "logs")                                      2645
def require_admin():                                                2646
        """非管理员直接 403，避免误操作基础数据。"""               2647
        if current_user.role != "admin":                            2648
            abort(403)                                              2649
def _intro_data():                                                  2650
```

```
    """登录页下方的系统介绍所用的静态内容。"""
    from app.services import dashboard as dash
    return {"modules": dash.MODULES}
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html",
    dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "toast-error")
            else:
                login_user(user, remember=form.remember.data
    )
                # 用 toast 类别：渲染为页面中上方的浮层，淡入停留后淡出。
                # 登录成功只是告知性反馈，不需要用户停下来阅读。
                flash("登录成功", "toast")
                return redirect(url_for("auth.index"))
        else:
            flash("账户名称或密码错误", "toast-error")
    # 字段为空之类的校验失败，错误留在表单里逐项显示，不走浮层：
    # 那类消息要对着输入框看，一闪而过反而让人以为「点了没反应」。
    return render_template("auth/login.html", form=form,
    **_intro_data())
@bp.route("/register", methods=["GET", "POST"])
def register():
    """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = RegisterForm()
    if form.validate_on_submit():
        username = form.username.data.strip()
        if User.query.filter_by(username=username).first():
            form.username.errors.append("该账户名称已被使用")
        else:
            user = User(username=username,
                        role=DEFAULT_ROLE,
                        password_hash=generate_password_hash
    (form.password.data),
```

```
                display_name=(form.display_name.data
    or "").strip() or None,
                gender=(form.gender.data or
    "").strip() or None,
                email=(form.email.data or
    "").strip() or None,
                phone=(form.phone.data or
    "").strip() or None)
            rel, err = _save_avatar(request.files.get("avata
    r"))
            if err:
                flash(err, "error")
            else:
                user.avatar = rel
                db.session.add(user)
                db.session.commit()
                login_user(user)
                flash("注册成功，欢迎使用", "toast")
                return redirect(url_for("auth.index"))
    return render_template("auth/register.html", form=form)
@bp.route("/logout")
@login_required
def logout():
    logout_user()
    flash("已退出登录", "ok")
    return redirect(url_for("auth.login"))
# ------------------------------ 个人设置 ------------------------
    ----
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。
@bp.route("/profile", methods=["GET", "POST"])
@login_required
def profile():
    is_admin = current_user.role == "admin"
    tab = request.values.get("tab") or "info"
    if tab not in PROFILE_TABS or (tab in ADMIN_TABS and
    not is_admin):
        tab = "info"
    # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名
    info_form = ProfileForm(prefix="info", obj=current_user)
    pwd_form = PasswordForm(prefix="pwd")
    if request.method == "POST":
        action = request.form.get("action")
        # 校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，
        # 否则错误提示会出现在另一个分类里，用户根本看不到。
        if action == "password":
            tab = "security"
        elif action == "info":
            tab = "info"
        if action == "info" and info_form.validate_on_submit
```

```
    ():                                                                 2751
            current_user.display_name =                                 2752
    (info_form.display_name.data or "").strip() or None                 2753
            current_user.gender = (info_form.gender.data or             2754
    "").strip() or None                                                 2755
            current_user.email = (info_form.email.data or               2756
    "").strip() or None                                                 2757
            current_user.phone = (info_form.phone.data or               2758
    "").strip() or None                                                 2759
            db.session.commit()                                         2760
            flash("个人资料已保存", "ok")                                 2761
            return redirect(url_for("auth.profile",                     2762
    tab="info"))                                                        2763
        if action == "password" and pwd_form.validate_on_sub            2764
    mit():                                                              2765
            if not check_password_hash(current_user.password            2766
    _hash,                                                              2767
                                                                        2768
    pwd_form.old_password.data):                                        2769
                pwd_form.old_password.errors.append("当前密码不正确          2770
    ")                                                                  2771
            else:                                                       2772
                current_user.password_hash =                            2773
    generate_password_hash(                                             2774
                    pwd_form.new_password.data)                         2775
                db.session.commit()                                     2776
                flash("密码已修改", "ok")                                 2777
                return redirect(url_for("auth.profile",                 2778
    tab="security"))                                                    2779
    ctx = dict(info_form=info_form, pwd_form=pwd_form,                  2780
    avatar_form=EmptyForm(),                                            2781
                tab=tab, role_labels=ROLE_LABELS,                       2782
    genders=GENDERS)                                                    2783
    if tab == "users":                                                  2784
        ctx.update(_users_context())                                    2785
    elif tab == "logs":                                                 2786
        ctx.update(_logs_context())                                     2787
    return render_template("auth/profile.html", **ctx)                  2788
def _users_context():                                                   2789
    """管理员分类「账号」所需的数据。"""                                      2790
    return {"users": User.query.order_by(User.id).all(),                2791
            "form": EmptyForm()}                                        2792
def _logs_context():                                                    2793
    """管理员分类「操作日志」所需的数据。"""                                  2794
    per_page = 50                                                       2795
    q = OperationLog.query.order_by(OperationLog.id.desc())             2796
    total = q.count()                                                   2797
    pages = max(1, (total + per_page - 1) // per_page)                  2798
    # 页码先夹到 [1, pages] 再算 offset：手工把 page 写成天文数字时，                2799
    # (page-1)*per_page 会超出 SQLite 能绑定的整数范围，直接 500。              2800
```

```
    try:                                                                2801
        page = int(request.args.get("page", 1))                         2802
    except (TypeError, ValueError):                                     2803
        page = 1                                                        2804
    page = max(1, min(page, pages))                                     2805
    records = q.offset((page - 1) * per_page).limit(per_page2806
    ).all()                                                             2807
    users_map = {u.id: u.username for u in User.query.all()}2808
    return {"records": records, "page": page, "pages":                  2809
    pages,                                                              2810
            "total": total, "users_map": users_map}                     2811
# ---------------------------- 头像 ----------------------------2812
    --                                                                  2813
def _avatar_length_ok(file_storage):                                    2814
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""                2815
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *  2816
    1024 * 1024)                                                        2817
    stream = file_storage.stream                                        2818
    pos = stream.tell()                                                 2819
    stream.seek(0, 2)                                                   2820
    size = stream.tell()                                                2821
    stream.seek(pos)                                                    2822
    return size <= limit, size                                          2823
def _save_avatar(file):                                                 2824
    """保存头像文件，返回 (相对路径, 错误信息)。没有文件时返回 (None, None)。"""          2825
    if file is None or not file.filename:                               2826
        return None, None                                               2827
    ok, size = _avatar_length_ok(file)                                  2828
    if not ok:                                                          2829
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"              2830
                      % (current_app.config["MAX_AVATAR_LENG2831
    TH"] / 1024 / 1024,                                                 2832
                         size / 1024 / 1024))                           2833
    return up.save_upload(file, "avatars",                              2834
                          current_app.config["ALLOWED_AVATAR2835
    _EXT"],                                                             2836
                          current_app.config["UPLOAD_FOLDER"2837
    ])                                                                  2838
@bp.route("/profile/avatar", methods=["POST"])                          2839
@login_required                                                         2840
def upload_avatar():                                                    2841
    form = EmptyForm()                                                  2842
    back = url_for("auth.profile", tab=request.form.get("tab2843
    ") or "info")                                                       2844
    if not form.validate_on_submit():                                   2845
        flash("表单校验失败，请重试", "error")                                2846
        return redirect(back)                                           2847
    rel, err = _save_avatar(request.files.get("avatar"))                2848
    if err:                                                             2849
        flash(err, "error")                                             2850
```

```
            return redirect(back)                                          2851
        if rel is None:                                                    2852
            flash("请选择头像图片", "error")                                2853
            return redirect(back)                                          2854
        # 换新头像时删掉旧文件，避免残留                                    2855
        up.remove_upload(current_user.avatar,                              2856
        current_app.config["UPLOAD_FOLDER"])                               2857
        current_user.avatar = rel                                          2858
        db.session.commit()                                                2859
        flash("头像已更新", "ok")                                          2860
        return redirect(back)                                              2861
@bp.route("/profile/avatar/delete", methods=["POST"])                      2862
@login_required                                                            2863
def delete_avatar():                                                       2864
        form = EmptyForm()                                                 2865
        back = url_for("auth.profile", tab=request.form.get("tab2866
        ") or "info")                                                      2867
        if form.validate_on_submit():                                      2868
            up.remove_upload(current_user.avatar,                          2869
        current_app.config["UPLOAD_FOLDER"])                               2870
            current_user.avatar = None                                     2871
            db.session.commit()                                            2872
            flash("头像已移除", "ok")                                      2873
        return redirect(back)                                              2874
# ---------------------------- 账号维护（仅管理员）                        2875
        ------------------------------                                     2876
@bp.route("/users")                                                        2877
@login_required                                                            2878
def users():                                                               2879
        """旧地址保留，统一并入「个人设置 · 账号」。"""                    2880
        return redirect(url_for("auth.profile", tab="users"))              2881
@bp.route("/users/add", methods=["GET", "POST"])                           2882
@login_required                                                            2883
def add_user():                                                            2884
        require_admin()                                                    2885
        if request.method == "POST":                                       2886
            username = request.form.get("username", "").strip()            2887
            password = request.form.get("password", "")                    2888
            role = request.form.get("role", DEFAULT_ROLE)                  2889
            if role not in ROLE_LABELS:                                    2890
                role = DEFAULT_ROLE                                        2891
            if not username or not password:                               2892
                flash("账户名称和密码不能为空", "error")                   2893
                return redirect(url_for("auth.add_user"))                  2894
            if User.query.filter_by(username=username).first():            2895
                flash("账户名称已存在", "error")                           2896
                return redirect(url_for("auth.add_user"))                  2897
            db.session.add(User(                                           2898
                username=username, role=role,                              2899
                password_hash=generate_password_hash(password),            2900
```

```
            display_name=request.form.get("display_name",
    "").strip() or None,
            gender=request.form.get("gender", "").strip()
    or None,
            email=request.form.get("email", "").strip() or
    None,
            phone=request.form.get("phone", "").strip() or
    None))
        db.session.commit()
        flash("账号已添加", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/add_user.html",
    role_labels=ROLE_LABELS,
                           genders=GENDERS,
    default_role=DEFAULT_ROLE,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        if role not in ROLE_LABELS:
            role = user.role
        active = request.form.get("active") == "1"
        #  保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,

    User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
    user_id=user.id))
        #  个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
    user_id=user.id))
        user.role = role
        user.active = active
        user.display_name = request.form.get("display_name",
    "").strip() or None
        user.gender = request.form.get("gender",
```

```
        "").strip() or None                                                    2951
            user.email = email or None                                         2952
            user.phone = request.form.get("phone", "").strip()                 2953
        or None                                                                2954
            new_password = request.form.get("password",                        2955
        "").strip()                                                            2956
            if new_password:                                                   2957
                user.password_hash = generate_password_hash(new_               2958
        password)                                                              2959
            db.session.commit()                                                2960
            flash("账号信息已保存", "ok")                                          2961
            return redirect(url_for("auth.profile",                            2962
        tab="users"))                                                          2963
        return render_template("auth/edit_user.html", user=user,               2964
                               role_labels=ROLE_LABELS,                        2965
        genders=GENDERS,                                                       2966
                               form=EmptyForm())                               2967
@bp.route("/users/<int:user_id>/delete", methods=["POST"])                     2968
@login_required                                                                2969
def delete_user(user_id):                                                      2970
        require_admin()                                                        2971
        user = db.get_or_404(User, user_id)                                    2972
        if user.id == current_user.id:                                         2973
            flash("不能删除当前登录的账号", "error")                                2974
            return redirect(url_for("auth.profile",                            2975
        tab="users"))                                                          2976
        if user.role == "admin":                                               2977
            others = User.query.filter(User.role == "admin",                   2978
                                       User.id != user.id,                     2979
                                       User.active.is_(True)).co               2980
        unt()                                                                  2981
            if others == 0:                                                    2982
                flash("系统至少需保留一名启用状态的管理员", "error")                    2983
                return redirect(url_for("auth.profile",                        2984
        tab="users"))                                                          2985
        db.session.delete(user)                                                2986
        db.session.commit()                                                    2987
        flash("账号「%s」已删除" % user.username, "ok")                           2988
        return redirect(url_for("auth.profile", tab="users"))                  2989
# ------------------------------ 操作日志（仅管理员）                                2990
        ------------------------------                                         2991
@bp.route("/logs")                                                             2992
@login_required                                                                2993
def logs():                                                                    2994
        """旧地址保留，统一并入「个人设置 · 操作日志」。"""                                2995
        return redirect(url_for("auth.profile", tab="logs"))                   2996
# ==== app/views/project.py ====                                               2997
# -*- coding: utf-8 -*-                                                        2998
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""                              2999
from datetime import datetime                                                  3000
```

```
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
from app.services import cleanup
from app.utils.parsing import FieldError, parse_float
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
    "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
            db.session.commit()
            flash("项目已创建", "ok")
            return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form,
    project=None)
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])
@login_required
def edit(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p.name = request.form.get("name", "").strip()
        p.code = request.form.get("code", "").strip() or
```

```
    None                                                                3051
        p.description = request.form.get("description",                 3052
    "").strip() or None                                                 3053
        p.status = request.form.get("status", "进行中")                  3054
        db.session.commit()                                             3055
        flash("项目已保存", "ok")                                         3056
        return redirect(url_for("project.index"))                       3057
    return render_template("project/edit.html", form=form,              3058
    project=p)                                                          3059
@bp.route("/<int:pid>/delete", methods=["POST"])                        3060
@login_required                                                         3061
def delete(pid):                                                        3062
    p = db.get_or_404(Project, pid)                                     3063
    # 项目下的图斑由 ORM 级联删除，但图斑下面的指标数据、限制性判定、                3064
    # 评价结果、得分明细、参照生态系统不会跟着走，必须先逐个图斑清理，              3065
    # 否则每删一个项目都在库里留下一批孤儿数据（见 app/services/cleanup.py）。   3066
    cleanup.delete_project(p)                                           3067
    flash("项目已删除", "ok")                                             3068
    return redirect(url_for("project.index"))                           3069
# ---------------------------- 矿山图斑 ----------------------------    3070
    ----                                                                3071
@bp.route("/<int:pid>/patches")                                         3072
@login_required                                                         3073
def patches(pid):                                                       3074
    p = db.get_or_404(Project, pid)                                     3075
    return render_template("project/patches.html",                      3076
    project=p,                                                          3077
                           form=EmptyForm())                            3078
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])             3079
@login_required                                                         3080
def patch_delete(patch_id):                                             3081
    """删除矿山图斑，及其微地质单元、指标数据与评价记录。"""                        3082
    patch = db.get_or_404(MinePatch, patch_id)                          3083
    pid = patch.project_id                                              3084
    # 清理逻辑统一放在 app/services/cleanup.py：项目删除走的是同一个函数，        3085
    # 免得两条删除路径各写一份、漏掉某张子表（曾经就漏过）。                       3086
    deleted, _counts = cleanup.delete_patch(patch.id)                   3087
    flash("图斑「%s」及其数据已删除" % (deleted.name if deleted else      3088
    patch.name),                                                        3089
          "ok")                                                         3090
    return redirect(url_for("project.patches", pid=pid))                3091
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])              3092
@login_required                                                         3093
def patch_add(pid):                                                     3094
    p = db.get_or_404(Project, pid)                                     3095
    form = EmptyForm()                                                  3096
    if request.method == "POST" and form.validate_on_submit(            3097
    ):                                                                  3098
        def fv(name, cast=str):                                         3099
            v = request.form.get(name, "").strip()                      3100
```

```
            return cast(v) if v != "" else None
        try:
            area_ha = parse_float(request.form.get("area_ha"
), "图斑面积（公顷）",
                                  minimum=0)
            stop_years = parse_float(request.form.get("stop_
years"), "停产年限",
                                     minimum=0)
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("project.patch_add",
pid=p.id))
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=area_ha,
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=stop_years,
            responsible_entity_missing=bool(request.form.get
("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
            # 参照生态系统默认取论文值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        # 数值字段统一走 parse_float：填了非数字给中文提示，而不是 500
```

```
        try:                                                              3151
            area_ha = parse_float(request.form.get("area_ha"3152
    ), "图斑面积（公顷）",                                                3153
                                  minimum=0)                              3154
            stop_years = parse_float(request.form.get("stop_3155
    years"), "停产年限",                                                  3156
                                    minimum=0)                            3157
            cov = parse_float(request.form.get("coverage_ben3158
    chmark"),                                                             3159
                              "植被覆盖度基准（%）", minimum=0,           3160
    maximum=100)                                                          3161
            sha = parse_float(request.form.get("shannon_benc3162
    hmark"),                                                              3163
                              "Shannon 基准", minimum=0)                  3164
        except FieldError as e:                                           3165
            flash(str(e), "error")                                        3166
            return redirect(url_for("project.patch_edit",                 3167
    patch_id=patch.id))                                                   3168
        patch.name = request.form.get("name", "").strip()                 3169
        patch.admin_region = fv("admin_region")                           3170
        patch.area_ha = area_ha                                           3171
        patch.mineral_type = fv("mineral_type")                           3172
        patch.mining_method = fv("mining_method")                         3173
        patch.stop_years = stop_years                                     3174
        patch.responsible_entity_missing =                                3175
    bool(request.form.get("responsible_entity_missing"))                  3176
        # 参照生态系统                                                    3177
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa3178
    tch.id).first()                                                       3179
        if ref is None:                                                   3180
            ref = ReferenceEcosystem(patch_id=patch.id)                   3181
            db.session.add(ref)                                           3182
        # 基准值：用户填 0 是合法输入（不能写成 `or 75.0`，那样 0 会被悄悄换成默认值）  3183
        ref.coverage_benchmark = 75.0 if cov is None else                 3184
    cov                                                                   3185
        ref.shannon_benchmark = 0.7 if sha is None else sha               3186
        ref.dominant_species = fv("dominant_species")                     3187
        patch.image_note = fv("image_note")                               3188
        db.session.commit()                                               3189
        flash("图斑已保存", "ok")                                         3190
        return redirect(url_for("project.patches",                        3191
    pid=patch.project_id))                                                3192
    return render_template("project/patch_edit.html",                     3193
    form=form,                                                            3194
                           project=patch.project,                         3195
    patch=patch,                                                          3196
                           reference=ReferenceEcosystem.quer3197
    y.filter_by(                                                          3198
                               patch_id=patch.id).first())                3199
# ---------------------------- 遥感影像挂接                               3200
```

```
    ------------------------------                                    3201
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])            3202
@login_required                                                       3203
def upload_image(patch_id):                                           3204
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                    3205
    from app.utils import uploads as up                               3206
    patch = db.get_or_404(MinePatch, patch_id)                        3207
    form = EmptyForm()                                                3208
    if not form.validate_on_submit():                                 3209
        flash("表单校验失败，请重试", "error")                        3210
        return redirect(url_for("project.patch_edit",                 3211
    patch_id=patch.id))                                               3212
    rel, err = up.save_upload(request.files.get("image"),             3213
    "patches",                                                        3214
                              current_app.config["ALLOWED_IM3215
    AGE_EXT"],                                                        3216
                              current_app.config["UPLOAD_FOL3217
    DER"])                                                            3218
    if err:                                                           3219
        flash(err, "error")                                           3220
        return redirect(url_for("project.patch_edit",                 3221
    patch_id=patch.id))                                               3222
    # 替换旧影像时清理原文件，避免残留                                3223
    if patch.image_path:                                              3224
        up.remove_upload(patch.image_path,                            3225
    current_app.config["UPLOAD_FOLDER"])                              3226
    patch.image_path = rel                                            3227
    patch.image_uploaded_at = datetime.utcnow()                       3228
    note = request.form.get("image_note", "").strip()                 3229
    if note:                                                          3230
        patch.image_note = note                                       3231
    db.session.commit()                                               3232
    flash("遥感影像已挂接", "ok")                                     3233
    return redirect(url_for("project.patch_edit",                     3234
    patch_id=patch.id))                                               3235
@bp.route("/patch/<int:patch_id>/image/delete",                       3236
    methods=["POST"])                                                 3237
@login_required                                                       3238
def delete_image(patch_id):                                           3239
    from app.utils import uploads as up                               3240
    patch = db.get_or_404(MinePatch, patch_id)                        3241
    form = EmptyForm()                                                3242
    if form.validate_on_submit():                                     3243
        up.remove_upload(patch.image_path,                            3244
    current_app.config["UPLOAD_FOLDER"])                              3245
        patch.image_path = None                                       3246
        patch.image_uploaded_at = None                                3247
        db.session.commit()                                           3248
        flash("遥感影像已移除", "ok")                                 3249
    return redirect(url_for("project.patch_edit",                     3250
```

```
    patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import UNIT_TYPES
from app.utils.parsing import FieldError, parse_float
bp = Blueprint("unit", __name__, url_prefix="/unit")
def recalc_ratio(patch_id):
    """按面积重算该图斑各单元面积占比。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    if total > 0:
        for u in units:
            u.ratio = round((u.area or 0.0) / total, 6)
    return total
@bp.route("/")
@login_required
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all
    ()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
```

```
                                unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        try:
            area_value = parse_float(area, "面积（公顷）",
    minimum=0)
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=area_value,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
```

```
        new_type = request.form.get("unit_type",                    3351
    u.unit_type)                                                    3352
        if new_type not in UNIT_TYPES:                              3353
            flash("微地质单元类型不正确", "error")                  3354
            return redirect(url_for("unit.edit",                    3355
    unit_id=u.id))                                                  3356
        #  改类型时同样不允许与本图斑已有的其他单元重复            3357
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,        3358
                                                                    3359
    unit_type=new_type).first()                                     3360
        if dup is not None and dup.id != u.id:                      3361
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,  3362
    "error")                                                        3363
            return redirect(url_for("unit.edit",                    3364
    unit_id=u.id))                                                  3365
        u.unit_type = new_type                                      3366
        area = request.form.get("area", "").strip()                 3367
        try:                                                        3368
            u.area = parse_float(area, "面积（公顷）", minimum=0)   3369
        except FieldError as e:                                     3370
            flash(str(e), "error")                                  3371
            return redirect(url_for("unit.edit",                    3372
    unit_id=u.id))                                                  3373
        u.description = request.form.get("description",             3374
    "").strip() or None                                             3375
        recalc_ratio(u.patch_id)                                    3376
        db.session.commit()                                         3377
        flash("微地质单元已保存", "ok")                             3378
        return redirect(url_for("unit.index",                       3379
    patch_id=u.patch_id))                                           3380
    return render_template("unit/edit.html", unit=u,                3381
    patch=u.patch,                                                  3382
                                unit_types=UNIT_TYPES, form=form)   3383
@bp.route("/delete/<int:unit_id>", methods=["POST"])                3384
@login_required                                                     3385
def delete(unit_id):                                                3386
    u = db.get_or_404(MicroUnit, unit_id)                           3387
    pid = u.patch_id                                                3388
    name = u.unit_type                                              3389
    #  一并删除该单元的指标数据，避免残留                          3390
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch        3391
    ronize_session=False)                                           3392
    db.session.delete(u)                                            3393
    db.session.flush()                                              3394
    recalc_ratio(pid)                                               3395
    db.session.commit()                                             3396
    flash("微地质单元「%s」已删除" % name, "ok")                    3397
    return redirect(url_for("unit.index", patch_id=pid))            3398
#  ==== app/views/standard.py ====                                  3399
#  -*- coding: utf-8 -*-                                           3400
```

```
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, Standard, StandardRule
bp = Blueprint("standard", __name__, url_prefix="/standard")
@bp.route("/")
@login_required
def index():
    standards = Standard.query.order_by(Standard.category,
    Standard.province).all()
    local = [s for s in standards if s.category == "地方标准"]
    policy = [s for s in standards if s.category != "地方标准"]
    return render_template("standard/index.html",
    local=local, policy=policy,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s = Standard(
            province=request.form.get("province",
    "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
        else:
            db.session.add(s)
            db.session.commit()
            flash("标准已添加", "ok")
            return redirect(url_for("standard.index"))
    return render_template("standard/edit.html", form=form,
    standard=None)
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])
@login_required
def edit(sid):
    s = db.get_or_404(Standard, sid)
```

```
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s.province = request.form.get("province",
    "").strip()
        s.name = request.form.get("name", "").strip()
        s.doc_number = request.form.get("doc_number",
    "").strip() or None
        s.publish_date = request.form.get("publish_date",
    "").strip() or None
        s.implement_date = request.form.get("implement_date"
    , "").strip() or None
        s.category = request.form.get("category", "地方标准")
        s.summary = request.form.get("summary", "").strip()
    or None
        db.session.commit()
        flash("已保存", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    return render_template("standard/edit.html", form=form,
    standard=s)
@bp.route("/<int:sid>/delete", methods=["POST"])
@login_required
def delete(sid):
    s = db.get_or_404(Standard, sid)
    StandardRule.query.filter_by(standard_id=sid).delete()
    db.session.delete(s)
    db.session.commit()
    flash("已删除", "ok")
    return redirect(url_for("standard.index"))
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])
@login_required
def rules(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        r = StandardRule(
            standard_id=s.id,
            item=request.form.get("item", "").strip(),
            operator=request.form.get("operator", ">="),
            threshold=request.form.get("threshold",
    "").strip() or None,
            condition_type=request.form.get("condition_type"
    , "必要条件"),
            region=request.form.get("region", "").strip()
    or None,
            note=request.form.get("note", "").strip() or
    None,
        )
        if not r.item:
```

```
                flash("判定项不能为空", "error")                                3501
            else:                                                               3502
                db.session.add(r)                                               3503
                db.session.commit()                                             3504
                flash("规则已添加", "ok")                                       3505
            return redirect(url_for("standard.rules", sid=s.id))                3506
        rs = StandardRule.query.filter_by(standard_id=s.id).all(                3507
        )                                                                       3508
        return render_template("standard/rules.html",                           3509
        form=form, standard=s, rules=rs)                                        3510
@bp.route("/rule/<int:rid>/delete", methods=["POST"])                           3511
@login_required                                                                 3512
def rule_delete(rid):                                                           3513
        r = db.get_or_404(StandardRule, rid)                                    3514
        sid = r.standard_id                                                     3515
        db.session.delete(r)                                                    3516
        db.session.commit()                                                     3517
        flash("规则已删除", "ok")                                               3518
        return redirect(url_for("standard.rules", sid=sid))                     3519
@bp.route("/compare")                                                           3520
@login_required                                                                 3521
def compare():                                                                  3522
        """省际口径比对：同一指标在不同省份的阈值横向对比。"""                  3523
        rules = StandardRule.query.join(Standard).all()                         3524
        groups = {}                                                             3525
        for r in rules:                                                         3526
            groups.setdefault(r.item, []).append(r)                             3527
        return render_template("standard/compare.html",                         3528
        groups=groups)                                                          3529
#  ==== app/views/constraint.py ====                                            3530
#  -*- coding: utf-8 -*-                                                        3531
"""限制性指标判定 —— 一票否决判定。                                             3532
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。 3533
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。                   3534
"""                                                                             3535
from flask import Blueprint, render_template, request,                          3536
    redirect, url_for, flash                                                    3537
from flask_login import login_required                                          3538
from app.forms import EmptyForm                                                 3539
from app.models import db, MinePatch, ConstraintCheck                           3540
from app.utils.constants import CONSTRAINT_ITEMS                                3541
bp = Blueprint("constraint", __name__,                                          3542
    url_prefix="/constraint")                                                   3543
#  字段名 -> 中文名                                                             3544
FIELDS = [                                                                      3545
    ("policy_pass", "政策约束"),                                                3546
    ("location_pass", "区位约束"),                                              3547
    ("pollution_pass", "土壤污染"),                                             3548
    ("geo_pass", "地质灾害"),                                                   3549
    ("farmland_pass", "耕地属性"),                                              3550
```

```
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    latest = {}
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch_id=p.id) \
            .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit():
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
        for field, _ in FIELDS:
            setattr(record, field, request.form.get(field) == "1")
        record.passed = all(getattr(record, f) for f, _ in FIELDS)
        failed = [name for f, name in FIELDS if not getattr(record, f)]
        record.reason = ("全部通过" if record.passed
                         else "未通过：" + "、".join(failed))
        db.session.commit()
        flash("判定结果已保存", "ok")
        return redirect(url_for("constraint.check", patch_id=patch.id))
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
    field_states = [(f, name, None if record is None else getattr(record, f))
                    for f, name in FIELDS]
    return render_template("constraint/check.html",
    patch=patch, form=form,
                           fields=FIELDS,
    field_states=field_states,
                           record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
```

```
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
import math
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
```

```
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 （指标名称列表， 默认权重列表）。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    # level 是手改地址栏就能改的参数，写成 abc 不该把页面打成 500
    try:
        level = int(request.values.get("level", 1))
```

```
except (TypeError, ValueError):
    level = 1
if level not in (1, 2):
    level = 1
l1 = normalize_l1(unit_type, level,
request.values.get("l1"))
if unit_type not in DEFAULT_WEIGHTS:
    flash("无效的微地质单元类型", "error")
    return redirect(url_for("ahp.index"))
names, vec = indicator_list(unit_type, level, l1)
n = len(names)
form = EmptyForm()
# 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
default = [[round(vec[i] / vec[j], 4) for j in
range(n)] for i in range(n)]
def compute(upper):
    A = ahp.matrix_from_upper(upper)
    r = ahp.weights_and_consistency(A)
    bad = ahp.find_most_inconsistent(A)
    return {
        "weights": [round(v, 4) for v in r["weights"]],
        "lambda_max": round(r["lambda_max"], 4),
        "ci": r["ci"],
        "cr": r["cr"],
        "consistent": r["consistent"],
        "matrix": [[round(A[i, j], 4) for j in
range(n)] for i in range(n)],
        "most_inconsistent": bad,
    }
result = None
if request.method == "POST" and form.validate_on_submit(
):
    action = request.form.get("action", "calc")
    try:
        upper = [float(request.form.get("m_%d_%d" % (i,
j), "1"))
                 for i in range(n) for j in range(i +
1, n)]
        # 0 / 负数 / inf / nan 能过 float()，但会让 n 次方根算出 inf
或 nan，
        # 保存时直接撞上 NOT NULL 约束（weight 为 nan）→ 500
        if any((not math.isfinite(x)) or x <= 0 for x
in upper):
            raise ValueError("矩阵元素必须是正数")
        result = compute(upper)
    except (ValueError, ZeroDivisionError):
        flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
"error")
        return redirect(url_for("ahp.matrix",
unit_type=unit_type,
```

```
                                                level=level, l1=l1))
        if action == "save":
            if not result["consistent"]:
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                      "请调整后再保存" % result["cr"], "error")
            else:
                pairs = to_code_pairs(unit_type, level, l1,
    names,
                                      result["weights"])
                scheme, cnt = ws.save_weights(unit_type,
    pairs)
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                      "此后该单元类型的评价将以此权重计算。"
                      % (unit_type, cnt, result["cr"]),
    "ok")
                return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                        level=level, l1=l1))
    elif request.values.get("calc") == "1":
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性
        result = compute([default[i][j] for i in range(n)
    for j in range(i + 1, n)])
    _, source_label, custom_scheme = ws.effective_weights(un
    it_type)
    return render_template(
        "ahp/matrix.html",
        form=form, unit_type=unit_type, level=level, l1=l1,
        names=names, vec=vec, default=default,
    result=result,
        source_label=source_label, custom_scheme=custom_sche
    me,
    )
@bp.route("/weights/reset", methods=["POST"])
def weights_reset():
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""
    unit_type = request.form.get("unit_type")
    form = EmptyForm()
    if form.validate_on_submit():
        if unit_type not in DEFAULT_WEIGHTS:
            flash("无效的微地质单元类型", "error")
        else:
            n = ws.reset_weights(unit_type)
            if n:
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,
    "ok")
            else:
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,
    "ok")
    return redirect(url_for("ahp.index"))
# ---------------------------- 指标维护 ----------------------------
```

```
    ----                                                                3801
@bp.route("/indicators")                                                3802
def indicators():                                                       3803
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                3804
    if Indicator.query.count() == 0:                                    3805
        ind_svc.seed_indicators()                                       3806
    all_ind = Indicator.query.order_by(Indicator.level,                 3807
    Indicator.id).all()                                                 3808
    by_parent = {}                                                      3809
    for i in all_ind:                                                   3810
        by_parent.setdefault(i.parent_code, []).append(i)               3811
    return render_template("ahp/indicators.html",                       3812
    by_parent=by_parent,                                                3813
                                  all_ind=all_ind,                      3814
    unit_types=UNIT_TYPES,                                              3815
                                  type_label=ind_svc.ideal_type_lab3816
    el,                                                                 3817
                                  form=EmptyForm())                     3818
@bp.route("/indicators/<code>/edit", methods=["GET",                    3819
    "POST"])                                                            3820
def indicator_edit(code):                                               3821
    ind = Indicator.query.filter_by(code=code).first_or_404(3822
    )                                                                   3823
    form = EmptyForm()                                                  3824
    if request.method == "POST" and form.validate_on_submit(3825
    ):                                                                  3826
        name = request.form.get("name", "").strip()                     3827
        if not name:                                                    3828
            flash("指标名称不能为空", "error")                           3829
            return redirect(url_for("ahp.indicator_edit",               3830
    code=code))                                                         3831
        ind.name = name                                                 3832
        ind.unit = request.form.get("unit", "").strip() or              3833
    None                                                                3834
        ind.ideal_value = request.form.get("ideal_value",               3835
    "").strip() or None                                                 3836
        ind.ideal_type = request.form.get("ideal_type") or              3837
    ind.ideal_type                                                      3838
        units = request.form.getlist("enabled_units")                   3839
        ind.enabled_units = ",".join(units) if units else               3840
    None                                                                3841
        ind.is_constraint = request.form.get("is_constraint"3842
    ) == "1"                                                            3843
        ind.description = request.form.get("description",               3844
    "").strip() or None                                                 3845
        db.session.commit()                                             3846
        flash("指标「%s」已保存" % ind.name, "ok")                       3847
        return redirect(url_for("ahp.indicators"))                      3848
    return render_template("ahp/indicator_edit.html",                   3849
    ind=ind,                                                            3850
```

```
                                    unit_types=UNIT_TYPES,                3851
                                    type_label=ind_svc.ideal_type_lab3852
    el, form=form)                                                        3853
@bp.route("/indicators/reset", methods=["POST"])                          3854
def indicators_reset():                                                   3855
    """按论文取值重置指标定义。"""                                          3856
    form = EmptyForm()                                                    3857
    if form.validate_on_submit():                                         3858
        n = ind_svc.seed_indicators()                                     3859
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")                    3860
    return redirect(url_for("ahp.indicators"))                            3861
# ------------------------------ 多专家矩阵 ------------------------------3862
    -----                                                                 3863
def expert_scope():                                                       3864
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""                  3865
    unit_type = request.values.get("unit_type") or                        3866
    UNIT_TYPES[0]                                                         3867
    if unit_type not in DEFAULT_WEIGHTS:                                  3868
        unit_type = UNIT_TYPES[0]                                         3869
    try:                                                                  3870
        level = int(request.values.get("level", 1))                      3871
    except (TypeError, ValueError):                                       3872
        level = 1                                                         3873
    if level not in (1, 2):                                               3874
        level = 1                                                         3875
    l1 = normalize_l1(unit_type, level,                                   3876
    request.values.get("l1"))                                             3877
    return unit_type, level, l1                                           3878
def expert_query(unit_type, level, l1):                                   3879
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,                 3880
    level=level)                                                          3881
    if level == 2:                                                        3882
        q = q.filter_by(l1_name=l1)                                       3883
    else:                                                                 3884
        q = q.filter(ExpertMatrix.l1_name.is_(None))                      3885
    return q.order_by(ExpertMatrix.id)                                    3886
def default_scheme(unit_type, level, l1=None):                            3887
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""               3888
    name = "%s · %s%s" % (unit_type,                                      3889
                          "一级指标" if level == 1 else "二级指标",        3890
                          "(%s)" % l1 if l1 else "")                      3891
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ3892
    e,                                                                    3893
                                          level=level,                    3894
    name=name).first()                                                    3895
    if scheme is None:                                                    3896
        scheme = WeightScheme(unit_type=unit_type,                        3897
    level=level, name=name,                                               3898
                              is_default=False)                           3899
        db.session.add(scheme)                                            3900
```

```
        db.session.flush()
    return scheme
@bp.route("/experts", methods=["GET", "POST"])
def experts():
    """多专家判断矩阵录入与汇总。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action")
        if action == "add":
            expert_name = request.form.get("expert_name",
    "").strip() or "未具名专家"
            try:
                upper = [float(request.form.get("m_%d_%d" %
    (i, j), "1"))
                         for i in range(n) for j in range(i
    + 1, n)]
            except ValueError:
                flash("矩阵输入有误，请填入正数", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
            # 0 / 负数 / inf / nan 都能过 float()，但会让 n 次方根算出
    inf 或 nan，
            # 于是库里存下 [[1,0],[Infinity,1]] 这种矩阵、CR 还显示
    0.0000。
            if any((not math.isfinite(x)) or x <= 0 for x
    in upper):
                flash("判断矩阵只能填正数（大于 0 的有限数字）", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
            A = ahp.matrix_from_upper(upper)
            r = ahp.weights_and_consistency(A)
            scheme = default_scheme(unit_type, level, l1)
            db.session.add(ExpertMatrix(
                scheme_id=scheme.id,
                unit_type=unit_type, l1_name=(l1 if level
    == 2 else None),
                expert_name=expert_name, level=level,
                matrix_json=json.dumps([list(map(float,
    row)) for row in A]),
                cr=round(r["cr"], 6)))
            db.session.commit()
```

```
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
    r["cr"]), "ok")
        elif action == "save_agg":
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算
            weights = _aggregate(unit_type, level, l1)
            if weights is None:
                flash("暂无可汇总的专家矩阵", "error")
            elif not weights["consistent"]:
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"
                      "请先检查或调整专家矩阵" % weights["cr"],
    "error")
            else:
                names_, _ = indicator_list(unit_type,
    level, l1)
                pairs = to_code_pairs(unit_type, level, l1,
    names_,
                                      weights["weights"])
                _, cnt = ws.save_weights(unit_type, pairs)
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"
                      % (unit_type, cnt, weights["cr"]),
    "ok")
        else:
            # action 缺失或值不认识时以前是静默跳回，用户只觉得「点了没反应」
            flash("未识别的操作，请从页面上的按钮重新提交", "error")
        return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
    unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
    default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
    unit_type, level, l1 = row.unit_type, row.level,
    row.l1_name
    form = EmptyForm()
    if form.validate_on_submit():
        db.session.delete(row)
        db.session.commit()
        flash("已删除专家矩阵", "ok")
    return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                            level=level, l1=l1))
def _aggregate(unit_type, level, l1):
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""
```

```
    rows = expert_query(unit_type, level, l1).all()
    mats = []
    for r in rows:
        try:
            mats.append([list(map(float, line))
                         for line in json.loads(r.matrix_jso
n)])
        except Exception:
            continue
    if not mats:
        return None
    agg = ahp.aggregate_matrices(mats)
    res = ahp.weights_and_consistency(agg)
    return {
        "count": len(mats),
        "weights": res["weights"],
        "lambda_max": res["lambda_max"],
        "ci": res["ci"], "cr": res["cr"],
        "consistent": res["consistent"],
        "most_inconsistent": ahp.find_most_inconsistent(agg)
,
        "expert_crs": [r.cr for r in rows if r.cr is not
None],
    }
@bp.route("/experts/aggregate")
def experts_aggregate():
    """按几何平均汇总多专家矩阵，并做一致性检验。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    rows = expert_query(unit_type, level, l1).all()
    result = _aggregate(unit_type, level, l1)
    if result is not None:
        result["weights"] = [round(v, 4) for v in
result["weights"]]
        result["lambda_max"] = round(result["lambda_max"],
4)
    return render_template("ahp/experts_aggregate.html",
    unit_type=unit_type,
                           level=level, l1=l1, names=names,
    rows=rows,
                           result=result, form=EmptyForm())
# ==== app/views/indicator.py ====
# -*- coding: utf-8 -*-
"""指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
```

```
import io
import json
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                       flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_ENABLED_INDICATORS
from app.utils.parsing import FieldError, parse_float
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
    url_prefix="/indicator")
# 指标字段定义：字段名 -> （中文名， 单位， 是否可选）
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True
    = 含重金属
    if "植被指标" in enabled:
        groups.append(("植被指标", VEG_FIELDS, False))
    if "地形指标" in enabled:
        groups.append(("地形指标", TERRAIN_FIELDS, False))
    return groups
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    return render_template("indicator/index.html",
    patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
```

```
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir
st()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit(
):
        # 各字段的合理范围（覆盖度是百分数、pH 0–14、坡度 0–90，其余非负）。
        # 以前这里直接 float(v)，用户填个「约 3.5」看到的就是英文 500 页。
        LIMITS = {
            "soil_ratio": ("土石比", 0, None),
            "thickness": ("土层厚度（cm）", 0, None),
            "ph": ("pH", 0, 14),
            "organic_matter": ("有机质（g/kg）", 0, None),
            "coverage": ("植被覆盖度（%）", 0, 100),
            "shannon": ("Shannon 指数", 0, None),
            "slope": ("坡度（°）", 0, 90),
        }
        try:
            values = {}
            for name, (label, lo, hi) in LIMITS.items():
                values[name] = parse_float(request.form.get(
name), label,
                                           minimum=lo,
maximum=hi)
            heavy = {}
            for m in HEAVY_METALS:
                v = parse_float(request.form.get("hm_" + m),
                                "重金属 %s（mg/kg）" % m,
minimum=0)
                if v is not None:
                    heavy[m] = v
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("indicator.edit",
unit_id=unit.id))
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        for name, value in values.items():
            setattr(ind, name, value)
        ind.heavy_metals = json.dumps(heavy,
ensure_ascii=False) if heavy else None
        ind.aspect = request.form.get("aspect", "").strip()
or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
```

```
            current = {                                                    4151
                "soil_ratio": ind.soil_ratio, "thickness":                 4152
        ind.thickness,                                                     4153
                "ph": ind.ph, "organic_matter":                            4154
        ind.organic_matter,                                                4155
                "coverage": ind.coverage, "shannon":                       4156
        ind.shannon,                                                       4157
                "slope": ind.slope, "aspect": ind.aspect,                  4158
            }                                                              4159
            current["heavy"] = ind.heavy_metals_dict()                     4160
        return render_template("indicator/edit.html",                      4161
                               unit=unit, form=form,                       4162
        groups=groups,                                                     4163
                               heavy_metals=HEAVY_METALS,                  4164
                               aspect_types=ASPECT_TYPES,                  4165
        current=current)                                                   4166
# ------------------------------ 批量导入 ------------------------------4167
        ----                                                               4168
@bp.route("/import", methods=["GET", "POST"])                              4169
def import_data():                                                         4170
        """批量导入指标数据（CSV / Excel）。"""                                4171
        patches = MinePatch.query.order_by(MinePatch.id.desc()).4172
        all()                                                              4173
        form = EmptyForm()                                                 4174
        report = None                                                      4175
        if request.method == "POST" and form.validate_on_submit(4176
        ):                                                                 4177
            patch_id = request.form.get("patch_id", type=int)              4178
            patch = db.session.get(MinePatch, patch_id) if                 4179
        patch_id else None                                                 4180
            if patch is None:                                              4181
                flash("请选择矿山图斑", "error")                                4182
                return redirect(url_for("indicator.import_data")4183
        )                                                                  4184
            file = request.files.get("datafile")                           4185
            if file is None or not file.filename:                          4186
                flash("请选择要导入的文件", "error")                            4187
                return redirect(url_for("indicator.import_data")4188
        )                                                                  4189
            ext = file.filename.rsplit(".", 1)[-1].lower()                 4190
            if ext not in ("csv", "xlsx", "xls"):                          4191
                flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")          4192
                return redirect(url_for("indicator.import_data")4193
        )                                                                  4194
            units = MicroUnit.query.filter_by(patch_id=patch.id)4195
        .all()                                                             4196
            unit_map = {}                                                  4197
            for u in units:                                                4198
                unit_map.setdefault(u.unit_type, u)                        4199
            if not unit_map:                                               4200
```

```
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
            return redirect(url_for("indicator.import_data"))
    )
        try:
            parsed, errors = imp.parse_upload(file.filename,
    file.stream, unit_map)
        except Exception as e:
            flash("文件解析失败：%s" % e, "error")
            return redirect(url_for("indicator.import_data"))
    )
        saved = 0
        for rec in parsed:
            unit = unit_map[rec["unit_type"]]
            ind = IndicatorData.query.filter_by(unit_id=unit
    .id).first()
            if ind is None:
                ind = IndicatorData(unit_id=unit.id)
                db.session.add(ind)
            f = rec["fields"]
            for key in ("soil_ratio", "thickness", "ph",
    "organic_matter",
                        "coverage", "shannon", "slope",
    "aspect"):
                if key in f:
                    setattr(ind, key, f[key])
            if rec["heavy_metals"]:
                ind.heavy_metals = json.dumps(rec["heavy_met
    als"],

    ensure_ascii=False)
            saved += 1
        db.session.commit()
        report = {"patch": patch.name, "saved": saved,
    "errors": errors,
                  "total": len(parsed) + len(
                      [e for e in errors if e.startswith("第
    ")])}
        if saved:
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")
        else:
            flash("没有可导入的数据行，请检查文件内容", "error")
    return render_template("indicator/import.html",
    patches=patches,
                           form=form, report=report,
    headers=imp.HEADERS)
@bp.route("/import/template")
def import_template():
    """下载导入模板（CSV，含一行示例）。"""
    csv_text = imp.build_template_csv()
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于
```

```
    Excel 正确识别中文
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="指标数据导入模板.csv",
                     mimetype="text/csv")
# ==== app/views/evaluate.py ====
# -*- coding: utf-8 -*-
"""适宜性定量评价引擎 —— 路由与页面逻辑。
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。
"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import EmptyForm
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail)
from app.services import evaluation_service as ev
from app.services import workflow as wf
from app.services import unit_config
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else
    0.0,
            "data": data,
```

```
        })                                                                  4301
    return result                                                           4302
def _round(v, n=6):                                                         4303
    return None if v is None else round(v, n)                               4304
def load_weights(units):                                                    4305
    """解析各单元类型当前生效的权重与来源说明。                                  4306
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。            4307
    来源一并记录，便于报告标注与事后复核。                                      4308
    """                                                                     4309
    from app.services import weight_service as ws                           4310
    weights, sources = {}, {}                                               4311
    for u in units:                                                         4312
        ut = u["unit_type"]                                                 4313
        if ut not in weights:                                               4314
            w, label, _ = ws.effective_weights(ut)                          4315
            weights[ut] = w                                                 4316
            sources[ut] = label                                             4317
    return weights, sources                                                 4318
def pack_units(unit_results, sources=None):                                 4319
    """把评价结果整理为可持久化的 JSON。                                        4320
    完整保留各单元得分、一级指标得分与逐条二级指标明细，                        4321
    使结果页与报告能够复现「已记录的那一次评价」，                              4322
    而不是每次打开时重新推算。                                                4323
    """                                                                     4324
    sources = sources or {}                                                 4325
    out = []                                                                4326
    for u in unit_results:                                                  4327
        out.append({                                                        4328
            "unit_type": u["unit_type"],                                    4329
            "ratio": _round(u["ratio"]),                                    4330
            "unit_score": _round(u["unit_score"]),                          4331
            "weighted": _round(u["weighted"]),                              4332
            "weight_source": sources.get(u["unit_type"]),                   4333
            "l1_scores": {k: _round(v) for k, v in                          4334
    u["l1_scores"].items()},                                                4335
            "details": [{                                                   4336
                "l1": d["l1"], "l2": d["l2"],                               4337
                "raw": None if d["raw"] is None else                        4338
    str(d["raw"]),                                                          4339
                "std_value": _round(d["std_value"]),                        4340
                "weight": _round(d["weight"]),                              4341
                "contribution": _round(d["contribution"]),                  4342
            } for d in u["details"]],                                       4343
        })                                                                  4344
    return out                                                              4345
def save_evaluation(patch_id, units):                                       4346
    """执行评价并把结果与逐层明细落库，返回评价记录。                           4347
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用                  4348
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、               4349
    一级指标得分与逐条二级指标明细，结果页与报告据此复现                        4350
```

```
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    #  标准化参数取自该图斑的参照生态系统与指标定义（界面上可改），
    #  没配过的项自动回落到代码常量，历史结果与验证算例不受影响。
    settings = unit_config.for_patch(patch_id)
    res = ev.score_mine(units, weights, settings)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
        grade=res["grade"],
        measures=res["grade_info"]["measure"],
        unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                               ensure_ascii=False),
    )
    db.session.add(saved)
    db.session.flush()        #  先取得 result.id，再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
                result_id=saved.id,
                unit_id=unit_ids.get(u["unit_type"]),
                l1_name=d["l1"], l2_name=d["l2"],
                raw_value=None if d["raw"] is None else
    str(d["raw"]),
                std_value=d["std_value"],
    weight=d["weight"],
                contribution=d["contribution"]))
    db.session.commit()
    return saved
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
```

```
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores,
                           undetected=wf.undetected_metals(p
    atch.id)
                           if patch else [])
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
```

```
from app.utils.timefmt import now_utc
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
        return patch, None, None, None, prog
    rec = prog["result"]
    result = {
        "final_score": rec.final_score,
        "grade": rec.grade,
        "grade_info": grade_of(rec.final_score),
        "unit_results": json.loads(rec.unit_scores) if
    rec.unit_scores else [],
    }
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
```

```
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    progress = {p.id: wf.report_progress(p.id) for p in patches}
    return render_template("report/index.html", patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog = _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    return render_template("report/detail.html",
    patch=patch, result=result,
                           record=record,
    reference=reference,
                           chart_units=chart_units,
    chart_weight=chart_weight,
                           undetected=wf.undetected_metals(patch.id))
@bp.route("/<int:patch_id>/word")
def word(patch_id):
    patch, reference, result, record, prog = _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    buf = rs.build_report(patch, result)
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name
    return send_file(buf, as_attachment=True,
    download_name=filename,
                     mimetype="application/vnd.openxmlformats-officedocument."
                              "wordprocessingml.document")
@bp.route("/<int:patch_id>/pdf")
def pdf(patch_id):
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""
    patch, reference, result, record, prog = _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    html = render_template(
        "report/print.html", patch=patch,
    reference=reference, result=result,
        chart_units=chart_units, chart_weight=chart_weight,
        generated_at=now_utc())
    data, err = browser.html_to_pdf(html)
```

```
    if err:                                                          4551
        flash("PDF 导出失败：%s" % err, "error")                      4552
        return redirect(url_for("report.detail",                     4553
    patch_id=patch_id))                                              4554
    return send_file(io.BytesIO(data), as_attachment=True,           4555
                     download_name="自然恢复适宜性评价报告_%s.pdf" %     4556
    patch.name,                                                      4557
                     mimetype="application/pdf")                     4558
# ==== app/views/search.py ====                                      4559
# -*- coding: utf-8 -*-                                              4560
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。                4561
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，                          4562
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。                      4563
"""                                                                  4564
from sqlalchemy import or_                                           4565
from flask import Blueprint, render_template, request                4566
from app.models import db, Project, MinePatch, Standard              4567
bp = Blueprint("search", __name__)                                   4568
LIMIT = 50              # 每类结果上限，避免一次渲染过多                  4569
@bp.route("/search")                                                 4570
def index():                                                         4571
    q = (request.args.get("q") or "").strip()                        4572
    projects, patches, standards = [], [], []                        4573
    if q:                                                            4574
        like = "%" + q + "%"                                         4575
        projects = Project.query.filter(or_(                         4576
            Project.name.like(like),                                 4577
            Project.code.like(like),                                 4578
            Project.description.like(like),                          4579
        )).order_by(Project.id.desc()).limit(LIMIT).all()            4580
        patches = MinePatch.query.filter(or_(                        4581
            MinePatch.name.like(like),                               4582
            MinePatch.admin_region.like(like),                       4583
            MinePatch.mineral_type.like(like),                       4584
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()          4585
        standards = Standard.query.filter(or_(                       4586
            Standard.name.like(like),                                4587
            Standard.doc_number.like(like),                          4588
            Standard.province.like(like),                            4589
            Standard.summary.like(like),                             4590
        )).order_by(Standard.id).limit(LIMIT).all()                  4591
    total = len(projects) + len(patches) + len(standards)            4592
    return render_template("search.html", q=q,                       4593
    projects=projects,                                               4594
                                patches=patches,                     4595
    standards=standards, total=total)                                4596
# ==== app/templates/base.html ====                                  4597
{% import "_macros.html" as ui %}<!DOCTYPE html>                     4598
<html lang="zh-CN">                                                  4599
<head>                                                               4600
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<link rel="icon" href="{{ url_for('static', filename='favicon.ico') }}">
<title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}</title>
<style>
  /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */
  :root {
    --accent: #2c6b5a;                 /* 主色：青绿，千里江山石绿偏青 */
    --accent-hover: #1e5548;
    --accent-soft: #e6f0ec;
    --side-1: #1a4a4a;                 /* 侧边栏：深青墨 */
    --side-2: #0f3338;
    --danger: #a0392a;                 /* 朱砂红 */
    --danger-soft: #fbeae6;
    --ok: #2c6b5a;
    --ok-soft: #e6f0ec;
    --todo: #c08a3e;                   /* 赭石色待办 */
    --ink: #2a2520;                    /* 深褐墨色 */
    --ink-soft: #6b5d4f;               /* 褐灰 */
    --line: #e0d8c8;                   /* 绢本边线 */
    --surface: #fdfaf2;                /* 暖白绢本 */
    --canvas: #f5efe0;                 /* 绢本米黄背景 */
    --radius: 10px;
    --shadow: 0 2px 8px rgba(42, 37, 32, .06);
    --header-h: 60px;
    --nav-w: 226px;
  }
  * { box-sizing: border-box; }
  html, body { margin: 0; padding: 0; }
  body {
    font-family: "Microsoft YaHei", "PingFang SC", "Noto Serif SC", "SimSun", serif;
    font-size: 14px;
    line-height: 1.6;
    color: var(--ink);
    background: var(--canvas);
  }
  /* ---------- 跳到主要内容（键盘可达性） ---------- */
  .skip-link {
    position: absolute;
    left: 12px;
    top: -60px;
    z-index: 30;
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
```

```
    text-decoration: none;                                                    4651
    transition: top .15s ease-out;                                            4652
}                                                                             4653
.skip-link:focus { top: 12px; }                                               4654
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------                    4655
   固定为一屏高，页面滚动交给右侧内容区，                                     4656
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */           4657
.app-shell {                                                                  4658
    display: flex;                                                            4659
    align-items: stretch;                                                     4660
    height: 100vh;                                                            4661
    height: 100dvh;                                                           4662
    overflow: hidden;                                                         4663
}                                                                             4664
/* ---------- 侧边栏 ---------- */                                           4665
.app-side {                                                                   4666
    flex: 0 0 var(--nav-w);                                                   4667
    width: var(--nav-w);                                                      4668
    display: flex;                                                            4669
    flex-direction: column;                                                   4670
    background: linear-gradient(180deg, var(--side-1) 0%,                    4671
var(--side-2) 100%);                                                          4672
    color: #fff;                                                              4673
}                                                                             4674
.app-brand {                                                                  4675
    display: flex;                                                            4676
    align-items: center;                                                      4677
    gap: 11px;                                                                4678
    padding: 17px 16px 15px;                                                  4679
    border-bottom: 1px solid rgba(255, 255, 255, .10);                        4680
}                                                                             4681
.app-brand__mark {                                                            4682
    flex: none;                                                               4683
    display: grid;                                                            4684
    place-items: center;                                                      4685
    width: 36px;                                                              4686
    height: 36px;                                                             4687
    border-radius: 10px;                                                      4688
    background: rgba(255, 255, 255, .15);                                     4689
}                                                                             4690
.app-brand__name {                                                            4691
    font-size: 14.5px;                                                        4692
    font-weight: 600;                                                         4693
    line-height: 1.35;                                                        4694
    letter-spacing: .2px;                                                     4695
}                                                                             4696
.app-brand__sub {                                                             4697
    font-size: 11.5px;                                                        4698
    line-height: 1.35;                                                        4699
    color: rgba(255, 255, 255, .58);                                          4700
```

```
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y: auto; }
.app-nav__group {
  margin: 12px 0 5px;
  padding: 0 10px;
  font-size: 11.5px;
  letter-spacing: .05em;
  color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 9px 11px;
  margin-bottom: 2px;
  border-radius: 8px;
  color: rgba(255, 255, 255, .80);
  text-decoration: none;
  transition: background-color .16s ease, color .16s ease;
}
.app-nav a:hover { background: rgba(255, 255, 255, .10); color: #fff; }
.app-nav a:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
  background: rgba(255, 255, 255, .17);
  color: #fff;
  font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
  margin-left: auto;
  min-width: 20px;
  padding: 0 6px;
  font-size: 11px;
  font-weight: 600;
  line-height: 18px;
  text-align: center;
  border-radius: 9px;
  background: var(--todo);
  color: #fff;
  font-variant-numeric: tabular-nums;
}
.app-user {
  display: flex;
```

```
    align-items: center;
    gap: 8px;
    padding: 10px 14px;
    border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
    display: flex;
    flex: 1;
    min-width: 0;
    align-items: center;
    gap: 10px;
    padding: 4px 6px;
    border-radius: 8px;
    color: inherit;
    text-decoration: none;
    transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255,
.10); }
.app-user__me:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-user__avatar {
    flex: none;
    display: grid;
    place-items: center;
    width: 32px;
    height: 32px;
    border-radius: 50%;
    background: rgba(255, 255, 255, .16);
    font-size: 13px;
    font-weight: 600;
    object-fit: cover;
}
.app-user__name {
    display: block;
    font-size: 13px;
    line-height: 1.3;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}
.app-user__state {
    display: flex;
    align-items: center;
    gap: 5px;
    font-size: 11.5px;
    line-height: 1.3;
    color: rgba(255, 255, 255, .60);
```

```
}
.app-user__dot {
  width: 6px;
  height: 6px;
  border-radius: 50%;
  background: #6ed49a;
}
.app-user__out {
  flex: none;
  padding: 6px;
  border-radius: 6px;
  color: rgba(255, 255, 255, .68);
  text-decoration: none;
}
.app-user__out:hover { color: #fff; background:
rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ----------
   整个骨架固定一屏高，滚动只发生在内容区内部：
   左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */
.app-content {
  flex: 1;
  min-width: 0;
  display: flex;
  flex-direction: column;
  overflow-y: auto;
  overflow-x: hidden;
  /* 细滚动条：默认半透明，悬停时加深，不抢视线 */
  scrollbar-width: thin;
  scrollbar-color: rgba(31, 107, 69, .28) transparent;
}
.app-content::-webkit-scrollbar { width: 12px; }
.app-content::-webkit-scrollbar-track { background:
transparent; }
.app-content::-webkit-scrollbar-thumb {
  background: rgba(31, 107, 69, .26);
  border: 3px solid transparent;
  background-clip: content-box;
  border-radius: 999px;
}
.app-content::-webkit-scrollbar-thumb:hover {
  background: rgba(31, 107, 69, .46);
  background-clip: content-box;
}
/* ---------- 顶栏：面包屑 + 搜索 ----------
   粘在内容区顶部，滚动时面包屑与搜索框始终可用 */
.app-header {
  position: sticky;
  top: 0;
  z-index: 20;
  display: flex;
```

```
    align-items: center;
    gap: 14px;
    height: var(--header-h);
    padding: 0 22px;
    background: var(--surface);
    border-bottom: 1px solid var(--line);
}
.crumb {
    display: flex;
    align-items: center;
    gap: 7px;
    font-size: 13px;
    color: var(--ink-soft);
    white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
    display: flex;
    align-items: center;
    gap: 8px;
    width: 268px;
    padding: 7px 12px;
    background: var(--canvas);
    border: 1px solid var(--line);
    border-radius: 9px;
    color: var(--ink-soft);
    transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
    flex: 1;
    min-width: 0;
    border: none;
    outline: none;
    background: transparent;
    font-family: inherit;
    font-size: 13px;
    color: var(--ink);
}
.search-box button {
    display: grid;
    place-items: center;
    padding: 0;
```

```
    border: none;                                                    4901
    background: transparent;                                         4902
    color: inherit;                                                  4903
    cursor: pointer;                                                 4904
}                                                                    4905
.icon-btn {                                                          4906
    display: grid;                                                   4907
    place-items: center;                                             4908
    width: 34px;                                                     4909
    height: 34px;                                                    4910
    border-radius: 9px;                                              4911
    border: 1px solid transparent;                                   4912
    color: var(--ink-soft);                                          4913
    text-decoration: none;                                           4914
    transition: background-color .16s ease, color .16s               4915
ease,                                                                4916
                border-color .16s ease;                              4917
}                                                                    4918
.icon-btn:hover {                                                    4919
    color: var(--accent);                                            4920
    background: var(--accent-soft);                                  4921
    border-color: rgba(31, 107, 69, .18);                            4922
}                                                                    4923
.app-main {                                                          4924
    flex: 1;                                                         4925
    min-width: 0;                                                    4926
    padding: 22px 26px 26px;                                         4927
}                                                                    4928
.container { max-width: none; width: 100%; }                         4929
/* ---------- 页脚 ---------- */                                     4930
.app-foot {                                                          4931
    display: flex;                                                   4932
    align-items: center;                                             4933
    justify-content: space-between;                                  4934
    gap: 14px;                                                       4935
    padding: 0 26px 18px;                                            4936
    font-size: 12px;                                                 4937
    color: var(--ink-soft);                                          4938
}                                                                    4939
/* ---------- 未登录时的裸页面（登录页） ---------- */               4940
.login-shell {                                                       4941
    display: flex;                                                   4942
    flex-direction: column;                                          4943
    min-height: 100dvh;                                              4944
}                                                                    4945
/* ---------- 卡片 ---------- */                                     4946
.card {                                                              4947
    background: var(--surface);                                      4948
    border: 1px solid var(--line);                                   4949
    border-radius: var(--radius);                                    4950
```

```
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 16px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
}
.card > h3 {
    margin: 20px 0 10px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0 16px;
    font-variant-numeric: tabular-nums;
}
caption {
    text-align: left;
    color: var(--ink-soft);
    font-size: 12px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
    padding: 8px 12px;
    text-align: center;
    vertical-align: middle;
}
th {
    background: var(--accent-soft);
    font-weight: 600;
    white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
    display: block;
    margin-bottom: 4px;
    color: var(--ink-soft);
    font-size: 13px;
```

```
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
  padding: 6px 9px;
  font-family: inherit;
  font-size: 14px;
  color: var(--ink);
  background: #fff;
  border: 1px solid #c6ccd4;
  border-radius: 4px;
  max-width: 100%;
}
input[type="number"] {
  width: 104px;
  text-align: center;
  font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
  border-color: var(--accent);
}
.checkbox {
  display: inline-flex;
  align-items: center;
  gap: 7px;
  cursor: pointer;
  user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 18px;
  font-family: inherit;
  font-size: 14px;
  line-height: 1.4;
  color: #fff;
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
```

```
  cursor: pointer;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.btn:hover {
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
.btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.btn--ghost {
  color: var(--accent);
  background: rgba(31, 107, 69, .06);
  border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
}
.btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 13px; border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
  background: rgba(31, 107, 69, .80);
  border-color: rgba(31, 107, 69, .30);
  box-shadow: none;
}
.btn--ghost:disabled:hover { background: rgba(31, 107, 69, .06); color: var(--accent); }
/* ---------- 表格操作列的行内小按钮 ----------
   取代原来「文字 | 文字 | 文字」的写法：
   圆角矩形 + 半透明底色，悬停时变实色。 */
```

```
.row-actions {
  display: inline-flex;
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */
  align-items: center;
  gap: 6px;
  white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
  display: inline-flex;
  align-items: center;
  padding: 4px 11px;
  font-family: inherit;
  font-size: 13px;
  line-height: 1.5;
  color: var(--accent);
  background: rgba(31, 107, 69, .07);
  border: 1px solid rgba(31, 107, 69, .20);
  border-radius: 7px;
  cursor: pointer;
  text-decoration: none;
  white-space: nowrap;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.link-btn:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px); box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.link-btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
```

```
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 浮层提示（登录成功、各类操作完成、登录失败） ---------- */
   外层 toast-stack 固定在页面中上方，内层每条各自淡入 → 停留 → 淡出，
   多条提示会依次排开而不重叠。
   底色是浅绿半透明：既有一层底色把文字托住，又能透出后面的内容，
   不用高饱和色块，提示不该比页面内容更抢眼。
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 16px;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 6px;
  transform: translateX(-50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(620px, 82vw);
  padding: 8px 22px;
  border-radius: 999px;
  /* 浅绿、半透明：不遮住背后的内容，也比纯透明更托得住文字 */
  background: rgba(223, 241, 231, .76);
  border: 1px solid rgba(31, 107, 69, .14);
  color: #2f4a3c;
  font-size: 16px;
  font-weight: 500;
  line-height: 1.65;
  letter-spacing: .03em;
  text-align: center;
  text-wrap: pretty;
```

```
    backdrop-filter: blur(6px) saturate(1.1);
    -webkit-backdrop-filter: blur(6px) saturate(1.1);
    box-shadow: 0 4px 16px rgba(23, 32, 27, .07);
    animation: toast-fade 2.6s ease-in-out forwards;
}
/* 失败类提示：同一套观感，换成浅红底，一眼能区分成败 */
.toast-center--error {
    background: rgba(252, 233, 230, .78);
    border-color: rgba(163, 43, 34, .16);
    color: #9c2b23;
}
@keyframes toast-fade {
    0%          { opacity: 0; transform: translateY(-6px); }
    14%, 74%    { opacity: 1; transform: none; }
    100%        { opacity: 0; transform: none; }
}
/* 系统设置里要求「减少动态效果」时，去掉位移，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
    .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
    @keyframes toast-fade-plain {
        0%          { opacity: 0; }
        14%, 74%    { opacity: 1; }
        100%        { opacity: 0; }
    }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
    display: inline-block;
    padding: 1px 9px;
    font-size: 12.5px;
    border-radius: 10px;
    border: 1px solid;
    white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
```

```
.tag--warn { color: var(--danger); background:                    5251
var(--danger-soft); border-color: #eec4c0; }                      5252
.tag--muted { color: var(--ink-soft); background:                 5253
#f1f3f6; border-color: #dde1e6; }                                 5254
/* ---------- 登录页 ---------- */                                 5255
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，    5256
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */    5257
.body--login {                                                    5258
  background:                                                     5259
    radial-gradient(1100px 640px at 50% 8%,                       5260
                    #edf6f1 0%, rgba(237, 246, 241, 0)            5261
72%),                                                             5262
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);            5263
}                                                                 5264
.body--login .app-main { padding: 0; }                            5265
.body--login .container { max-width: none; width: 100%;           5266
}                                                                 5267
/* 仅供读屏软件的隐藏文本 */                                       5268
.visually-hidden {                                                5269
  position: absolute !important;                                  5270
  width: 1px;                                                     5271
  height: 1px;                                                    5272
  margin: -1px;                                                   5273
  padding: 0;                                                     5274
  overflow: hidden;                                               5275
  clip: rect(0 0 0 0);                                            5276
  white-space: nowrap;                                            5277
  border: 0;                                                      5278
}                                                                 5279
/* ---------- 工作台：页首横幅 ---------- */                       5280
.hero {                                                           5281
  position: relative;                                             5282
  overflow: hidden;                                               5283
  min-height: 176px;                                              5284
  padding: 26px 28px;                                             5285
  margin-bottom: 16px;                                            5286
  border-radius: 12px;                                            5287
  /* 左侧压深保证文字可读，右侧透出远山插画；                      5288
     背景按底部对齐，避免远山被居中裁掉 */                         5289
  background:                                                     5290
    linear-gradient(100deg, rgba(15, 40, 45, .90) 0%,             5291
                            rgba(20, 55, 58, .55) 45%,            5292
                            rgba(20, 55, 58, .12) 100%),          5293
    url("{{ url_for('static', filename='hero_landscape.j          5294
pg') }}")                                                         5295
    center center / cover no-repeat;                              5296
  color: #fff;                                                    5297
}                                                                 5298
.hero__body { max-width: 720px; }                                 5299
.hero__title {                                                    5300
```

```
    margin: 0 0 10px;
    font-size: 21px;
    font-weight: 700;
    line-height: 1.4;
    letter-spacing: .3px;
}
.hero__desc {
    margin: 0;
    max-width: 640px;
    font-size: 12.5px;
    line-height: 1.85;
    color: rgba(255, 255, 255, .86);
}
.hero__actions {
    display: flex;
    flex-wrap: wrap;
    gap: 10px;
    margin-top: 18px;
}
.hero-btn {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    padding: 7px 15px;
    font-size: 13px;
    border-radius: 8px;
    text-decoration: none;
    transition: background-color .16s ease, border-color .16s ease,
                color .16s ease;
}
.hero-btn--solid {
    color: #14432f;
    background: rgba(255, 255, 255, .92);
    border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
    color: #fff;
    background: rgba(255, 255, 255, .14);
    border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(190px,
```

```
1fr));                                                                    5351
  gap: 12px;                                                              5352
  margin-bottom: 22px;                                                    5353
}                                                                         5354
.stat-card {                                                              5355
  display: block;                                                         5356
  padding: 15px 17px;                                                     5357
  background: var(--surface);                                             5358
  border: 1px solid var(--line);                                          5359
  border-radius: var(--radius);                                           5360
  box-shadow: var(--shadow);                                              5361
  color: var(--ink);                                                      5362
  text-decoration: none;                                                  5363
  transition: border-color .16s ease, box-shadow .16s                     5364
ease;                                                                     5365
}                                                                         5366
.stat-card:hover {                                                        5367
  border-color: rgba(31, 107, 69, .45);                                   5368
  box-shadow: 0 3px 12px rgba(19, 60, 43, .08);                           5369
}                                                                         5370
.stat-card__label {                                                       5371
  display: block;                                                         5372
  font-size: 12.5px;                                                      5373
  color: var(--ink-soft);                                                 5374
}                                                                         5375
.stat-card__value {                                                       5376
  display: block;                                                         5377
  margin: 3px 0 2px;                                                      5378
  font-size: 26px;                                                        5379
  font-weight: 700;                                                       5380
  line-height: 1.25;                                                      5381
  color: var(--accent);                                                   5382
  font-variant-numeric: tabular-nums;                                     5383
}                                                                         5384
.stat-card__unit {                                                        5385
  margin-left: 3px;                                                       5386
  font-size: 13px;                                                        5387
  font-weight: 500;                                                       5388
  color: var(--ink-soft);                                                 5389
}                                                                         5390
.stat-card__note {                                                        5391
  display: block;                                                         5392
  font-size: 12px;                                                        5393
  color: var(--ink-soft);                                                 5394
}                                                                         5395
/* ---------- 工作台：分区标题 ---------- */                              5396
.section { margin-bottom: 22px; }                                         5397
.section__head {                                                          5398
  display: flex;                                                          5399
  align-items: baseline;                                                  5400
```

```
    flex-wrap: wrap;                                                    5401
    gap: 6px 14px;                                                      5402
    margin-bottom: 12px;                                                5403
}                                                                       5404
.section__title { margin: 0; font-size: 16px; }                         5405
.section__desc { margin: 0; font-size: 12.5px; color:                   5406
var(--ink-soft); }                                                      5407
/* ---------- 工作台：流程步骤卡 ---------- */                           5408
.step-grid {                                                            5409
    display: grid;                                                      5410
    grid-template-columns: repeat(auto-fill,                            5411
minmax(238px, 1fr));                                                    5412
    gap: 12px;                                                          5413
}                                                                       5414
.step-card {                                                            5415
    position: relative;                                                 5416
    display: block;                                                     5417
    padding: 16px 16px 15px;                                            5418
    background: var(--surface);                                         5419
    border: 1px solid var(--line);                                      5420
    border-radius: var(--radius);                                       5421
    box-shadow: var(--shadow);                                          5422
    color: var(--ink);                                                  5423
    text-decoration: none;                                              5424
    transition: border-color .16s ease, box-shadow .16s                 5425
ease,                                                                   5426
                transform .16s ease;                                    5427
}                                                                       5428
.step-card:hover {                                                      5429
    border-color: rgba(31, 107, 69, .55);                               5430
    box-shadow: 0 5px 16px rgba(19, 60, 43, .10);                       5431
    transform: translateY(-1px);                                        5432
}                                                                       5433
.step-card:focus-visible { outline: 2px solid                           5434
var(--accent); outline-offset: 2px; }                                   5435
.step-card__no {                                                        5436
    position: absolute;                                                 5437
    top: 13px;                                                          5438
    right: 15px;                                                        5439
    font-size: 17px;                                                    5440
    font-weight: 700;                                                   5441
    color: #d7e3db;                                                     5442
    font-variant-numeric: tabular-nums;                                 5443
}                                                                       5444
.step-card__icon {                                                      5445
    display: grid;                                                      5446
    place-items: center;                                                5447
    width: 32px;                                                        5448
    height: 32px;                                                       5449
    margin-bottom: 10px;                                                5450
```

```
    border-radius: 9px;
    background: var(--accent-soft);
    color: var(--accent);
}
.step-card__title {
    display: block;
    margin-bottom: 5px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.step-card__desc {
    display: block;
    font-size: 12.5px;
    line-height: 1.65;
    color: var(--ink-soft);
    text-wrap: pretty;
}
.step-card__badge {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #fdf1e2;
    color: #9a5a12;
    border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
    background: #fafbfa;
    border-color: #eceff0;
    box-shadow: none;
    cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
    background: #f1f3f2;
    color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #f1f3f2;
    color: #8b948f;
```

```
}
/* ---------- 结果摘要 ---------- */
.stat-row {
  display: flex;
  flex-wrap: wrap;
  gap: 28px;
  margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
  display: flex; align-items: center; gap: 7px;
  padding: 8px 12px; font-size: 13px;
  background: #fff; border: 1px solid var(--line);
  border-radius: var(--radius);
}
.step__no {
  display: inline-flex; align-items: center;
justify-content: center;
  width: 20px; height: 20px; border-radius: 50%;
  background: #eef1f5; color: var(--ink-soft);
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块（中性底色，不用整块警示色压页面） ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  background: var(--surface);
  border: 1px solid var(--line);
```

```
    border-left: 3px solid var(--todo);
    border-radius: var(--radius);
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
/* ---------- 前置条件表（与页面主表同款，只把待处理的行标成浅红） ---------- */
.prereq-table { margin: 0 0 18px; }
.prereq-table > caption { padding-bottom: 7px; }
.prereq-row > td { background: var(--danger-soft); }
.prereq-row:hover > td { background: #f6dcd7; }
.prereq-flag { color: var(--danger); font-weight: 600; }
.prereq-row .muted { color: #9a6b62; }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
    border: 1px solid var(--line);
    border-radius: var(--radius);
    background: var(--surface);
    padding: 14px 16px;
    margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin: 0 0 12px;
    padding-bottom: 9px;
    border-bottom: 1px solid var(--line);
}
.patch-panel__name {
    margin: 0;
    font-size: 15px;
    font-weight: 600;
}
.patch-panel__meta {
    font-size: 12.5px;
    color: var(--ink-soft);
}
.patch-panel__meta a { color: var(--accent); }
.patch-panel__tail { margin-left: auto; }
.patch-panel .notice { margin: 0; }
.patch-panel table { margin: 0; }
/* ---------- 窄屏适配 ---------- */
@media (max-width: 960px) {
    /* 上下布局时取消「一屏高 + 内容区内部滚动」，
       否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */
    .app-shell {
        flex-direction: column;
        height: auto;
        min-height: 100vh;
```

```
            min-height: 100dvh;
            overflow: visible;
        }
        .app-content { overflow: visible; }
        .app-header { position: static; }
        .app-side { width: auto; flex: none; }
        .app-brand { padding: 12px 14px; }
        .app-nav { padding: 4px 10px 10px; }
        .app-nav ul { display: flex; flex-wrap: wrap; gap:
    4px; }
        .app-nav a { margin-bottom: 0; }
        .app-nav__group { margin: 8px 0 4px; }
        .app-user { padding: 10px 14px; }
        .app-header { padding: 0 14px; gap: 10px; }
        .search-box { width: 180px; }
        .app-main { padding: 16px 14px 20px; }
        .app-foot { padding: 0 14px 14px; }
    }
    @media (max-width: 640px) {
        .crumb { display: none; }
        .search-box { width: auto; flex: 1; }
    }
  </style>
</head>
<body class="{% if not current_user.is_authenticated
    %}body--login{% endif %}">
  <a class="skip-link" href="#main">跳到主要内容</a>
  {# 取一次即可（同一请求内重复调用返回相同内容）。
     类别为 toast / ok / toast-error 的渲染成中上方浮层，出现后自动淡出；
     其余仍是页面顶部的提示条 ——
     需要对照着改表单的错误，留在页面上比一闪而过更有用。 #}
  {% set flash_msgs = get_flashed_messages(with_categories=t
    rue) %}
  <div class="toast-stack">
    {% for category, message in flash_msgs if category in
    ("toast", "ok", "toast-error") %}
    <div class="toast-center{% if category == 'toast-error'
    %} toast-center--error{% endif %}"
         role="status" aria-live="polite">{{ message
    }}</div>
    {% endfor %}
  </div>
  {% if current_user.is_authenticated %}
  <div class="app-shell">
    <aside class="app-side">
      <div class="app-brand">
        <span class="app-brand__mark">{{
    ui.icon("mountain", 21) }}</span>
        <span>
          <span class="app-brand__name">矿山自然恢复评价</span><br>
```

```
        <span class="app-brand__sub">适宜性评价系统 V1.0</span>
      </span>
    </div>
    <nav class="app-nav" aria-label="功能导航">
      <ul>
        <li>
          <a href="/"{% if home_active %}
aria-current="page"{% endif %}>
            {{ ui.icon("home") }}<span>工作台</span>
          </a>
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for item in g.links %}
        <li>
          <a href="{{ item.href }}"{% if item.active %}
aria-current="page"{% endif %}>
            {{ ui.icon(item.icon) }}<span>{{ item.label
}}</span>
            {% if item.badge %}<span
class="app-nav__badge">{{ item.badge }}</span>{% endif
%}
          </a>
        </li>
        {% endfor %}
      </ul>
      {% endfor %}
    </nav>
    <div class="app-user">
      <a class="app-user__me" href="/profile"
title="个人设置">
        {% if current_user.avatar %}
        <img class="app-user__avatar"
             src="{{ url_for('static',
filename=current_user.avatar) }}" alt="">
        {% else %}
        <span class="app-user__avatar"
aria-hidden="true">{{ current_user.initial }}</span>
        {% endif %}
        <span style="min-width:0;">
          <span class="app-user__name">{{
current_user.shown_name }}</span>
          <span class="app-user__state">
            <span class="app-user__dot"
aria-hidden="true"></span>
            {{ role_labels.get(current_user.role,
current_user.role) }}
          </span>
```

```
            </span>
          </a>
          <a class="app-user__out" href="/logout" title="退出登录"
             aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>
        </div>
      </aside>
      <div class="app-content">
        <header class="app-header">
          <nav class="crumb" aria-label="当前位置">
            <a href="/">首页</a>
            {{ ui.icon("chevron", 13) }}
            <strong>{{ page_title }}</strong>
          </nav>
          <span class="app-header__spacer"></span>
          <form class="search-box" method="get"
      action="/search" role="search">
            {{ ui.icon("search", 16) }}
            <label for="q" class="visually-hidden">搜索项目、图斑、标准</
      label>
            <input id="q" type="search" name="q"
      autocomplete="off"
                   placeholder="搜索项目、图斑、标准" value="{{
      search_q }}">
            <button type="submit" aria-label="搜索">{{
      ui.icon("chevron", 15) }}</button>
          </form>
          {% if current_user.is_admin %}
          <a class="icon-btn" href="/profile?tab=logs"
      title="操作日志"
             aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>
          {% endif %}
        </header>
  {% else %}
  <div class="login-shell">
  {% endif %}
      <main id="main" class="app-main">
        <div class="container">
          {% for category, message in flash_msgs if category
      not in ("toast", "ok", "toast-error") %}
          <p class="flash {{ category }}" role="status">{{
      message }}</p>
          {% endfor %}
          {% block content %}{% endblock %}
        </div>
      </main>
  {% if current_user.is_authenticated %}
        <footer class="app-foot">
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０</span>
          <span>数据截止 {{ today }}</span>
        </footer>
```

```
    </div>
  </div>
  {% else %}
  </div>
  {% endif %}
  {% block scripts %}{% endblock %}
</body>
</html>
# ==== app/templates/_macros.html ====
{# 共用宏：图标、流程状态面板 #}
{# ============================================================
   ================== 
   图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）
   不引图标字体或外部图片，避免额外请求与资源依赖；
   stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。
   ============================================================
   ================== #}
{% set ICONS = {
  "dot": '<circle cx="12" cy="12" r="3"/>',
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
    17h5"/>',
  "database": '<ellipse cx="12" cy="5.5" rx="8"
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2"/>',
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
  "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
    21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
    4.4V21"/>',
  "mail": '<rect x="3" y="5" width="18" height="14"
```

```
      rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>',
  "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0
      0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6
      16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>',
  "lock": '<rect x="4" y="10.4" width="16" height="10.6"
      rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>',
  "briefcase": '<rect x="3" y="7.4" width="18"
      height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1
      10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>',
  "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
      20.5-4-4"/>',
  "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
  "plus": '<path d="M12 5v14M5 12h14"/>',
  "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
  "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
  "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
  "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
  "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
     fill="none" stroke="currentColor" stroke-width="1.7"
     stroke-linecap="round" stroke-linejoin="round"
     aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
```

```
      </span>                                                                  5851
    </div>                                                                     5852
    <div class="step {% if progress.units.done %}step--done{%                 5853
      else %}step--todo{% endif %}">                                           5854
      <span class="step__no">2</span>                                          5855
      <span class="step__name">划分微地质单元</span>                              5856
      <span class="step__state">                                               5857
        {% if progress.units.done %}已完成{% else %}待完成{% endif             5858
      %}                                                                       5859
      </span>                                                                  5860
    </div>                                                                     5861
    <div class="step {% if progress.data.ready %}step--done{%                  5862
      else %}step--todo{% endif %}">                                           5863
      <span class="step__no">3</span>                                          5864
      <span class="step__name">录入指标数据</span>                               5865
      <span class="step__state">                                               5866
        {% if progress.data.ready %}已完成{% else %}待完成{% endif             5867
      %}                                                                       5868
      </span>                                                                  5869
    </div>                                                                     5870
    <div class="step step--next">                                              5871
      <span class="step__no">4</span>                                          5872
      <span class="step__name">执行适宜性评价</span>                             5873
      <span class="step__state">当前步骤</span>                                  5874
    </div>                                                                     5875
</div>                                                                         5876
{% endmacro %}                                                                 5877
{# 前置条件未满足时的提示：做成与页面主表同款的表格，只把待处理的那几行标成浅红，                  5878
   说明写在 caption 里，缺什么、去哪补都在行内。 #}                              5879
{% macro workflow_blocking(progress) %}                                        5880
{% if progress.blocking %}                                                     5881
<table class="prereq-table">                                                   5882
  <caption>本步骤的前置条件尚未满足，按论文 4.1 节的流程顺序先补上：</caption>          5883
  <thead>                                                                      5884
    <tr>                                                                       5885
      <th scope="col">前置步骤</th>                                            5886
      <th scope="col">说明</th>                                                5887
      <th scope="col">操作</th>                                                5888
    </tr>                                                                      5889
  </thead>                                                                     5890
  <tbody>                                                                      5891
    {% for b in progress.blocking %}                                           5892
    <tr class="prereq-row">                                                    5893
      <td>{{ b.label }}</td>                                                   5894
      <td class="muted">{{ b.reason }}</td>                                    5895
      <td><a class="btn btn--sm" href="{{ b.fix_url }}">{{                     5896
    b.fix_label }}</a></td>                                                    5897
    </tr>                                                                      5898
    {% endfor %}                                                               5899
  </tbody>                                                                     5900
```

```
</table>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   只在流程里**第一个**按图斑操作的页面（限制性指标判定）列一次，后面各页
   不再重复；建好图斑才会出现在下面的表里，但离「能进入本页流程」还差别的步骤，
   所以措辞只说到「出现在这里」为止。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<table class="prereq-table">
  <caption>以下项目下尚未建立矿山图斑，本页按图斑组织数据，建好图斑后才会出现在这里：</caption>
  <thead>
    <tr>
      <th scope="col">所属项目</th>
      <th scope="col">矿山图斑</th>
      <th scope="col">状态</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in projects_without_patch %}
    <tr class="prereq-row">
      <td>项目「{{ p.name }}」</td>
      <td class="muted">未建立</td>
      <td><span class="prereq-flag">缺少矿山图斑</span></td>
      <td>
        <a class="btn btn--sm" href="/project/{{ p.id }}/patch/add">
          {{ icon("plus", 14) }}去新建图斑
        </a>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endif %}
{% endmacro %}
# ==== app/templates/_login_badge.svg ====
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg" viewBox="0 0 240 240" role="img"
```

```
  aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
<title>系统徽标</title>
<g class="lp-ring" style="--k: 2.7">
  <circle cx="162.00" cy="120.00" r="2.60"
  fill="#c4f5ab" />
  <circle cx="157.84" cy="138.22" r="2.97"
  fill="#b8f5ab" />
  <circle cx="146.19" cy="152.84" r="3.06"
  fill="#acf5ab" />
  <circle cx="129.35" cy="160.95" r="2.80"
  fill="#abf5b7" />
  <circle cx="110.65" cy="160.95" r="2.40"
  fill="#abf5c2" />
  <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce" />
  <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da" />
  <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6" />
  <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da" />
  <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce" />
  <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2" />
  <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7" />
  <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab" />
  <circle cx="157.84" cy="101.78" r="2.23"
  fill="#b8f5ab" />
</g>
<g class="lp-ring" style="--k: 2.15">
  <circle cx="177.29" cy="129.07" r="3.92"
  fill="#a4e28d" />
  <circle cx="171.68" cy="146.33" r="4.01"
  fill="#9be28d" />
  <circle cx="161.01" cy="161.01" r="3.87"
  fill="#91e28d" />
  <circle cx="146.33" cy="171.68" r="3.55"
  fill="#8de292" />
  <circle cx="129.07" cy="177.29" r="3.17"
  fill="#8de29c" />
  <circle cx="110.93" cy="177.29" r="2.88"
  fill="#8de2a6" />
  <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af" />
  <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9" />
  <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2" />
  <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc" />
  <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc" />
  <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2" />
  <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9" />
  <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af" />
  <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6" />
  <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c" />
  <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292" />
  <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d" />
  <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d" />
```

```
    <circle cx="177.29" cy="110.93" r="3.63"                    6001
    fill="#a4e28d"/>                                            6002
  </g>                                                          6003
  <g class="lp-ring" style="--k: 1.85">                         6004
    <circle cx="194.00" cy="120.00" r="4.94"                    6005
    fill="#91cf72"/>                                            6006
    <circle cx="191.85" cy="137.71" r="4.92"                    6007
    fill="#88cf72"/>                                            6008
    <circle cx="185.52" cy="154.39" r="4.73"                    6009
    fill="#80cf72"/>                                            6010
    <circle cx="175.39" cy="169.07" r="4.42"                    6011
    fill="#78cf72"/>                                            6012
    <circle cx="162.04" cy="180.90" r="4.06"                    6013
    fill="#72cf73"/>                                            6014
    <circle cx="146.24" cy="189.19" r="3.73"                    6015
    fill="#72cf7b"/>                                            6016
    <circle cx="128.92" cy="193.46" r="3.51"                    6017
    fill="#72cf83"/>                                            6018
    <circle cx="111.08" cy="193.46" r="3.45"                    6019
    fill="#72cf8b"/>                                            6020
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93"/>    6021
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c"/>    6022
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4"/>    6023
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac"/>    6024
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4"/>    6025
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc"/>    6026
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4"/>    6027
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac"/>     6028
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4"/>     6029
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c"/>     6030
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93"/>     6031
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b"/>    6032
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83"/>    6033
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b"/>    6034
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73"/>    6035
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72"/>    6036
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72"/>    6037
    <circle cx="191.85" cy="102.29" r="4.80"                    6038
    fill="#88cf72"/>                                            6039
  </g>                                                          6040
  <g class="lp-ring" style="--k: 1.62">                         6041
    <circle cx="209.57" cy="128.82" r="5.68"                    6042
    fill="#76bb59"/>                                            6043
    <circle cx="206.12" cy="146.13" r="5.40"                    6044
    fill="#6fbb59"/>                                            6045
    <circle cx="199.37" cy="162.43" r="5.06"                    6046
    fill="#68bb59"/>                                            6047
    <circle cx="189.57" cy="177.10" r="4.71"                    6048
    fill="#61bb59"/>                                            6049
    <circle cx="177.10" cy="189.57" r="4.41"                    6050
```

```
    fill="#5abb59" />                                                    6051
    <circle cx="162.43" cy="199.37" r="4.19"                             6052
    fill="#59bb5f" />                                                    6053
    <circle cx="146.13" cy="206.12" r="4.10"                             6054
    fill="#59bb66" />                                                    6055
    <circle cx="128.82" cy="209.57" r="4.15"                             6056
    fill="#59bb6d" />                                                    6057
    <circle cx="111.18" cy="209.57" r="4.32"                             6058
    fill="#59bb73" />                                                    6059
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a" />            6060
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81" />            6061
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88" />            6062
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f" />            6063
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96" />            6064
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d" />            6065
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4" />            6066
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4" />            6067
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d" />             6068
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96" />             6069
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f" />             6070
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88" />             6071
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81" />             6072
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a" />             6073
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73" />            6074
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d" />            6075
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66" />            6076
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f" />            6077
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59" />            6078
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59" />            6079
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59" />            6080
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59" />            6081
    <circle cx="209.57" cy="111.18" r="5.85"                             6082
    fill="#76bb59" />                                                    6083
  </g>                                                                   6084
  <g class="lp-ring" style="--k: 1.46">                                  6085
    <circle cx="226.00" cy="120.00" r="5.94"                             6086
    fill="#65a843" />                                                    6087
    <circle cx="224.55" cy="137.45" r="5.62"                             6088
    fill="#5fa843" />                                                    6089
    <circle cx="220.26" cy="154.42" r="5.29"                             6090
    fill="#59a843" />                                                    6091
    <circle cx="213.22" cy="170.45" r="5.00"                             6092
    fill="#53a843" />                                                    6093
    <circle cx="203.65" cy="185.11" r="4.77"                             6094
    fill="#4da843" />                                                    6095
    <circle cx="191.79" cy="197.99" r="4.63"                             6096
    fill="#47a843" />                                                    6097
    <circle cx="177.98" cy="208.74" r="4.59"                             6098
    fill="#43a846" />                                                    6099
    <circle cx="162.58" cy="217.07" r="4.67"                             6100
```

```
      fill="#43a84c"/>                                                    6101
      <circle cx="146.02" cy="222.76" r="4.85"                            6102
      fill="#43a852"/>                                                    6103
      <circle cx="128.75" cy="225.64" r="5.10"                            6104
      fill="#43a858"/>                                                    6105
      <circle cx="111.25" cy="225.64" r="5.42"                            6106
      fill="#43a85e"/>                                                    6107
      <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>            6108
      <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>            6109
      <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>            6110
      <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>            6111
      <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>            6112
      <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>            6113
      <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>            6114
      <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>            6115
      <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>            6116
      <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>            6117
      <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>             6118
      <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>             6119
      <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>             6120
      <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>             6121
      <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>             6122
      <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>             6123
      <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>             6124
      <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>            6125
      <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>            6126
      <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>            6127
      <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>            6128
      <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>            6129
      <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>            6130
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>            6131
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>            6132
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>            6133
      <circle cx="224.55" cy="102.55" r="6.23"                            6134
      fill="#5fa843"/>                                                    6135
    </g>                                                                  6136
    <g class="lp-badge__mark">                                            6137
      <circle cx="120" cy="120" r="27" fill="#14432f"/>                   6138
      <g transform="translate(120 120) scale(1.5)                         6139
      translate(-12 -12)" fill="none" stroke="#ffffff"                    6140
      stroke-width="1.9" stroke-linecap="round"                           6141
      stroke-linejoin="round">                                            6142
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>                       6143
        <circle cx="17" cy="6" r="2.2"/>                                  6144
      </g>                                                                6145
    </g>                                                                  6146
</svg>                                                                    6147
# ==== app/templates/error.html ====                                      6148
{% extends "base.html" %}                                                 6149
{% block title %}{{ code }} · {{ message }}{% endblock %}                 6150
```

```
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>{{ code }} · {{ message }}</h2>
  <p class="muted">
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
  </p>
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ==================
     纯白底、居中、大量留白；色彩只用在徽标与按钮上。
     首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
     随滚动逐条弹入、往回滚时弹回。 */
  .lp {
    color: #1d1d1f;
    background:
      linear-gradient(180deg, rgba(245, 239, 224, .88) 0%,
                              rgba(245, 239, 224, .92)
      100%),
      url("../static/hero_qinglv.jpg")
      center top / cover no-repeat fixed;
  }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
    position: relative;
    background:
      linear-gradient(180deg, rgba(15, 40, 45, .72) 0%,
                              rgba(20, 55, 58, .50) 50%,
                              rgba(15, 40, 45, .76) 100%),
      url("../static/hero_qinglv.jpg")
      center center / cover no-repeat;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
```

```
.lp-badge {
  width: 212px;
  height: 212px;
  margin-bottom: 36px;
  --p: 0;
  animation: lp-in .7s ease both;
}
.lp-badge__svg {
  display: block;
  width: 100%;
  height: 100%;
  overflow: visible;                    /* 散开时圆点不被画布裁掉 */
  opacity: calc(1 - var(--p) * .94);
  filter: blur(calc(var(--p) * 2.6px));
}
.lp-ring {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
}
.lp-badge__mark {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + .18 * var(--p)));
}
.lp-title {
  margin: 0 0 18px;
  font-size: clamp(28px, 4.2vw, 42px);
  font-weight: 700;
  line-height: 1.3;
  letter-spacing: -.4px;
  color: #fff;
  text-shadow: 0 2px 12px rgba(0, 0, 0, .3);
  animation: lp-in .7s ease .06s both;
}
.lp-cta {
  padding: 13px 52px;
  font-family: inherit;
  font-size: 17px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
  border-radius: 980px;
  cursor: pointer;
  animation: lp-in .7s ease .18s both;
  transition: background-color .2s ease, box-shadow .2s ease,
              transform .2s ease;
}
```

```
.lp-cta:hover {
  background: var(--accent-hover);
  box-shadow: 0 10px 26px rgba(44, 107, 90, .4);
}
.lp-cta:active { transform: translateY(1px); box-shadow: none; }
.lp-cta:focus-visible { outline: 3px solid rgba(31, 107, 69, .40); outline-offset: 3px; }
.lp-more {
  margin-top: 38px;
  font-size: 13px;
  color: rgba(255, 255, 255, .7);
  text-decoration: none;
  animation: lp-in .7s ease .24s both;
}
.lp-more:hover { color: #fff; }
@keyframes lp-in {
  from { opacity: 0; transform: translateY(12px); }
  to   { opacity: 1; transform: none; }
}
/* ---------- 滚动进场 ---------- */
   进入视口时从屏外飞入，离开视口时飞回，可反复。
   交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
.lp-reveal {
  opacity: 0;
  transition: opacity .7s ease,
              transform 1.05s cubic-bezier(.18, 1.04, .3, 1),
              border-color .62s ease, background-color .62s ease;
  transition-delay: 0s;
  will-change: opacity, transform;
}
/* 起始位置整体在屏外：横向按视口宽度取位移，纵向按视口高度取位移，
   再带一点缩小，落位时按贝塞尔曲线的过冲轻微回弹。 */
.lp-reveal[data-dir="left"]  { transform: translate3d(-56vw, 5vh, 0) scale(.9); }
.lp-reveal[data-dir="right"] { transform: translate3d(56vw, 5vh, 0) scale(.9); }
.lp-reveal[data-dir="up"]    { transform: translate3d(0, 24vh, 0) scale(.94); }
.lp-reveal.is-in {
  opacity: 1;
  transform: none;
  transition-delay: calc(var(--i, 0) * 110ms);
}
/* 标题颜色由淡到深，强化「浮现」的层次感 */
.lp-reveal h2,
.lp-reveal h3,
.lp-reveal .lp-mod__name { color: #c2d2c8; transition:
```

```
    color .7s ease; }
.lp-reveal.is-in h3,
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }
.lp-reveal.is-in h2 { color: #1d1d1f; }
/* ---------- 系统介绍 ---------- */
.lp-intro {
  max-width: none;
  margin: 0;
  padding: 56px 24px 72px;
  background: transparent;
  /* 卡片从屏外飞入，横向超出的部分裁掉，免得出现横向滚动条；
     用 clip 而不是 hidden，纵向滚动仍由页面本身负责 */
  overflow-x: clip;
}
.lp-intro > * { max-width: 1040px; margin-left: auto;
  margin-right: auto; }
.lp-intro__title {
  margin: 0 auto 26px;
  font-size: clamp(22px, 2.6vw, 30px);
  text-align: center;
  letter-spacing: -.3px;
}
/* 三张特点卡与九个模块卡都按三等分排版，并整体居中 */
.lp-points {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 30px 26px;
  background: rgba(253, 250, 242, .82);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  border: 1px solid rgba(224, 216, 200, .6);
  border-radius: 22px;
  box-shadow: 0 24px 60px rgba(42, 37, 32, .10), 0 8px
  20px rgba(42, 37, 32, .05);
  transition: transform .25s ease, box-shadow .25s ease;
}
.lp-point:hover {
  transform: translateY(-4px);
  box-shadow: 0 32px 72px rgba(42, 37, 32, .14), 0 12px
  28px rgba(42, 37, 32, .07);
}
.lp-point__icon {
  display: grid;
  place-items: center;
  width: 40px;
  height: 40px;
```

```
    margin-bottom: 16px;
    border-radius: 12px;
    background: rgba(44, 107, 90, .1);
    color: var(--accent);
  }
  .lp-point h3 { margin: 0 0 8px; font-size: 16px; }
  .lp-point p {
    margin: 0;
    font-size: 13.5px;
    line-height: 1.85;
    color: #6e6e73;
    text-wrap: pretty;
  }
  .lp-sub-title {
    margin: 0 auto 20px;
    font-size: 17px;
    text-align: center;
  }
  .lp-mods {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 12px;
    margin-bottom: 46px;
    align-items: start;
  }
  /* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */
  .lp-mod {
    position: relative;
    padding: 17px 18px 17px 48px;
    background: rgba(253, 250, 242, .75);
    backdrop-filter: blur(8px);
    -webkit-backdrop-filter: blur(8px);
    border: 1px solid rgba(224, 216, 200, .6);
    border-radius: 14px;
    outline: none;
    transition: border-color .3s ease, box-shadow .3s ease;
  }
  .lp-mod:hover,
  .lp-mod:focus-within,
  .lp-mod:focus {
    border-color: rgba(44, 107, 90, .5);
    box-shadow: 0 8px 22px rgba(42, 37, 32, .08);
  }
  .lp-mod__no {
    position: absolute;
    left: 17px;
    top: 17px;
    font-size: 13px;
    font-weight: 700;
    color: #c8d6cd;
```

```
    font-variant-numeric: tabular-nums;
  }
  .lp-mod__name {
    display: block;
    font-size: 14px;
    font-weight: 600;
  }
  /* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
  .lp-mod__body {
    display: grid;
    grid-template-rows: 0fr;
    transition: grid-template-rows .34s ease, margin-top
    .34s ease;
  }
  .lp-mod:hover .lp-mod__body,
  .lp-mod:focus-within .lp-mod__body {
    grid-template-rows: 1fr;
    margin-top: 7px;
  }
  .lp-mod__desc {
    overflow: hidden;
    margin: 0;
    font-size: 12.5px;
    line-height: 1.7;
    color: #6e6e73;
    opacity: 0;
    transition: opacity .3s ease;
    text-wrap: pretty;
  }
  .lp-mod:hover .lp-mod__desc,
  .lp-mod:focus-within .lp-mod__desc { opacity: 1; }
  /* 触摸设备没有悬停，直接把说明摊开 */
  @media (hover: none) {
    .lp-mod__body { grid-template-rows: 1fr; margin-top:
    7px; }
    .lp-mod__desc { opacity: 1; }
  }
  .lp-foot {
    margin: 0;
    padding-top: 26px;
    border-top: 1px solid #e6e6e8;
    text-align: center;
    font-size: 12.5px;
    color: #86868b;
  }
  /* ---------- 登录面板：点击后渐变浮现 ---------- */
  .lp-auth {
    position: fixed;
    inset: 0;
    z-index: 60;
```

```
    display: grid;                                                    6451
    place-items: center;                                              6452
    padding: 24px;                                                    6453
    opacity: 0;                                                       6454
    visibility: hidden;                                               6455
    transition: opacity .42s ease, visibility 0s linear               6456
    .42s;                                                             6457
}                                                                     6458
.lp-auth.is-open {                                                    6459
    opacity: 1;                                                       6460
    visibility: visible;                                              6461
    transition: opacity .42s ease, visibility 0s;                     6462
}                                                                     6463
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                    6464
.lp-auth__scrim {                                                     6465
    position: absolute;                                               6466
    inset: 0;                                                         6467
    background:                                                       6468
      radial-gradient(1000px 640px at 50% 0%,                         6469
                      rgba(31, 107, 69, .22), rgba(31, 107,           6470
    69, 0) 70%),                                                      6471
      linear-gradient(180deg, rgba(255, 255, 255, .84),               6472
                                rgba(246, 250, 247, .95));            6473
    backdrop-filter: blur(16px) saturate(1.25);                       6474
    -webkit-backdrop-filter: blur(16px) saturate(1.25);               6475
}                                                                     6476
.lp-auth__panel {                                                     6477
    position: relative;                                               6478
    width: 100%;                                                      6479
    max-width: 396px;                                                 6480
    padding: 36px 36px 26px;                                          6481
    background: #fff;                                                 6482
    border-radius: 20px;                                              6483
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                    6484
    opacity: 0;                                                       6485
    transform: translateY(18px) scale(.975);                          6486
    transition: transform .55s cubic-bezier(.16, .84, .34,            6487
    1) .05s,                                                          6488
                opacity .55s ease .05s;                               6489
}                                                                     6490
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:             6491
    none; }                                                           6492
.lp-auth__close {                                                     6493
    position: absolute;                                               6494
    top: 14px;                                                        6495
    right: 14px;                                                      6496
    display: grid;                                                    6497
    place-items: center;                                              6498
    width: 30px;                                                      6499
    height: 30px;                                                     6500
```

```
    color: #86868b;
    background: transparent;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    transition: background-color .18s ease, color .18s ease;
  }
  .lp-auth__close:hover { background: #f5f5f7; color:
    #1d1d1f; }
  .lp-auth__title {
    margin: 0 0 6px;
    text-align: center;
    font-size: 24px;
    font-weight: 700;
    letter-spacing: -.2px;
  }
  .lp-auth__sub {
    margin: 0 0 26px;
    text-align: center;
    font-size: 13px;
    color: #86868b;
  }
  /* 字段校验错误：贴着输入框显示，不做成会消失的浮层 ——
     这类消息要对着输入框看，一闪而过会让人以为「点了没反应」 */
  .lp-field__error {
    margin: 7px 0 0;
    font-size: 12.5px;
    color: #9c2b23;
  }
  .lp-field input[aria-invalid="true"] {
    background: #fdeeec;
    border-color: rgba(156, 43, 35, .35);
  }
  .lp-field { margin-bottom: 15px; }
  .lp-field label {
    display: block;
    margin-bottom: 7px;
    font-size: 13px;
    color: #6e6e73;
  }
  .lp-field input {
    width: 100%;
    padding: 13px 16px;
    font-family: inherit;
    font-size: 16px;                /* 不小于 16px，避免手机端聚焦时自动放大 */
    color: #1d1d1f;
    background: #f5f5f7;
    border: 1px solid transparent;
    border-radius: 12px;
    outline: none;
```

```
    transition: background-color .18s ease, border-color .18s ease,
                box-shadow .18s ease;
}
.lp-field input::placeholder { color: #a1a1a6; }
.lp-field input:focus {
  background: #fff;
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
.lp-remember {
  display: flex;
  align-items: center;
  gap: 8px;
  margin: 4px 0 20px;
  font-size: 13px;
  color: #6e6e73;
  cursor: pointer;
}
.lp-remember input {
  width: 16px;
  height: 16px;
  margin: 0;
  accent-color: var(--accent);
  cursor: pointer;
}
.lp-submit {
  width: 100%;
  padding: 13px 20px;
  font-family: inherit;
  font-size: 16px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
  border-radius: 980px;
  cursor: pointer;
  transition: background-color .2s ease, box-shadow .2s ease,
              transform .2s ease;
}
.lp-submit:hover {
  background: var(--accent-hover);
  box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
}
.lp-submit:active { transform: translateY(1px);
  box-shadow: none; }
.lp-submit:focus-visible { outline: 3px solid rgba(31, 107, 69, .40); outline-offset: 3px; }
.lp-auth__hint {
```

```
      margin: 18px 0 0;
      text-align: center;
      font-size: 12.5px;
      color: #a1a1a6;
    }
    .lp-auth__reg {
      color: var(--accent);
      font-weight: 500;
      text-decoration: none;
      border-bottom: 1px solid rgba(31, 107, 69, .35);
      transition: border-color .18s ease;
    }
    .lp-auth__reg:hover { border-bottom-color: var(--accent); }
    /* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */
    @media (prefers-reduced-motion: reduce) {
      .lp-badge, .lp-title, .lp-cta, .lp-more { animation: none; }
      .lp-badge__svg { opacity: 1; filter: none; }
      .lp-ring, .lp-badge__mark { transform: none; }
      .lp-reveal { opacity: 1; transform: none !important; }
      .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name { color: inherit; }
      .lp-reveal.is-in h2 { color: #1d1d1f; }
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name { color: var(--accent); }
      .lp-auth, .lp-auth__panel { transition: opacity .2s linear; }
      .lp-auth__panel { transform: none; }
    }
    /* 窄屏退成两列、一列，避免卡片被挤变形 */
    @media (max-width: 900px) {
      .lp-points, .lp-mods { grid-template-columns: repeat(2, minmax(0, 1fr)); }
    }
    @media (max-width: 620px) {
      .lp-points, .lp-mods { grid-template-columns: minmax(0, 1fr); }
    }
    @media (max-width: 560px) {
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }
      .lp-badge { width: 160px; height: 160px; margin-bottom: 26px; }
      .lp-auth__panel { padding: 30px 24px 22px; }
    }
</style>
<div class="lp">
  {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口 ---------------- #}
  <section class="lp-hero">
```

```
    <div class="lp-badge" id="lp-badge" aria-hidden="true">
      {% include "_login_badge.svg" %}
    </div>
    <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>
    <button type="button" class="lp-cta" id="lp-open"
            aria-expanded="false" aria-controls="lp-auth">登录
    </button>
    <a class="lp-more" href="#intro">
      向下滑动，了解系统 <span aria-hidden="true">↓</span>
    </a>
  </section>
  {# ---------------- 系统介绍：随滚动逐条弹入 ---------------- #}
  <section class="lp-intro" id="intro">
    <h2 class="lp-intro__title lp-reveal"
    data-dir="up">一套把评价流程固定下来的工具</h2>
    <div class="lp-points">
      {% set pdirs = ['left', 'up', 'right'] %}
      {% set points = [
        ('sliders', '方法可溯',
         '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回
    看。'),
        ('shield', '流程可控',
         '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重
    建或人工辅助的结论。'),
        ('file', '结果可复算',
         '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
      ] %}
      {% for ic, title, text in points %}
      <div class="lp-point lp-reveal" data-dir="{{
    pdirs[loop.index0 % 3] }}"
           style="--i: {{ loop.index0 }}">
        <span class="lp-point__icon">{{ icon(ic, 20)
    }}</span>
        <h3>{{ title }}</h3>
        <p>{{ text }}</p>
      </div>
      {% endfor %}
    </div>
    <h3 class="lp-sub-title lp-reveal"
    data-dir="up">九个功能模块</h3>
    <div class="lp-mods">
      {% set mdirs = ['left', 'up', 'right'] %}
      {% for no, name, desc in modules %}
      <div class="lp-mod lp-reveal" data-dir="{{
    mdirs[loop.index0 % 3] }}"
           style="--i: {{ loop.index0 % 3 }}" tabindex="0">
        <span class="lp-mod__no">{{ no }}</span>
        <span class="lp-mod__name">{{ name }}</span>
        <div class="lp-mod__body"><p class="lp-mod__desc">{{
    desc }}</p></div>
```

```
        </div>                                                                  6701
        {% endfor %}                                                            6702
    </div>                                                                      6703
    <p class="lp-foot lp-reveal" data-dir="up">                                 6704
        历史遗留废弃矿山自然恢复适宜性评价系统 V1.0 ·                                  6705
        运行于浏览器，数据保存在本机 SQLite 数据库                                     6706
    </p>                                                                        6707
</section>                                                                      6708
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}                  6709
<div class="lp-auth" id="lp-auth" aria-hidden="true">                           6710
    <div class="lp-auth__scrim" data-lp-close></div>                            6711
    <div class="lp-auth__panel" role="dialog"                                   6712
    aria-modal="true"                                                           6713
          aria-labelledby="lp-auth-title">                                      6714
        <button type="button" class="lp-auth__close"                            6715
    data-lp-close                                                               6716
                aria-label="关闭登录框">                                          6717
            <svg width="16" height="16" viewBox="0 0 24 24"                     6718
    fill="none"                                                                 6719
                stroke="currentColor" stroke-width="2"                          6720
    stroke-linecap="round"                                                      6721
                aria-hidden="true"><path d="M6 6l12 12M18 6L6                   6722
    18"/></svg>                                                                 6723
        </button>                                                               6724
        <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>                   6725
        <p class="lp-auth__sub">使用系统账号登录</p>                                 6726
        <form method="post" novalidate>                                         6727
            {{ form.hidden_tag() }}                                             6728
            <div class="lp-field">                                              6729
                <label for="{{ form.username.id }}">账户名称</label>              6730
                {{ form.username(id=form.username.id,                           6731
    autocomplete="username",                                                    6732
                                 placeholder="请输入账户名称",                     6733
                                 **({'aria-invalid': 'true'} if                 6734
    form.username.errors else {})) }}                                           6735
                {% for e in form.username.errors %}                             6736
                <p class="lp-field__error">{{ e }}</p>                          6737
                {% endfor %}                                                    6738
            </div>                                                              6739
            <div class="lp-field">                                              6740
                <label for="{{ form.password.id }}">密码</label>                6741
                {{ form.password(id=form.password.id,                           6742
    autocomplete="current-password",                                            6743
                                 placeholder="请输入密码",                         6744
                                 **({'aria-invalid': 'true'} if                 6745
    form.password.errors else {})) }}                                           6746
                {% for e in form.password.errors %}                             6747
                <p class="lp-field__error">{{ e }}</p>                          6748
                {% endfor %}                                                    6749
            </div>                                                              6750
```

```
          <label class="lp-remember" for="{{ form.remember.id
    }}">
            {{ form.remember(id=form.remember.id) }}
            <span>记住我（下次自动登录，一个月内有效）</span>
          </label>
          {{ form.submit(class_="lp-submit") }}
        </form>
        <p class="lp-auth__hint">
          还没有账号？<a class="lp-auth__reg"
                         href="{{ url_for('auth.register')
    }}">点我注册</a>
        </p>
        <p class="lp-auth__hint" style="margin-top:10px;">
          演示账号：admin 密码：admin123
        </p>
      </div>
    </div>
  </div>
{% endblock %}
{% block scripts %}
<script>
(function () {
  var reduce = window.matchMedia &&
    window.matchMedia('(prefers-reduced-motion:
    reduce)').matches;
  /* ---------- 登录面板 ---------- */
  var btn = document.getElementById('lp-open');
  var auth = document.getElementById('lp-auth');
  var lastFocus = null;
  function openPanel() {
    lastFocus = document.activeElement;
    auth.classList.add('is-open');
    auth.setAttribute('aria-hidden', 'false');
    btn.setAttribute('aria-expanded', 'true');
    document.body.style.overflow = 'hidden';
    // 记到地址栏，直接访问 /login#login 也能展开登录框
    if (location.hash !== '#login') {
      history.replaceState(null, '', location.pathname +
    '#login');
    }
    var first = auth.querySelector('input');
    if (first) window.setTimeout(function () {
    first.focus(); }, 300);
  }
  function closePanel() {
    auth.classList.remove('is-open');
    auth.setAttribute('aria-hidden', 'true');
    btn.setAttribute('aria-expanded', 'false');
    document.body.style.overflow = '';
    if (location.hash === '#login') {
```

```
      history.replaceState(null, '', location.pathname);
    }
    if (lastFocus && lastFocus.focus) lastFocus.focus();
  }
  if (btn && auth) {
    btn.addEventListener('click', openPanel);
    Array.prototype.forEach.call(
      auth.querySelectorAll('[data-lp-close]'),
      function (el) { el.addEventListener('click',
    closePanel); }
    );
    document.addEventListener('keydown', function (e) {
      if (e.key === 'Escape' && auth.classList.contains('is-
    open')) closePanel();
    });
    // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框。
    // 失败可能落在两处：字段校验错误在表单里，凭据错误在浮层里。
    if (location.hash === '#login'
        || document.querySelector('.lp-field__error')
        || document.querySelector('.toast-center')) {
      openPanel();
    }
  }
  /* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */
     原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
     内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
     与徽标用同一次滚动回调，行为可预期、也便于排查。
     判断「进没进视野」用的是**布局位置**，不是 getBoundingClientRect()：
     进场动画本身就是位移（左右两列横向 56vw、中列自下方 24vh 升起），
     用视觉位置判断时，中列的盒子一直比自己的位置低一截，
     于是它总要再往下滚一段才触发，看起来比左右两列慢半拍。
     transform 不影响 offsetTop，所以这里沿 offsetParent 把布局位置累加出来，
     只在加载时算一次；滚动过程中不再测量，也就没有读写交替带来的重排。 */
  var items = document.querySelectorAll('.lp-reveal');
  var badge = document.getElementById('lp-badge');
  var ticking = false;
  var slots = [];
  function layoutTop(el) {
    var top = 0;
    var node = el;
    while (node) {
      top += node.offsetTop || 0;
      node = node.offsetParent;
    }
    return top;
  }
  if (!reduce) {
    for (var k = 0; k < items.length; k++) {
      slots.push({
        el: items[k],
```

```
          top: layoutTop(items[k]),
          height: items[k].offsetHeight
        });
      }
    }
    function applyScroll() {
      ticking = false;
      var vh = window.innerHeight || 800;
      var y = window.scrollY || window.pageYOffset || 0;
      /* 徽标：滚动前约三分之一屏内完成散开淡出 */
      if (badge) {
        var range = vh * 0.34;
        var p = range > 0 ? y / range : 0;
        badge.style.setProperty('--p', Math.min(1,
      Math.max(0, p)).toFixed(3));
      }
      if (reduce || !slots.length) return;
      for (var j = 0; j < slots.length; j++) {
        var s = slots[j];
        var viewTop = s.top - y;
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内
        var shown = viewTop < vh * 0.88 && viewTop + s.height
      > vh * 0.06;
        if (shown) s.el.classList.add('is-in');
        else s.el.classList.remove('is-in');
      }
    }
    function onScroll() {
      if (!ticking) {
        ticking = true;
        window.requestAnimationFrame(applyScroll);
      }
    }
    // 减少动态效果时不做滚动特效，内容直接显示
    if (reduce) {
      Array.prototype.forEach.call(items, function (el) {
        el.classList.add('is-in');
      });
      if (badge) badge.style.setProperty('--p', '0');
    } else {
      window.addEventListener('scroll', onScroll, { passive:
      true });
      window.addEventListener('resize', onScroll);
      applyScroll();
    }
  })();
  </script>
  {% endblock %}
  # ==== app/templates/auth/register.html ====
  {% extends "base.html" %}
```

```
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 注册页 ==================
     与登录页同一套观感：浅绿渐变底 + 白色卡片。
     必填只有账户名称与密码，其余收在「选填」分隔线以下，
     这样首次注册不会被一堆输入框挡住。 */
  .rg {
    min-height: 100dvh;
    display: grid;
    place-items: center;
    padding: 40px 20px 56px;
    background:
      radial-gradient(900px 520px at 50% 0%,
                      rgba(31, 107, 69, .14), rgba(31, 107,
    69, 0) 72%),
      #f6faf7;
  }
  .rg-card {
    width: 100%;
    max-width: 470px;
    padding: 38px 40px 30px;
    background: #fff;
    border-radius: 20px;
    box-shadow: 0 26px 70px rgba(14, 46, 32, .13);
    animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)
    both;
  }
  @keyframes rg-in {
    from { opacity: 0; transform: translateY(16px)
    scale(.98); }
    to   { opacity: 1; transform: none; }
  }
  .rg-title {
    margin: 0 0 6px;
    text-align: center;
    font-size: 25px;
    font-weight: 700;
    letter-spacing: -.2px;
  }
  .rg-sub {
    margin: 0 0 28px;
    text-align: center;
    font-size: 13px;
    color: #86868b;
  }
  .rg-field { margin-bottom: 15px; }
  .rg-field > label {
    display: block;
    margin-bottom: 7px;
```

```
    font-size: 13px;                                                    6951
    color: #6e6e73;                                                     6952
  }                                                                     6953
  .rg-field > label .rg-opt {                                           6954
    margin-left: 5px;                                                   6955
    font-size: 12px;                                                    6956
    color: #a1a1a6;                                                     6957
  }                                                                     6958
  .rg-field input[type="text"],                                         6959
  .rg-field input[type="password"] {                                    6960
    width: 100%;                                                        6961
    padding: 12px 15px;                                                 6962
    font-family: inherit;                                               6963
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */    6964
    color: #1d1d1f;                                                     6965
    background: #f5f5f7;                                                6966
    border: 1px solid transparent;                                      6967
    border-radius: 12px;                                                6968
    outline: none;                                                      6969
    transition: background-color .18s ease, border-color                6970
    .18s ease,                                                          6971
                box-shadow .18s ease;                                   6972
  }                                                                     6973
  .rg-field input::placeholder { color: #a1a1a6; }                      6974
  .rg-field input:focus {                                               6975
    background: #fff;                                                   6976
    border-color: rgba(31, 107, 69, .55);                               6977
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                       6978
  }                                                                     6979
  .rg-field input[aria-invalid="true"] {                                6980
    border-color: rgba(180, 60, 50, .55);                               6981
    background: #fff;                                                   6982
  }                                                                     6983
  .rg-field select {                                                    6984
    width: 100%;                                                        6985
    padding: 12px 15px;                                                 6986
    font-family: inherit;                                               6987
    font-size: 15px;                                                    6988
    color: #1d1d1f;                                                     6989
    background: #f5f5f7;                                                6990
    border: 1px solid transparent;                                      6991
    border-radius: 12px;                                                6992
    outline: none;                                                      6993
  }                                                                     6994
  .rg-field select:focus {                                              6995
    background: #fff;                                                   6996
    border-color: rgba(31, 107, 69, .55);                               6997
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                       6998
  }                                                                     6999
  .rg-err {                                                             7000
```

```
    margin: 6px 0 0;
    font-size: 12.5px;
    color: var(--danger);
  }
  /* 「选填」分隔线 */
  .rg-split {
    display: flex;
    align-items: center;
    gap: 12px;
    margin: 24px 0 18px;
    font-size: 12.5px;
    color: #a1a1a6;
  }
  .rg-split::before,
  .rg-split::after {
    content: "";
    flex: 1;
    height: 1px;
    background: #ececee;
  }
  .rg-pair {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 0 14px;
  }
  @media (max-width: 480px) { .rg-pair {
    grid-template-columns: 1fr; } }
  /* 头像：圆形预览 + 选择文件 */
  .rg-avatar {
    display: flex;
    align-items: center;
    gap: 14px;
    margin-bottom: 15px;
  }
  .rg-avatar__face {
    display: grid;
    place-items: center;
    flex: none;
    width: 54px;
    height: 54px;
    border-radius: 50%;
    background: #eef2f0;
    color: #9aa8a1;
    font-size: 12px;
    overflow: hidden;
    background-size: cover;
    background-position: center;
  }
  .rg-avatar__pick { min-width: 0; }
  .rg-avatar__pick input[type="file"] {
```

```
      max-width: 100%;
      font-family: inherit;
      font-size: 12.5px;
      color: #6e6e73;
    }
    .rg-avatar__tip {
      margin: 4px 0 0;
      font-size: 12px;
      color: #a1a1a6;
    }
    .rg-submit {
      width: 100%;
      margin-top: 24px;
      padding: 13px 20px;
      font-family: inherit;
      font-size: 16px;
      font-weight: 500;
      color: #fff;
      background: var(--accent);
      border: none;
      border-radius: 980px;
      cursor: pointer;
      transition: background-color .2s ease, box-shadow .2s
      ease,
                  transform .2s ease;
    }
    .rg-submit:hover {
      background: var(--accent-hover);
      box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
    }
    .rg-submit:active { transform: translateY(1px);
      box-shadow: none; }
    .rg-submit:focus-visible { outline: 3px solid rgba(31,
      107, 69, .40); outline-offset: 3px; }
    .rg-foot {
      margin: 20px 0 0;
      text-align: center;
      font-size: 13px;
      color: #86868b;
    }
    .rg-foot a { color: var(--accent); text-decoration: none;
      }
    .rg-foot a:hover { text-decoration: underline; }
    @media (prefers-reduced-motion: reduce) {
      .rg-card { animation: none; }
    }
</style>
<div class="rg">
    <div class="rg-card">
      <h1 class="rg-title">创建账号</h1>
```

```
<p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>
<form method="post" enctype="multipart/form-data"
novalidate>
  {{ form.hidden_tag() }}
  <div class="rg-field">
    <label for="{{ form.username.id }}">账户名称</label>
    {{ form.username(id=form.username.id,
autocomplete="username",
                     placeholder="2–64 个字符",
                     **({'aria-invalid': 'true'} if
form.username.errors else {})) }}
    {% for e in form.username.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
  </div>
  <div class="rg-field">
    <label for="{{ form.password.id }}">密码</label>
    {{ form.password(id=form.password.id,
autocomplete="new-password",
                     placeholder="至少 6 位",
                     **({'aria-invalid': 'true'} if
form.password.errors else {})) }}
    {% for e in form.password.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
  </div>
  <div class="rg-field">
    <label for="{{ form.confirm.id }}">确认密码</label>
    {{ form.confirm(id=form.confirm.id,
autocomplete="new-password",
                    placeholder="再次输入密码",
                    **({'aria-invalid': 'true'} if
form.confirm.errors else {})) }}
    {% for e in form.confirm.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
  </div>
  <p class="rg-split">以下均为选填</p>
  <div class="rg-avatar">
    <span class="rg-avatar__face" id="rg-face"
aria-hidden="true">头像</span>
    <div class="rg-avatar__pick">
      <label class="visually-hidden"
for="rg-avatar">选择头像图片</label>
      <input id="rg-avatar" type="file" name="avatar"
             accept=".png,.jpg,.jpeg,.gif,.webp">
      <p class="rg-avatar__tip">PNG / JPG / GIF /
WebP，不超过 4 MB</p>
    </div>
  </div>
  <div class="rg-pair">
    <div class="rg-field">
      <label for="{{ form.display_name.id }}">昵称</label>
```

```
            {{ form.display_name(id=form.display_name.id,
                                 placeholder="留空则显示账户名称") }}
          </div>
          <div class="rg-field">
            <label for="{{ form.gender.id }}">性别</label>
            {{ form.gender(id=form.gender.id) }}
          </div>
        </div>
        <div class="rg-field">
          <label for="{{ form.email.id }}">邮箱</label>
          {{ form.email(id=form.email.id,
      autocomplete="email",
                        placeholder="用于联系，可留空",
                        **({'aria-invalid': 'true'} if
      form.email.errors else {})) }}
          {% for e in form.email.errors %}<p
      class="rg-err">{{ e }}</p>{% endfor %}
        </div>
        <div class="rg-field" style="margin-bottom:0;">
          <label for="{{ form.phone.id }}">手机号</label>
          {{ form.phone(id=form.phone.id, autocomplete="tel",
      placeholder="可留空") }}
        </div>
        {{ form.submit_reg(class_="rg-submit") }}
      </form>
      <p class="rg-foot">已有账号？<a href="{{
      url_for('auth.login') }}">返回登录</a></p>
    </div>
  </div>
{% endblock %}
{% block scripts %}
<script>
// 选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了
(function () {
  var input = document.getElementById('rg-avatar');
  var face = document.getElementById('rg-face');
  if (!input || !face || !window.FileReader) return;
  input.addEventListener('change', function () {
    var file = input.files && input.files[0];
    if (!file) return;
    var reader = new FileReader();
    reader.onload = function (e) {
      face.textContent = '';
      face.style.backgroundImage = 'url("' +
    e.target.result + '")';
    };
    reader.readAsDataURL(file);
  });
})();
</script>
```

```
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <p class="hero__desc">
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
    </p>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
      <a class="hero-btn hero-btn--solid"
    href="/project/add">
        {{ icon("plus", 16) }}新建评价项目
      </a>
    </div>
  </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">评价项目</span>
    <span class="stat-card__value">{{ dash.stats.projects
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {%- if projects_without_patch -%}
        其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
      {%- elif dash.progress.patches -%}
        已建立 {{ dash.progress.patches }} 个矿山图斑
      {%- else -%}
        尚未建立矿山图斑
      {%- endif -%}
    </span>
  </a>
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">矿山图斑</span>
    <span class="stat-card__value">{{ dash.stats.patches
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {% if dash.progress.ready %}其中 {{ dash.progress.ready
    }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
```

```
      </span>                                                                  7251
    </a>                                                                       7252
    <a class="stat-card" href="/unit/">                                        7253
      <span class="stat-card__label">微地质单元</span>                          7254
      <span class="stat-card__value">{{ dash.stats.unit_types                  7255
      }}<span class="stat-card__unit">类</span></span>                         7256
      <span class="stat-card__note">                                           7257
        {% if dash.progress.units %}共划分 {{                                  7258
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}           7259
      </span>                                                                  7260
    </a>                                                                       7261
    <a class="stat-card" href="/standard/">                                    7262
      <span class="stat-card__label">标准与政策</span>                          7263
      <span class="stat-card__value">{{ dash.stats.standards                   7264
      }}<span class="stat-card__unit">项</span></span>                         7265
      <span class="stat-card__note">地方标准与政策文件</span>                   7266
    </a>                                                                       7267
</section>                                                                     7268
{# ---------- 评价流程 ---------- #}                                          7269
<section class="section">                                                      7270
    <div class="section__head">                                                7271
      <h2 class="section__title">评价流程</h2>                                 7272
      <p class="section__desc">                                                7273
        按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算                7274
      </p>                                                                     7275
    </div>                                                                     7276
    <div class="step-grid">                                                    7277
      {% for s in dash.steps %}                                                7278
      {% if s.available %}                                                     7279
      <a class="step-card" href="{{ s.href }}">                                7280
        <span class="step-card__no">{{ s.no }}</span>                          7281
        <span class="step-card__icon">{{ icon(s.icon, 18)                      7282
      }}</span>                                                                7283
        <span class="step-card__title">{{ s.title }}</span>                    7284
        <span class="step-card__desc">{{ s.desc }}</span>                      7285
        {% if s.pending %}                                                     7286
        <span class="step-card__badge">待处理 {{ s.pending                     7287
      }}</span>                                                                7288
        {% endif %}                                                            7289
      </a>                                                                     7290
      {% else %}                                                               7291
      <div class="step-card step-card--locked"                                 7292
      aria-disabled="true">                                                    7293
        <span class="step-card__no">{{ s.no }}</span>                          7294
        <span class="step-card__icon">{{ icon(s.icon, 18)                      7295
      }}</span>                                                                7296
        <span class="step-card__title">{{ s.title }}</span>                    7297
        <span class="step-card__desc">{{ s.desc }}</span>                      7298
        <span class="step-card__lock">{{ s.note }}</span>                      7299
      </div>                                                                   7300
```

```
      {% endif %}                                                    7301
      {% endfor %}                                                   7302
    </div>                                                           7303
</section>                                                           7304
{# ---------- 系统管理 ---------- #}                                 7305
{# 个人设置、账号与操作日志的入口都在左侧导航栏，工作台不再重复列出 #}  7306
{% endblock %}                                                       7307
# ==== app/templates/auth/add_user.html ====                         7308
{% extends "base.html" %}                                            7309
{% block title %}新增账号{% endblock %}                              7310
{% block content %}                                                  7311
<div class="card" style="max-width:560px;">                          7312
  <h2>新增账号</h2>                                                  7313
  <form method="post">                                               7314
    {{ form.hidden_tag() }}                                          7315
    <div class="field">                                              7316
      <label class="field__label" for="username">账户名称</label     7317
    >                                                                7318
      <input id="username" type="text" name="username"               7319
    required                                                         7320
             maxlength="64" autocomplete="off"                       7321
    style="width:280px;">                                            7322
    </div>                                                           7323
    <div class="field">                                              7324
      <label class="field__label" for="password">密码</label>        7325
      <input id="password" type="password" name="password"           7326
    required                                                         7327
             autocomplete="new-password"                             7328
    style="width:280px;">                                            7329
    </div>                                                           7330
    <div class="field">                                              7331
      <label class="field__label" for="role">职务</label>            7332
      <select id="role" name="role">                                 7333
        {% for code, label in role_labels.items() %}                 7334
        <option value="{{ code }}" {% if code ==                     7335
    default_role %}selected{% endif %}>{{ label }}</option>          7336
        {% endfor %}                                                 7337
      </select>                                                      7338
      <p class="muted" style="margin:6px 0 0;">                      7339
        管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。  7340
      </p>                                                           7341
    </div>                                                           7342
    <hr style="margin:22px 0;border:none;border-top:1px              7343
    solid var(--line);">                                             7344
    <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3 7345
    >                                                                7346
    <div class="field">                                              7347
      <label class="field__label" for="display_name">昵称</lab       7348
    el>                                                              7349
      <input id="display_name" type="text"                           7350
```

```
      name="display_name" maxlength="64"                                7351
                placeholder="留空则显示账户名称">                          7352
      </div>                                                            7353
      <div class="field">                                               7354
        <label class="field__label" for="gender">性别</label>           7355
        <select id="gender" name="gender">                              7356
          <option value="">不填</option>                                7357
          {% for g in genders %}<option value="{{ g }}">{{ g            7358
      }}</option>{% endfor %}                                           7359
        </select>                                                       7360
      </div>                                                            7361
      <div class="field">                                               7362
        <label class="field__label" for="email">邮箱</label>            7363
        <input id="email" type="text" name="email"                      7364
      maxlength="128">                                                  7365
      </div>                                                            7366
      <div class="field">                                               7367
        <label class="field__label" for="phone">手机号</label>          7368
        <input id="phone" type="text" name="phone"                      7369
      maxlength="32">                                                   7370
      </div>                                                            7371
      <p class="actions" style="margin-top:18px;">                      7372
        <button type="submit" class="btn">保存</button>                 7373
        <a class="btn btn--ghost" href="{{                              7374
      url_for('auth.profile', tab='users') }}">返回</a>                 7375
      </p>                                                              7376
    </form>                                                             7377
</div>                                                                  7378
{% endblock %}                                                          7379
# ==== app/templates/auth/edit_user.html ====                           7380
{% extends "base.html" %}                                               7381
{% block title %}编辑账号 · {{ user.username }}{% endblock %}           7382
{% block content %}                                                     7383
<div class="card" style="max-width:560px;">                             7384
  <h2>编辑账号 · {{ user.username }}</h2>                               7385
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>             7386
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"            7387
    >                                                                   7388
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。       7389
  </p>                                                                  7390
  <form method="post">                                                  7391
    {{ form.hidden_tag() }}                                             7392
    <div class="field">                                                 7393
      <label class="field__label" for="display_name">昵称</lab          7394
    el>                                                                 7395
      <input id="display_name" type="text"                              7396
    name="display_name" maxlength="64"                                  7397
             value="{{ user.display_name or '' }}"                      7398
             placeholder="留空则显示账户名称">                          7399
    </div>                                                              7400
```

```
<div class="field">
  <label class="field__label" for="gender">性别</label>
  <select id="gender" name="gender">
    <option value="">不填</option>
    {% for g in genders %}
    <option value="{{ g }}" {% if user.gender == g %}selected{% endif %}>{{ g }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="email">邮箱</label>
  <input id="email" type="text" name="email" maxlength="128"
         value="{{ user.email or '' }}">
</div>
<div class="field">
  <label class="field__label" for="phone">手机号</label>
  <input id="phone" type="text" name="phone" maxlength="32"
         value="{{ user.phone or '' }}">
</div>
<hr style="margin:22px 0;border:none;border-top:1px solid var(--line);">
<h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>
<p class="muted" style="margin:0 0 14px;font-size:12.5px;">
  职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志；
  操作员可使用全部业务功能。
</p>
<div class="field">
  <label class="field__label" for="role">职务</label>
  <select id="role" name="role">
    {% for code, label in role_labels.items() %}
    <option value="{{ code }}" {% if user.role == code %}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="checkbox" for="active">
    <input id="active" type="checkbox" name="active" value="1"
           {% if user.active %}checked{% endif %}>
    <span>启用该账号</span>
  </label>
  <p class="muted" style="margin:6px 0 0;">
    停用后该账号无法登录，但其历史操作记录会保留。
    系统至少需保留一名启用状态的管理员。
  </p>
```

```
      </div>                                                                    7451
      <div class="field">                                                       7452
        <label class="field__label" for="password">重置密码（留空则不修改 7453
      ）</label>                                                                7454
        <input id="password" type="password" name="password"                    7455
               autocomplete="new-password"                                      7456
      style="width:280px;"                                                      7457
               placeholder="留空表示保持原密码">                                7458
      </div>                                                                    7459
      <p class="actions" style="margin-top:18px;">                              7460
        <button type="submit" class="btn">保存</button>                         7461
        <a class="btn btn--ghost" href="{{                                      7462
      url_for('auth.profile', tab='users') }}">返回</a>                         7463
      </p>                                                                      7464
    </form>                                                                     7465
</div>                                                                          7466
{% endblock %}                                                                  7467
# ==== app/templates/auth/profile.html ====                                     7468
{% extends "base.html" %}                                                       7469
{% from "_macros.html" import icon %}                                           7470
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}    7471
{% block content %}                                                             7472
<style>                                                                         7473
  /* ================== 个人设置（仿 Apple 账户页） ==================          7474
     左栏是身份与分类，右栏是该分类下的功能卡。                                  7475
     每张卡默认只显示当前值，点开才展开对应的修改表单 ——                        7476
     不点就不会误改，页面也不会一上来堆满输入框。 */                            7477
  .acct {                                                                       7478
    display: grid;                                                              7479
    grid-template-columns: 244px 1fr;                                           7480
    gap: 40px;                                                                  7481
    align-items: start;                                                         7482
  }                                                                             7483
  @media (max-width: 900px) {                                                   7484
    .acct { grid-template-columns: 1fr; gap: 24px; }                            7485
  }                                                                             7486
  /* ---------- 左栏 ---------- */                                              7487
  .acct-rail__avatar {                                                          7488
    display: grid;                                                              7489
    place-items: center;                                                        7490
    width: 88px;                                                                7491
    height: 88px;                                                               7492
    border-radius: 50%;                                                         7493
    background: #e9edeb;                                                        7494
    color: #97a09b;                                                             7495
    font-size: 32px;                                                            7496
    font-weight: 500;                                                           7497
    object-fit: cover;                                                          7498
  }                                                                             7499
  .acct-rail__name {                                                            7500
```

```
    margin: 16px 0 2px;
    font-size: 16px;
    font-weight: 600;
  }
  .acct-rail__sub {
    margin: 0 0 24px;
    font-size: 13px;
    color: #86868b;
    word-break: break-all;
  }
  .acct-tabs {
    display: flex;
    flex-direction: column;
    gap: 2px;
    margin-bottom: 24px;
  }
  .acct-tabs a {
    padding: 7px 12px;
    border: 1px solid transparent;
    border-radius: 7px;
    font-size: 15px;
    color: #1d1d1f;
    text-decoration: none;
    transition: background-color .16s ease, border-color
    .16s ease;
  }
  .acct-tabs a:hover { background: #f5f5f7; }
  .acct-tabs a.is-active {
    font-weight: 600;
    border-color: #bcc9c1;
    background: #fff;
  }
  /* 更换头像：折叠在左栏，点开才出现文件选择 */
  .acct-avatar-edit > summary {
    display: inline-block;
    padding: 5px 12px;
    font-size: 13px;
    color: var(--accent);
    background: var(--accent-soft);
    border-radius: 7px;
    cursor: pointer;
    list-style: none;
  }
  .acct-avatar-edit > summary::-webkit-details-marker {
    display: none; }
  .acct-avatar-edit > summary:hover { background: #dcebe2; }
  .acct-avatar-edit__body { margin-top: 12px; }
  .acct-avatar-edit__body .btn { width: 100%;
    margin-bottom: 8px; }
  .acct-avatar-edit__file {
```

```
    margin: 0 0 10px;
    font-size: 12px;
    color: var(--ink-soft);
    word-break: break-all;
  }
  /* ---------- 右栏 ---------- */
  .acct-title {
    margin: 0 0 8px;
    font-size: 26px;
    font-weight: 700;
    letter-spacing: -.3px;
  }
  .acct-lead {
    margin: 0 0 26px;
    font-size: 14px;
    color: #6e6e73;
  }
  /* 固定两列，与 Apple 账户页的 2×2 一致；
     卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
  .acct-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 16px;
    align-items: start;
  }
  @media (max-width: 760px) {
    .acct-grid { grid-template-columns: 1fr; }
  }
  /* 功能卡：details/summary 实现，点开才展开表单 */
  .acct-card {
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    transition: border-color .2s ease, box-shadow .2s ease;
  }
  .acct-card[open] {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
  }
  .acct-card > summary {
    position: relative;
    display: block;
    padding: 18px 56px 18px 20px;
    border-radius: 14px;
    cursor: pointer;
    list-style: none;
  }
  .acct-card > summary::-webkit-details-marker { display:
    none; }
  .acct-card > summary:hover { background: #fafbfa; }
```

```
.acct-card > summary:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 2px;
}
.acct-card__title {
  display: block;
  font-size: 15px;
  font-weight: 600;
}
.acct-card__value {
  display: block;
  margin-top: 3px;
  font-size: 14px;
  color: #6e6e73;
  overflow-wrap: anywhere;
}
.acct-card__icon {
  position: absolute;
  top: 17px;
  right: 17px;
  display: grid;
  place-items: center;
  width: 27px;
  height: 27px;
  border-radius: 8px;
  background: var(--accent-soft);
  color: var(--accent);
}
.acct-card__body {
  padding: 4px 20px 20px;
  border-top: 1px solid #f0f1f2;
  margin-top: 2px;
}
.acct-card__body .field:first-child { margin-top: 16px; }
.acct-card__body .btn { width: 100%; }
/* 只读信息列表 */
.acct-kv {
  display: grid;
  grid-template-columns: 88px 1fr;
  gap: 7px 12px;
  margin: 16px 0 0;
  font-size: 13.5px;
}
.acct-kv dt { color: var(--ink-soft); }
.acct-kv dd { margin: 0; }
.field__error {
  margin: 5px 0 0;
  font-size: 12.5px;
  color: var(--danger);
}
```

```
  /* 管理员分类（账号 / 操作日志）里的表格面板 */
  .acct-panel {
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    overflow: hidden;
  }
  .acct-panel__head {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 12px;
    padding: 13px 20px;
    border-bottom: 1px solid #f0f1f2;
  }
  .acct-panel__count {
    margin: 0;
    font-size: 13.5px;
    color: #6e6e73;
  }
  .acct-panel table { margin: 0; }
  .acct-panel th,
  .acct-panel td { border: 0; border-bottom: 1px solid
    #f0f1f2; }
  .acct-panel thead th { background: #f7f9f8; }
  .acct-panel tbody tr:last-child td { border-bottom: 0; }
  .acct-panel tbody tr:hover { background: #fafbfa; }
</style>
<div class="acct">
  {# ---------------- 左栏：身份 + 分类 ---------------- #}
  <aside class="acct-rail">
    {% if current_user.avatar %}
    <img class="acct-rail__avatar"
         src="{{ url_for('static', filename=current_user.ava
    tar) }}"
         alt="{{ current_user.shown_name }} 的头像">
    {% else %}
    <span class="acct-rail__avatar" aria-hidden="true">{{
    current_user.initial }}</span>
    {% endif %}
    <p class="acct-rail__name">{{ current_user.shown_name
    }}</p>
    <p class="acct-rail__sub">
      {{ role_labels.get(current_user.role,
    current_user.role) }}
      {% if current_user.email %} · {{ current_user.email
    }}{% endif %}
    </p>
    <nav class="acct-tabs" aria-label="个人设置分类">
      <a href="{{ url_for('auth.profile', tab='info') }}"
```

```
        {% if tab == 'info' %}class="is-active"                    7701
    aria-current="page"{% endif %}>个人信息</a>                    7702
      <a href="{{ url_for('auth.profile', tab='security')          7703
    }}"                                                            7704
        {% if tab == 'security' %}class="is-active"                7705
    aria-current="page"{% endif %}>登录与安全性</a>                7706
      {% if current_user.is_admin %}                               7707
      <a href="{{ url_for('auth.profile', tab='users') }}"         7708
        {% if tab == 'users' %}class="is-active"                   7709
    aria-current="page"{% endif %}>账号</a>                        7710
      <a href="{{ url_for('auth.profile', tab='logs') }}"          7711
        {% if tab == 'logs' %}class="is-active"                    7712
    aria-current="page"{% endif %}>操作日志</a>                    7713
      {% endif %}                                                  7714
    </nav>                                                         7715
    <details class="acct-avatar-edit">                             7716
      <summary>更换头像</summary>                                  7717
      <div class="acct-avatar-edit__body">                         7718
        <form method="post" action="/profile/avatar"               7719
              enctype="multipart/form-data">                       7720
          {{ avatar_form.hidden_tag() }}                           7721
          <input type="hidden" name="tab" value="{{ tab }}">       7722
          <label class="visually-hidden"                           7723
    for="avatar-file">选择头像图片</label>                         7724
          <input id="avatar-file" type="file" name="avatar"        7725
                 accept=".png,.jpg,.jpeg,.gif,.webp">              7726
          <p class="acct-avatar-edit__file">PNG / JPG / GIF        7727
    / WebP，不超过 4 MB</p>                                        7728
          <button type="submit" class="btn                         7729
    btn--sm">上传头像</button>                                     7730
        </form>                                                    7731
        {% if current_user.avatar %}                               7732
        <form method="post" action="/profile/avatar/delete"        7733
              onsubmit="return confirm('确定移除当前头像吗？');">  7734
          {{ avatar_form.hidden_tag() }}                           7735
          <input type="hidden" name="tab" value="{{ tab }}">       7736
          <button type="submit" class="btn btn--sm                 7737
    btn--ghost">移除头像</button>                                  7738
        </form>                                                    7739
        {% endif %}                                                7740
      </div>                                                       7741
    </details>                                                     7742
  </aside>                                                         7743
  {# ---------------- 右栏：功能卡 ---------------- #}             7744
  <div class="acct-main">                                          7745
    {% if tab == 'security' %}                                     7746
    <h1 class="acct-title">登录与安全性</h1>                       7747
    <p class="acct-lead">管理与登录账号、账号安全有关的设置。</p> 7748
    <div class="acct-grid">                                        7749
      <details class="acct-card"                                   7750
```

```
            {% if pwd_form.old_password.errors or                    7751
pwd_form.new_password.errors or pwd_form.confirm.errors              7752
%}open{% endif %}>                                                   7753
    <summary>                                                        7754
      <span class="acct-card__title">密码</span>                      7755
      <span class="acct-card__value">修改登录密码</span>             7756
      <span class="acct-card__icon">{{ icon("lock", 15)              7757
}}</span>                                                            7758
    </summary>                                                       7759
    <div class="acct-card__body">                                    7760
      <form method="post">                                           7761
        {{ pwd_form.hidden_tag() }}                                  7762
        <input type="hidden" name="action"                           7763
value="password">                                                    7764
        <input type="hidden" name="tab"                              7765
value="security">                                                    7766
        <div class="field">                                          7767
          <label class="field__label" for="{{                        7768
pwd_form.old_password.id }}">当前密码</label>                        7769
          {{ pwd_form.old_password(id=pwd_form.old_passw             7770
ord.id,                                                              7771
                                                                     7772
autocomplete="current-password") }}                                  7773
          {% for e in pwd_form.old_password.errors %}                7774
          <p class="field__error">{{ e }}</p>                        7775
          {% endfor %}                                               7776
        </div>                                                       7777
        <div class="field">                                          7778
          <label class="field__label" for="{{                        7779
pwd_form.new_password.id }}">新密码</label>                          7780
          {{ pwd_form.new_password(id=pwd_form.new_passw             7781
ord.id,                                                              7782
                                                                     7783
autocomplete="new-password") }}                                      7784
          {% for e in pwd_form.new_password.errors %}                7785
          <p class="field__error">{{ e }}</p>                        7786
          {% endfor %}                                               7787
        </div>                                                       7788
        <div class="field">                                          7789
          <label class="field__label" for="{{                        7790
pwd_form.confirm.id }}">确认新密码</label>                           7791
          {{ pwd_form.confirm(id=pwd_form.confirm.id,                7792
                              autocomplete="new-password             7793
") }}                                                                7794
          {% for e in pwd_form.confirm.errors %}                     7795
          <p class="field__error">{{ e }}</p>                        7796
          {% endfor %}                                               7797
        </div>                                                       7798
        <p class="actions">{{ pwd_form.submit_pwd(class_             7799
="btn") }}</p>                                                       7800
```

```
        </form>                                                    7801
      </div>                                                       7802
    </details>                                                     7803
    <details class="acct-card">                                    7804
      <summary>                                                    7805
        <span class="acct-card__title">登录账号</span>             7806
        <span class="acct-card__value">{{                          7807
current_user.username }}</span>                                    7808
        <span class="acct-card__icon">{{ icon("user", 15)          7809
}}</span>                                                          7810
      </summary>                                                   7811
      <div class="acct-card__body">                                7812
        <dl class="acct-kv">                                       7813
          <dt>登录账号</dt><dd class="mono">{{                     7814
current_user.username }}</dd>                                      7815
          <dt>职务</dt>                                            7816
          <dd>{{ role_labels.get(current_user.role,                7817
current_user.role) }}</dd>                                         7818
          <dt>状态</dt>                                            7819
          <dd>                                                     7820
            {% if current_user.active %}                           7821
            <span class="tag tag--ok">启用</span>                  7822
            {% else %}                                             7823
            <span class="tag tag--warn">停用</span>                7824
            {% endif %}                                            7825
          </dd>                                                    7826
          <dt>创建时间</dt>                                        7827
          <dd class="mono">{{ current_user.created_at |            7828
localdt }}</dd>                                                    7829
        </dl>                                                      7830
      </div>                                                       7831
    </details>                                                     7832
</div>                                                             7833
{% elif tab == 'users' %}                                          7834
<h1 class="acct-title">账号</h1>                                   7835
<p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p 7836
>                                                                  7837
{% include "auth/_users.html" %}                                   7838
{% elif tab == 'logs' %}                                           7839
<h1 class="acct-title">操作日志</h1>                               7840
<p class="acct-lead">记录各账号在本系统中的关键操作。</p>         7841
{% include "auth/_logs.html" %}                                    7842
{% else %}                                                         7843
<h1 class="acct-title">个人信息</h1>                               7844
<p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>       7845
<form method="post">                                               7846
  {{ info_form.hidden_tag() }}                                     7847
  <input type="hidden" name="action" value="info">                 7848
  <input type="hidden" name="tab" value="info">                    7849
  <div class="acct-grid">                                          7850
```

```
      <details class="acct-card" {% if                                  7851
info_form.display_name.errors %}open{% endif %}>                        7852
        <summary>                                                       7853
          <span class="acct-card__title">昵称</span>                    7854
          <span class="acct-card__value">                               7855
            {{ current_user.display_name or "未设置" }}                 7856
          </span>                                                       7857
          <span class="acct-card__icon">{{ icon("user",                 7858
15) }}</span>                                                           7859
        </summary>                                                      7860
        <div class="acct-card__body">                                   7861
          <div class="field">                                           7862
            <label class="field__label" for="{{                         7863
info_form.display_name.id }}">                                          7864
              昵称                                                      7865
            </label>                                                    7866
            {{ info_form.display_name(id=info_form.display              7867
_name.id,                                                               7868
                                                                        7869
placeholder="留空则显示账户名称") }}                                    7870
            {% for e in info_form.display_name.errors %}                7871
            <p class="field__error">{{ e }}</p>                         7872
            {% endfor %}                                                7873
          </div>                                                        7874
          <p class="actions">{{ info_form.submit_info(clas              7875
s_="btn") }}</p>                                                        7876
        </div>                                                          7877
      </details>                                                        7878
      <details class="acct-card" {% if                                  7879
info_form.gender.errors %}open{% endif %}>                              7880
        <summary>                                                       7881
          <span class="acct-card__title">性别</span>                    7882
          <span class="acct-card__value">{{                             7883
current_user.gender or "未设置" }}</span>                               7884
          <span class="acct-card__icon">{{ icon("user",                 7885
15) }}</span>                                                           7886
        </summary>                                                      7887
        <div class="acct-card__body">                                   7888
          <div class="field">                                           7889
            <label class="field__label" for="{{                         7890
info_form.gender.id }}">性别</label>                                    7891
            {{ info_form.gender(id=info_form.gender.id) }}              7892
            {% for e in info_form.gender.errors %}                      7893
            <p class="field__error">{{ e }}</p>                         7894
            {% endfor %}                                                7895
          </div>                                                        7896
          <p class="actions">{{ info_form.submit_info(clas              7897
s_="btn") }}</p>                                                        7898
        </div>                                                          7899
      </details>                                                        7900
```

```
          <details class="acct-card" {% if                    7901
      info_form.email.errors %}open{% endif %}>               7902
            <summary>                                         7903
              <span class="acct-card__title">邮箱</span>      7904
              <span class="acct-card__value">{{               7905
      current_user.email or "未设置" }}</span>                 7906
              <span class="acct-card__icon">{{ icon("mail",   7907
      15) }}</span>                                           7908
            </summary>                                        7909
            <div class="acct-card__body">                     7910
              <div class="field">                             7911
                <label class="field__label" for="{{           7912
      info_form.email.id }}">邮箱</label>                      7913
                {{ info_form.email(id=info_form.email.id) }}  7914
                {% for e in info_form.email.errors %}         7915
                <p class="field__error">{{ e }}</p>           7916
                {% endfor %}                                  7917
              </div>                                          7918
              <p class="actions">{{ info_form.submit_info(clas7919
      s_="btn") }}</p>                                        7920
            </div>                                            7921
          </details>                                          7922
          <details class="acct-card" {% if                    7923
      info_form.phone.errors %}open{% endif %}>               7924
            <summary>                                         7925
              <span class="acct-card__title">手机号</span>    7926
              <span class="acct-card__value">{{               7927
      current_user.phone or "未设置" }}</span>                 7928
              <span class="acct-card__icon">{{ icon("phone",  7929
      15) }}</span>                                           7930
            </summary>                                        7931
            <div class="acct-card__body">                     7932
              <div class="field">                             7933
                <label class="field__label" for="{{           7934
      info_form.phone.id }}">手机号</label>                    7935
                {{ info_form.phone(id=info_form.phone.id) }}  7936
                {% for e in info_form.phone.errors %}         7937
                <p class="field__error">{{ e }}</p>           7938
                {% endfor %}                                  7939
              </div>                                          7940
              <p class="actions">{{ info_form.submit_info(clas7941
      s_="btn") }}</p>                                        7942
            </div>                                            7943
          </details>                                          7944
        </div>                                                7945
      </form>                                                 7946
      {% endif %}                                             7947
    </div>                                                    7948
</div>                                                        7949
{% endblock %}                                                7950
```

```
{% block scripts %}
<script>
// 同时只展开一张卡，避免点开多张后页面被撑得很长
(function () {
  var cards = document.querySelectorAll('.acct-card');
  Array.prototype.forEach.call(cards, function (card) {
    card.addEventListener('toggle', function () {
      if (!card.open) return;
      Array.prototype.forEach.call(cards, function (other) {
        if (other !== card && other.open) other.open =
    false;
      });
    });
  });
})();
</script>
{% endblock %}
# ==== app/templates/auth/_users.html ====
{# 管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}
<div class="acct-panel">
  <div class="acct-panel__head">
    <p class="acct-panel__count">共 {{ users|length }}
    个账号</p>
    <a class="btn btn--sm" href="{{ url_for('auth.add_user')
    }}">新增账号</a>
  </div>
  <table>
    <thead>
      <tr>
        <th scope="col">ID</th>
        <th scope="col">账户名称</th>
        <th scope="col">昵称</th>
        <th scope="col">职务</th>
        <th scope="col">状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in users %}
      <tr>
        <td class="num">{{ u.id }}</td>
        <td class="mono">
          {{ u.username }}
          {% if u.id == current_user.id %}<span
    class="muted">（当前登录）</span>{% endif %}
        </td>
        <td style="text-align:left;">
          {% if u.display_name %}{{ u.display_name }}
          {% else %}<span class="muted">—</span>{% endif %}
        </td>
```

```
          <td>                                                                  8001
            <span class="tag {{ 'tag--ok' if u.role ==                          8002
      'admin' else 'tag--muted' }}">                                            8003
              {{ role_labels.get(u.role, u.role) }}                             8004
            </span>                                                             8005
          </td>                                                                 8006
          <td>                                                                  8007
            {% if u.active %}<span class="tag                                   8008
      tag--ok">启用</span>                                                      8009
            {% else %}<span class="tag tag--muted">停用</span>{%                8010
      endif %}                                                                  8011
          </td>                                                                 8012
          <td class="nowrap">                                                   8013
            <span class="row-actions">                                          8014
              <a class="link-btn" href="{{                                      8015
      url_for('auth.edit_user', user_id=u.id) }}">编辑</a>                      8016
              <form method="post"                                               8017
                    action="{{ url_for('auth.delete_user',                      8018
      user_id=u.id) }}"                                                         8019
                    onsubmit="return confirm('确定删除账号「{{                  8020
      u.username }}」吗？\n该操作无法恢复。');">                                8021
                {{ form.hidden_tag() }}                                         8022
                <button type="submit" class="link-btn                           8023
      link-btn--danger"                                                         8024
                        {% if u.id == current_user.id                           8025
      %}disabled                                                                8026
                        title="不能删除当前登录的账号"{% endif                  8027
      %}>删除</button>                                                          8028
              </form>                                                           8029
            </span>                                                             8030
          </td>                                                                 8031
        </tr>                                                                   8032
        {% endfor %}                                                            8033
      </tbody>                                                                  8034
    </table>                                                                    8035
</div>                                                                          8036
# ==== app/templates/auth/_logs.html ====                                       8037
{# 管理员的「操作日志」分类。由 auth/profile.html 内联引入，不单独成页。 #}    8038
<div class="acct-panel">                                                        8039
    <div class="acct-panel__head">                                              8040
      <p class="acct-panel__count">                                             8041
        共 {{ total }} 条记录，当前第 {{ page }} / {{ pages }} 页               8042
      </p>                                                                      8043
      {% if pages > 1 %}                                                        8044
      <nav class="row-actions" aria-label="日志分页">                           8045
        {% if page > 1 %}                                                       8046
        <a class="link-btn"                                                     8047
           href="{{ url_for('auth.profile', tab='logs',                         8048
      page=page - 1) }}">上一页</a>                                             8049
        {% endif %}                                                             8050
```

```
        {% if page < pages %}                                                  8051
        <a class="link-btn"                                                    8052
            href="{{ url_for('auth.profile', tab='logs',                       8053
    page=page + 1) }}">下一页</a>                                              8054
        {% endif %}                                                            8055
      </nav>                                                                   8056
      {% endif %}                                                              8057
    </div>                                                                     8058
    {% if not records %}                                                       8059
    <p class="muted" style="padding:18px 20px;">暂无操作记录。</p>             8060
    {% else %}                                                                 8061
    <table>                                                                    8062
      <thead>                                                                  8063
        <tr>                                                                   8064
          <th scope="col">时间</th>                                            8065
          <th scope="col">操作人</th>                                          8066
          <th scope="col">操作</th>                                            8067
          <th scope="col">对象</th>                                            8068
          <th scope="col">说明</th>                                            8069
        </tr>                                                                  8070
      </thead>                                                                 8071
      <tbody>                                                                  8072
        {% for r in records %}                                                 8073
        <tr>                                                                   8074
          <td class="mono nowrap">{{ r.created_at |                            8075
    localdt("%Y-%m-%d %H:%M:%S") }}</td>                                       8076
          <td>{{ users_map.get(r.user_id, "—") }}</td>                         8077
          <td style="text-align:left;">{{ r.action }}</td>                     8078
          <td>                                                                 8079
            {% if r.target_type %}{{ r.target_type }}{% if                     8080
    r.target_id %}#{{ r.target_id }}{% endif %}                                8081
            {% else %}—{% endif %}                                             8082
          </td>                                                                8083
          <td style="text-align:left;">{{ r.detail or "—"                      8084
    }}</td>                                                                    8085
        </tr>                                                                  8086
        {% endfor %}                                                           8087
      </tbody>                                                                 8088
    </table>                                                                   8089
    {% endif %}                                                                8090
</div>                                                                         8091
# ==== app/templates/project/index.html ====                                   8092
{% extends "base.html" %}                                                      8093
{% block title %}项目与矿山档案{% endblock %}                                  8094
{% block content %}                                                            8095
<div class="card">                                                             8096
  <h2>项目与矿山档案</h2>                                                      8097
  <p class="actions" style="margin-bottom:12px;">                              8098
    <a class="btn btn--sm" href="/project/add">新建项目</a>                    8099
  </p>                                                                         8100
```

```
{% if not projects %}
<p class="muted">
  暂无项目。点击上方「新建项目」开始建立评价项目，
  或运行 <span class="mono">python seed_demo.py</span>
  载入论文案例。
</p>
{% else %}
<table>
  <caption class="muted">共 {{ projects|length }} 个项目</caption>
  <thead>
    <tr>
      <th scope="col">项目名称</th>
      <th scope="col">编号</th>
      <th scope="col">状态</th>
      <th scope="col">图斑数</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in projects %}
    <tr>
      <td style="text-align:left;">{{ p.name }}</td>
      <td class="mono">{{ p.code or "—" }}</td>
      <td><span class="tag tag--muted">{{ p.status }}</span></td>
      <td class="num">
        {% if p.patches.count() %}
          {{ p.patches.count() }}
        {% else %}
          <span class="tag tag--warn">0</span>
        {% endif %}
      </td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/project/{{ p.id }}/patches">{% if p.patches.count() %}矿山图斑{% else %}+新建图斑{% endif %}</a>
          <a class="link-btn" href="/project/{{ p.id }}/edit">编辑</a>
          <form method="post" action="/project/{{ p.id }}/delete"
                onsubmit="return confirm('确定删除项目「{{ p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">
            {{ form.hidden_tag() }}
            <button type="submit" class="link-btn link-btn--danger">删除</button>
          </form>
        </span>
      </td>
```

```
        </tr>                                                                  8151
        {% endfor %}                                                           8152
      </tbody>                                                                 8153
    </table>                                                                   8154
    {% endif %}                                                                8155
</div>                                                                         8156
{% endblock %}                                                                 8157
# ==== app/templates/project/edit.html ====                                    8158
{% extends "base.html" %}                                                      8159
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%                  8160
    endblock %}                                                                8161
{% block content %}                                                            8162
<div class="card" style="max-width:640px;">                                    8163
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                         8164
  <form method="post">                                                         8165
    {{ form.hidden_tag() }}                                                    8166
    <div class="field">                                                        8167
      <label class="field__label" for="name">项目名称</label>                  8168
      <input id="name" type="text" name="name"                                 8169
    style="width:100%;" required                                               8170
             value="{{ project.name if project else '' }}">                    8171
    </div>                                                                     8172
    <div class="field">                                                        8173
      <label class="field__label" for="code">项目编号</label>                  8174
      <input id="code" type="text" name="code"                                 8175
    style="width:240px;"                                                       8176
             value="{{ project.code or '' if project else                      8177
    '' }}">                                                                    8178
    </div>                                                                     8179
    {% if project %}                                                           8180
    <div class="field">                                                        8181
      <label class="field__label" for="status">状态</label>                    8182
      <select id="status" name="status">                                       8183
        {% for s in ["进行中", "已完成", "已归档"] %}                          8184
        <option value="{{ s }}" {% if project.status == s                      8185
    %}selected{% endif %}>{{ s }}</option>                                     8186
        {% endfor %}                                                           8187
      </select>                                                                8188
    </div>                                                                     8189
    {% endif %}                                                                8190
    <div class="field">                                                        8191
      <label class="field__label" for="description">项目说明</la               8192
    bel>                                                                       8193
      <textarea id="description" name="description">{{                         8194
    project.description or '' if project else ''                               8195
    }}</textarea>                                                              8196
    </div>                                                                     8197
    <p class="actions" style="margin-top:18px;">                               8198
      <button type="submit" class="btn">保存</button>                          8199
      <a class="btn btn--ghost" href="/project/">返回</a>                      8200
```

```
      </p>                                                                    8201
    </form>                                                                   8202
</div>                                                                        8203
{% endblock %}                                                                8204
# ==== app/templates/project/patches.html ====                                8205
{% extends "base.html" %}                                                     8206
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}                  8207
{% block content %}                                                           8208
<div class="card">                                                            8209
  <h2>矿山图斑 · {{ project.name }}</h2>                                      8210
  <p class="actions" style="margin-bottom:12px;">                             8211
    <a class="btn btn--sm" href="/project/{{ project.id                       8212
    }}/patch/add">新建图斑</a>                                                8213
    <a class="btn btn--sm btn--ghost"                                         8214
    href="/project/">返回项目列表</a>                                         8215
  </p>                                                                        8216
  {% if not project.patches.count() %}                                        8217
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>  8218
  {% else %}                                                                  8219
  <table>                                                                     8220
    <caption class="muted">共 {{ project.patches.count() }}                   8221
    个图斑</caption>                                                          8222
    <thead>                                                                   8223
      <tr>                                                                    8224
        <th scope="col">图斑名称</th>                                         8225
        <th scope="col">行政区</th>                                           8226
        <th scope="col">面积（公顷）</th>                                     8227
        <th scope="col">开采矿种</th>                                         8228
        <th scope="col">微地质单元</th>                                       8229
        <th scope="col">操作</th>                                             8230
      </tr>                                                                   8231
    </thead>                                                                  8232
    <tbody>                                                                   8233
      {% for p in project.patches %}                                          8234
      <tr>                                                                    8235
        <td style="text-align:left;">{{ p.name }}</td>                        8236
        <td>{{ p.admin_region or "—" }}</td>                                  8237
        <td class="num">{{ p.area_ha if p.area_ha is not                      8238
    none else "—" }}</td>                                                     8239
        <td>{{ p.mineral_type or "—" }}</td>                                  8240
        <td class="num">{{ p.units.count() }}</td>                            8241
        <td class="nowrap">                                                   8242
          <span class="row-actions">                                          8243
            <a class="link-btn" href="/unit/{{ p.id                           8244
    }}">微地质单元</a>                                                       8245
            <a class="link-btn" href="/constraint/{{ p.id                     8246
    }}">限制性判定</a>                                                       8247
            <a class="link-btn" href="/project/patch/{{                       8248
    p.id }}/edit">编辑</a>                                                    8249
            <form method="post" action="/project/patch/{{                     8250
```

```
      p.id }}/delete"                                                    8251
                        onsubmit="return confirm('确定删除图斑「{{       8252
      p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  8253
                    {{ form.hidden_tag() }}                              8254
                    <button type="submit" class="link-btn                8255
      link-btn--danger">删除</button>                                    8256
                </form>                                                  8257
              </span>                                                    8258
            </td>                                                        8259
          </tr>                                                          8260
          {% endfor %}                                                   8261
      </tbody>                                                           8262
    </table>                                                             8263
    {% endif %}                                                          8264
</div>                                                                   8265
{% endblock %}                                                           8266
# ==== app/templates/project/patch_edit.html ====                        8267
{% extends "base.html" %}                                                8268
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%             8269
      endblock %}                                                        8270
{% block content %}                                                      8271
<div class="card" style="max-width:720px;">                              8272
    <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name      8273
      }}</h2>                                                            8274
    <form method="post">                                                 8275
      {{ form.hidden_tag() }}                                            8276
      <div class="field">                                                8277
        <label class="field__label" for="patch-name">图斑名称</lab       8278
      el>                                                                8279
        <input id="patch-name" type="text" name="name"                   8280
      style="width:100%;" required                                       8281
               value="{{ patch.name if patch else '' }}">                8282
      </div>                                                             8283
      <div class="field">                                                8284
        <label class="field__label" for="admin-region">所在行政区</       8285
      label>                                                             8286
        <input id="admin-region" type="text"                             8287
      name="admin_region" style="width:100%;"                            8288
               value="{{ patch.admin_region or '' if patch               8289
      else '' }}">                                                       8290
      </div>                                                             8291
      <div class="field">                                                8292
        <label class="field__label" for="area-ha">图斑面积（公顷）</la   8293
      bel>                                                               8294
        <input id="area-ha" type="number" step="any"                     8295
      name="area_ha"                                                     8296
               value="{{ patch.area_ha if patch and                      8297
      patch.area_ha is not none else '' }}">                             8298
      </div>                                                             8299
      <div class="field">                                                8300
```

```
    <label class="field__label" for="mineral-type">开采矿种</label>
    <input id="mineral-type" type="text" name="mineral_type" style="width:240px;"
           value="{{ patch.mineral_type or '' if patch else '' }}">
</div>
<div class="field">
    <label class="field__label" for="mining-method">开采方式</label>
    <input id="mining-method" type="text" name="mining_method" style="width:240px;"
           value="{{ patch.mining_method or '' if patch else '' }}">
</div>
<div class="field">
    <label class="field__label" for="stop-years">停产年限</label>
    <input id="stop-years" type="number" step="any" name="stop_years"
           value="{{ patch.stop_years if patch and patch.stop_years is not none else '' }}">
</div>
<div class="field">
    <label class="checkbox" for="responsible-missing">
      <input id="responsible-missing" type="checkbox" name="responsible_entity_missing" value="1"
             {% if not patch or patch.responsible_entity_missing %}checked{% endif %}>
      <span>责任主体缺失</span>
    </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
<p class="muted">
  论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
  该基准与图斑绑定管理，可按不同参照生态系统分别设定。
</p>
<div class="field">
    <label class="field__label" for="coverage-benchmark">植被覆盖度基准（%）</label>
    <input id="coverage-benchmark" type="number" step="any" name="coverage_benchmark"
           value="{{ reference.coverage_benchmark if reference else 75 }}">
</div>
<div class="field">
    <label class="field__label" for="shannon-benchmark">Shannon-Wiener 多样性指数基准</label>
    <input id="shannon-benchmark" type="number" step="any" name="shannon_benchmark"
```

```
                value="{{ reference.shannon_benchmark if
    reference else 0.7 }}">
    </div>
    <div class="field">
      <label class="field__label" for="dominant-species">建群种
    </label>
      <input id="dominant-species" type="text"
    name="dominant_species" style="width:100%;"
                value="{{ reference.dominant_species or '' if
    reference else '' }}">
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/project/{{
    project.id }}/patches">返回</a>
    </p>
  </form>
</div>
{% if patch %}
<div class="card" style="max-width:720px;">
  <h2>遥感影像挂接</h2>
  <p class="muted">
    挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
    支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
  </p>
  {% if patch.image_path %}
  <figure style="margin:0 0 16px;">
    <img src="{{ url_for('static', filename=patch.image_path
    ) }}"
         alt="图斑「{{ patch.name }}」已挂接的遥感影像"
         style="max-width:100%;border:1px solid
    var(--line);border-radius:6px;">
    <figcaption class="muted" style="margin-top:6px;">
      {{ patch.image_note or "遥感影像" }}
      {% if patch.image_uploaded_at %}
      · 上传于 {{ patch.image_uploaded_at | localdt }}
      {% endif %}
    </figcaption>
  </figure>
  <form method="post" action="/project/patch/{{ patch.id
    }}/image/delete"
        onsubmit="return confirm('确定移除该遥感影像吗？');">
    {{ form.hidden_tag() }}
    <button type="submit" class="btn btn--sm
    btn--danger">移除影像</button>
  </form>
  <hr style="margin:16px 0;border:none;border-top:1px solid
    var(--line);">
  {% endif %}
  <form method="post" action="/project/patch/{{ patch.id
```

```
      }}/image"                                                                    8401
           enctype="multipart/form-data">                                          8402
      {{ form.hidden_tag() }}                                                      8403
      <div class="field">                                                          8404
        <label class="field__label" for="image-file">                              8405
          {{ "替换影像文件" if patch.image_path else "选择影像文件" }}              8406
        </label>                                                                   8407
        <input id="image-file" type="file" name="image"                            8408
               accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t                     8409
      iff" required>                                                               8410
      </div>                                                                       8411
      <div class="field">                                                          8412
        <label class="field__label" for="image-note">影像说明（如来源、            8413
      时相）</label>                                                               8414
        <input id="image-note" type="text" name="image_note"                       8415
      style="width:100%;"                                                          8416
               placeholder="如：2025 年 6 月无人机正射影像"                        8417
               value="{{ patch.image_note or '' }}">                               8418
      </div>                                                                       8419
      <p class="actions">                                                          8420
        <button type="submit" class="btn">{{ "替换影像" if                        8421
      patch.image_path else "上传并挂接" }}</button>                               8422
      </p>                                                                         8423
    </form>                                                                        8424
</div>                                                                             8425
{% endif %}                                                                        8426
{% endblock %}                                                                     8427
# ==== app/templates/unit/list.html ====                                           8428
{% extends "base.html" %}                                                          8429
{% block title %}微地质单元{% endblock %}                                          8430
{% block content %}                                                                8431
<div class="card">                                                                 8432
  <h2>微地质单元</h2>                                                              8433
  {% if not rows %}                                                                8434
  <p class="muted">                                                                8435
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。                              8436
  </p>                                                                             8437
  {% else %}                                                                       8438
  <table>                                                                          8439
    <thead>                                                                        8440
      <tr>                                                                         8441
        <th scope="col">所属项目</th>                                              8442
        <th scope="col">矿山图斑</th>                                              8443
        <th scope="col">已划分单元</th>                                            8444
        <th scope="col">占比合计</th>                                              8445
        <th scope="col">操作</th>                                                  8446
      </tr>                                                                        8447
    </thead>                                                                       8448
    <tbody>                                                                        8449
      {% for r in rows %}                                                          8450
```

```
        <tr>                                                                    8451
          <td>{{ r.patch.project.name if r.patch.project else                   8452
      "—" }}</td>                                                               8453
          <td style="text-align:left;">{{ r.patch.name }}</td>                  8454
          <td class="num">{{ r.count }} 个</td>                                 8455
          <td class="num">                                                      8456
            {% if r.count %}                                                    8457
              {{ "%.2f%%"|format(r.ratio_sum * 100) }}                          8458
              {% if r.ratio_sum < 0.999 or r.ratio_sum >                        8459
      1.001 %}                                                                  8460
                <span class="warn">（不等于 100%）</span>                       8461
              {% endif %}                                                       8462
            {% else %}—{% endif %}                                              8463
          </td>                                                                 8464
          <td class="nowrap">                                                   8465
            <span class="row-actions">                                          8466
              <a class="link-btn" href="/unit/{{ r.patch.id                     8467
      }}">划分与维护</a>                                                        8468
              <a class="link-btn" href="/indicator/">录入指标数据</a            8469
      >                                                                         8470
            </span>                                                             8471
          </td>                                                                 8472
        </tr>                                                                   8473
        {% endfor %}                                                            8474
      </tbody>                                                                  8475
    </table>                                                                    8476
    {% endif %}                                                                 8477
</div>                                                                          8478
{% endblock %}                                                                  8479
# ==== app/templates/unit/index.html ====                                       8480
{% extends "base.html" %}                                                       8481
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}                    8482
{% block content %}                                                             8483
<div class="card">                                                              8484
  <h2>微地质单元 · {{ patch.name }}</h2>                                        8485
  {% if units %}                                                                8486
  <table>                                                                       8487
    <caption class="muted">                                                     8488
      共 {{ units|length }} 个单元；                                            8489
      面积合计 <span class="num">{{ "%.4f"|format(total_area)                   8490
    }}</span>；                                                                 8491
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *                 8492
    100) }}</span>                                                              8493
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >                     8494
    1.001) %}                                                                   8495
        <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>       8496
      {% endif %}                                                               8497
    </caption>                                                                  8498
    <thead>                                                                     8499
      <tr>                                                                      8500
```

```
          <th scope="col">微地质单元</th>
          <th scope="col">面积</th>
          <th scope="col">面积占比</th>
          <th scope="col">说明</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for u in units %}
        <tr>
          <td>{{ u.unit_type }}</td>
          <td class="num">{{ u.area if u.area is not none
      else "—" }}</td>
          <td class="num">
            {% if u.ratio is not none %}{{
      "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
          </td>
          <td style="text-align:left;">{{ u.description or
      "—" }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/unit/edit/{{ u.id
      }}">编辑</a>
              <form method="post" action="/unit/delete/{{
      u.id }}"
                    onsubmit="return confirm('确定删除微地质单元「{{
      u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
      link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% else %}
    <p class="muted">尚未划分微地质单元。请在下方添加。</p>
    {% endif %}
    <h3>添加微地质单元</h3>
    <form method="post" action="/unit/{{ patch.id }}/add">
      {{ form.hidden_tag() }}
      <div class="actions">
        <label class="field__label" for="add-unit-type"
      style="margin:0;">类型</label>
        <select id="add-unit-type" name="unit_type">
          {% for t in unit_types %}<option value="{{ t }}">{{
      t }}</option>{% endfor %}
        </select>
```

```
        <label class="field__label" for="add-unit-area"
      style="margin:0;">面积</label>
        <input id="add-unit-area" type="number" step="any"
      name="area">
        <label class="field__label" for="add-unit-desc"
      style="margin:0;">说明</label>
        <input id="add-unit-desc" type="text"
      name="description" style="width:220px;">
        <button type="submit" class="btn btn--sm">添加</button>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost" href="/project/{{
      patch.project_id }}/patches">返回图斑列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/indicator/">去录入指标数据</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/unit/edit.html ====
{% extends "base.html" %}
{% block title %}编辑微地质单元{% endblock %}
{% block content %}
<div class="card">
    <h2>编辑微地质单元 · {{ patch.name }}</h2>
    <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="unit-type">微地质单元类型</l
      abel>
        <select id="unit-type" name="unit_type">
          {% for t in unit_types %}
          <option value="{{ t }}" {% if unit.unit_type == t
      %}selected{% endif %}>{{ t }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="unit-area">面积</label>
        <input id="unit-area" type="number" step="any"
      name="area"
               value="{{ unit.area if unit.area is not none
      else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="unit-desc">说明</label>
        <input id="unit-desc" type="text" name="description"
      style="width:420px;"
               value="{{ unit.description or '' }}">
```

```
      </div>                                                                    8601
      <p class="actions" style="margin-top:16px;">                              8602
        <button type="submit" class="btn">保存</button>                        8603
        <a class="btn btn--ghost" href="/unit/{{                                8604
      unit.patch_id }}">返回</a>                                                8605
      </p>                                                                      8606
    </form>                                                                     8607
</div>                                                                          8608
{% endblock %}                                                                  8609
# ==== app/templates/standard/index.html ====                                   8610
{% extends "base.html" %}                                                       8611
{% block title %}政策与标准知识库{% endblock %}                                 8612
{% block content %}                                                             8613
<div class="card">                                                              8614
  <h2>政策与标准知识库</h2>  <p class="actions"                                 8615
    style="margin-bottom:14px;">                                                8616
    <a class="btn btn--sm" href="/standard/add">添加标准</a>                    8617
    <a class="btn btn--sm btn--ghost"                                           8618
    href="/standard/compare">省际口径比对</a>                                   8619
  </p>                                                                          8620
  <h3>地方标准</h3>                                                            8621
  {% if local %}                                                                8622
  <table>                                                                       8623
    <thead>                                                                     8624
      <tr>                                                                      8625
        <th scope="col">省份</th>                                               8626
        <th scope="col">名称</th>                                               8627
        <th scope="col">编号</th>                                               8628
        <th scope="col">规则数</th>                                             8629
        <th scope="col">操作</th>                                               8630
      </tr>                                                                     8631
    </thead>                                                                    8632
    <tbody>                                                                     8633
      {% for s in local %}                                                      8634
      <tr>                                                                      8635
        <td>{{ s.province }}</td>                                               8636
        <td style="text-align:left;">{{ s.name }}</td>                          8637
        <td class="mono">{{ s.doc_number or "—" }}</td>                         8638
        <td class="num">{{ s.rules|length }}</td>                               8639
        <td class="nowrap">                                                     8640
          <span class="row-actions">                                            8641
            <a class="link-btn" href="/standard/{{ s.id                         8642
    }}/rules">规则</a>                                                          8643
            <a class="link-btn" href="/standard/{{ s.id                         8644
    }}/edit">编辑</a>                                                           8645
            <form method="post" action="/standard/{{ s.id                       8646
    }}/delete"                                                                  8647
                  onsubmit="return confirm('确定删除「{{                        8648
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length              8649
    }} 条规则将一并删除，且无法恢复。');">                                      8650
```

```
                {{ form.hidden_tag() }}                                    8651
                <button type="submit" class="link-btn                      8652
  link-btn--danger">删除</button>                                          8653
            </form>                                                        8654
          </span>                                                          8655
        </td>                                                              8656
      </tr>                                                                8657
      {% endfor %}                                                         8658
  </tbody>                                                                 8659
</table>                                                                   8660
{% else %}                                                                 8661
<p class="muted">暂无地方标准。运行 <span class="mono">python             8662
  seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>          8663
{% endif %}                                                                8664
<h3>政策文件</h3>                                                          8665
{% if policy %}                                                            8666
<table>                                                                    8667
  <thead>                                                                  8668
    <tr>                                                                   8669
      <th scope="col">省份</th>                                            8670
      <th scope="col">名称</th>                                            8671
      <th scope="col">文号</th>                                            8672
      <th scope="col">规则数</th>                                          8673
      <th scope="col">操作</th>                                            8674
    </tr>                                                                  8675
  </thead>                                                                 8676
  <tbody>                                                                  8677
    {% for s in policy %}                                                  8678
    <tr>                                                                   8679
      <td>{{ s.province }}</td>                                            8680
      <td style="text-align:left;">{{ s.name }}</td>                       8681
      <td class="mono">{{ s.doc_number or "—" }}</td>                      8682
      <td class="num">{{ s.rules|length }}</td>                            8683
      <td class="nowrap">                                                  8684
        <span class="row-actions">                                         8685
          <a class="link-btn" href="/standard/{{ s.id                      8686
  }}/rules">规则</a>                                                       8687
          <a class="link-btn" href="/standard/{{ s.id                      8688
  }}/edit">编辑</a>                                                        8689
          <form method="post" action="/standard/{{ s.id                    8690
  }}/delete"                                                               8691
                onsubmit="return confirm('确定删除「{{                     8692
  s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length           8693
  }} 条规则将一并删除，且无法恢复。');">                                   8694
            {{ form.hidden_tag() }}                                        8695
            <button type="submit" class="link-btn                          8696
  link-btn--danger">删除</button>                                          8697
          </form>                                                          8698
        </span>                                                            8699
      </td>                                                                8700
```

```
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% else %}
    <p class="muted">暂无政策文件。</p>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/standard/edit.html ====
{% extends "base.html" %}
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{% endblock %}
{% block content %}
<div class="card" style="max-width:800px;">
    <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="province">省份</label>
        <input id="province" type="text" name="province"
      style="width:160px;" required
               value="{{ standard.province if standard else
      '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="std-name">名称</label>
        <input id="std-name" type="text" name="name"
      style="width:100%;" required
               value="{{ standard.name if standard else ''
      }}">
      </div>
      <div class="field">
        <label class="field__label" for="doc-number">标准编号 /
      文号</label>
        <input id="doc-number" type="text" name="doc_number"
      style="width:320px;"
               value="{{ standard.doc_number or '' if
      standard else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="category">类别</label>
        <select id="category" name="category">
          {% for c in ["地方标准", "政策文件"] %}
          <option value="{{ c }}" {% if standard and
      standard.category == c %}selected{% endif %}>{{ c
      }}</option>
          {% endfor %}
        </select>
      </div>
```

```
      <div class="field">
        <label class="field__label" for="publish-date">发布时间</l
      abel>
        <input id="publish-date" type="text"
      name="publish_date" style="width:200px;"
               placeholder="如 2023-03-30"
               value="{{ standard.publish_date or '' if
      standard else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="implement-date">实施时间<
      /label>
        <input id="implement-date" type="text"
      name="implement_date" style="width:200px;"
               placeholder="如 2023-04-30（未明确可留空）"
               value="{{ standard.implement_date or '' if
      standard else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="summary">特点 /
      判定标准</label>
        <textarea id="summary" name="summary"
      style="min-height:96px;">{{ standard.summary or '' if
      standard else '' }}</textarea>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        {% if standard %}
        <form method="post" action="/standard/{{ standard.id
      }}/delete" style="display:inline;"
              onsubmit="return confirm('确定删除「{{
      standard.province }} {{ standard.name
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn
      btn--danger">删除该标准</button>
        </form>
        {% endif %}
        <a class="btn btn--ghost" href="/standard/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
#  ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
     %}
{% block content %}
<div class="card">
  <h2>结构化规则 · {{ standard.province }} {{ standard.name
```

```
    }}</h2>
  {% if rules %}
  <table>
    <caption class="muted">共 {{ rules|length }}
    条规则</caption>
    <thead>
      <tr>
        <th scope="col">判定项</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">适用分区</th>
        <th scope="col">备注</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rules %}
      <tr>
        <td style="text-align:left;">{{ r.item }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
        <td>
          <form method="post" action="/standard/rule/{{
    r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item
    }} {{ r.operator }} {{ r.threshold or '' }}」吗?');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm
    btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}
  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
        <input id="r-item" type="text" name="item"
    placeholder="如：植被覆盖度" required>
```

```
        </div>                                                                  8851
        <div>                                                                   8852
          <label class="field__label" for="r-op">比较符</label>                 8853
          <select id="r-op" name="operator">                                    8854
            {% for o in [">=", ">", "<=", "<", "=="]                            8855
      %}<option value="{{ o }}">{{ o }}</option>{% endfor %}                    8856
          </select>                                                             8857
        </div>                                                                  8858
        <div>                                                                   8859
          <label class="field__label" for="r-thr">阈值</label>                  8860
          <input id="r-thr" type="text" name="threshold"                        8861
      placeholder="如：50" style="width:110px;">                                8862
        </div>                                                                  8863
        <div>                                                                   8864
          <label class="field__label" for="r-cond">条件类型</label8865
      >                                                                         8866
          <select id="r-cond" name="condition_type">                            8867
            <option value="必要条件">必要条件</option>                          8868
            <option value="充分条件">充分条件</option>                          8869
          </select>                                                             8870
        </div>                                                                  8871
        <div>                                                                   8872
          <label class="field__label" for="r-region">适用分区</lab8873
      el>                                                                       8874
          <input id="r-region" type="text" name="region"                        8875
      placeholder="如宁夏分区" style="width:130px;">                            8876
        </div>                                                                  8877
        <div>                                                                   8878
          <label class="field__label" for="r-note">备注</label>                 8879
          <input id="r-note" type="text" name="note"                            8880
      style="width:220px;">                                                     8881
        </div>                                                                  8882
        <div>                                                                   8883
          <button type="submit" class="btn">添加规则</button>                   8884
        </div>                                                                  8885
      </div>                                                                    8886
    </form>                                                                     8887
    <p class="actions" style="margin-top:20px;">                                8888
      <a class="btn btn--sm btn--ghost"                                         8889
      href="/standard/">返回标准列表</a>                                        8890
      <a class="btn btn--sm btn--ghost"                                         8891
      href="/standard/compare">省际口径比对</a>                                 8892
    </p>                                                                        8893
</div>                                                                          8894
{% endblock %}                                                                  8895
# ==== app/templates/standard/compare.html ====                                 8896
{% extends "base.html" %}                                                       8897
{% block title %}省际口径比对{% endblock %}                                     8898
{% block content %}                                                             8899
<div class="card">                                                              8900
```

```
  <h2>省际口径比对</h2>
  {% if not groups %}
  <p class="muted">
    暂无规则数据。请先在「标准知识库」中为标准录入规则，
    或运行 <span class="mono">python seed_demo.py</span> 载入论文表 1 数据。
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name }}</td>
        <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
  <h2>限制性指标判定</h2>
  {{ patch_guide(projects_without_patch) }}
```

```
  {% if not patches %}                                                    8951
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>          8952
  {% else %}                                                              8953
  <table>                                                                 8954
    <thead>                                                               8955
      <tr>                                                                8956
        <th scope="col">所属项目</th>                                      8957
        <th scope="col">矿山图斑</th>                                      8958
        <th scope="col">判定结果</th>                                      8959
        <th scope="col">操作</th>                                          8960
      </tr>                                                               8961
    </thead>                                                              8962
    <tbody>                                                               8963
      {% for p in patches %}                                              8964
      {% set r = latest.get(p.id) %}                                      8965
      <tr>                                                                8966
        <td>{{ p.project.name if p.project else "—" }}</td>               8967
        <td style="text-align:left;">{{ p.name }}</td>                    8968
        <td style="text-align:left;">                                     8969
          {% if r %}                                                      8970
            {% if r.passed %}                                             8971
            <span class="tag tag--ok">通过</span>                          8972
            {% else %}                                                    8973
            <span class="tag tag--warn">不通过</span>                      8974
            {% endif %}                                                   8975
            <span class="muted">{{ r.reason }}</span>                     8976
          {% else %}                                                      8977
          <span class="tag tag--muted">未判定</span>                      8978
          {% endif %}                                                     8979
        </td>                                                             8980
        <td class="nowrap">                                               8981
          <span class="row-actions">                                      8982
            <a class="link-btn" href="/constraint/{{ p.id                 8983
    }}">执行判定</a>                                                      8984
          </span>                                                         8985
        </td>                                                             8986
      </tr>                                                               8987
      {% endfor %}                                                        8988
    </tbody>                                                              8989
  </table>                                                                8990
  {% endif %}                                                             8991
</div>                                                                    8992
{% endblock %}                                                            8993
# ==== app/templates/constraint/check.html ====                           8994
{% extends "base.html" %}                                                 8995
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}           8996
{% block content %}                                                       8997
<div class="card" style="max-width:760px;">                               8998
  <h2>限制性条件判定 · {{ patch.name }}</h2>                              8999
  <form method="post">                                                    9000
```

```
      {{ form.hidden_tag() }}                                          9001
      {% for field, name, val in field_states %}                       9002
      <fieldset style="border:1px solid                                9003
      var(--line);border-radius:6px;padding:10px                       9004
      14px;margin:0 0 10px;">                                          9005
        <legend style="padding:0 6px;font-weight:600;">{{              9006
      name }}</legend>                                                 9007
        <div class="actions">                                          9008
          <label class="checkbox">                                     9009
            <input type="radio" name="{{ field }}" value="1"           9010
                   {% if val is none or val %}checked{% endif          9011
      %}>                                                              9012
            <span>通过</span>                                          9013
          </label>                                                     9014
          <label class="checkbox">                                     9015
            <input type="radio" name="{{ field }}" value="0"           9016
                   {% if val is not none and not val                   9017
      %}checked{% endif %}>                                            9018
            <span>不通过</span>                                        9019
          </label>                                                     9020
        </div>                                                         9021
      </fieldset>                                                      9022
      {% endfor %}                                                     9023
      <p class="actions" style="margin-top:16px;">                     9024
        <button type="submit" class="btn">保存判定</button>            9025
        <a class="btn btn--ghost" href="/constraint/">返回</a>         9026
      </p>                                                             9027
    </form>                                                            9028
    {% if record %}                                                    9029
    <div style="margin-top:18px;padding:14px;border-radius:6px9030
      ;border-left:4px solid;                                          9031
                 {% if record.passed %}background:var(--ok-soft9032
      );border-left-color:var(--ok);                                   9033
                 {% else %}background:var(--danger-soft);border9034
      -left-color:var(--danger);{% endif %}">                          9035
      <p style="margin:0 0 4px;font-weight:600;">                      9036
        判定结论：                                                     9037
        {% if record.passed %}                                         9038
        <span class="ok">通过，可进入定量评价</span>                   9039
        {% else %}                                                     9040
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>     9041
        {% endif %}                                                    9042
      </p>                                                             9043
      <p class="muted" style="margin:0;">{{ record.reason              9044
      }}</p>                                                           9045
    </div>                                                             9046
    {% endif %}                                                        9047
</div>                                                                 9048
{% endblock %}                                                         9049
#  ==== app/templates/ahp/index.html ====                             9050
```

```
{% extends "base.html" %}
{% block title %}AHP 权重计算{% endblock %}
{% block content %}
<div class="card" style="max-width:720px;">
  <h2>AHP 权重计算 · 选择指标层级</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="{{ form.unit_type.id }}">微地质单元类型</label>
      {{ form.unit_type(id=form.unit_type.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.level.id }}">指标层级</label>
      {{ form.level(id=form.level.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.l1.id }}">
        一级指标（层级选「二级指标」时需选择）
      </label>
      {{ form.l1(id=form.l1.id) }}
    </div>
    <p class="actions" style="margin-top:16px;">
      {{ form.submit(class_="btn") }}
      <a class="btn btn--ghost" href="/ahp/indicators">指标定义维护</a>
      <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a>
    </p>
  </form>
</div>
<div class="card">
  <h2>各微地质单元当前生效权重</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">当前权重来源</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in weight_rows %}
      <tr>
        <td style="text-align:left;">{{ r.unit_type }}</td>
        <td style="text-align:left;">
          {% if r.custom %}
          <span class="tag tag--warn">{{ r.label }}</span>
          {% else %}
```

```
            <span class="tag tag--muted">{{ r.label }}</span>
            {% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/ahp/matrix?unit_type={{ r.unit_type }}&level=1">一级指标权重</a>
              <a class="link-btn" href="/ahp/matrix?unit_type={{ r.unit_type }}&level=1&calc=1">查看一致性</a>
              {% if r.custom %}
              <form method="post" action="/ahp/weights/reset"
                    onsubmit="return confirm('确定把「{{ r.unit_type }}」恢复为论文表 3 的默认权重吗？');">
                <input type="hidden" name="csrf_token" value="{{ csrf_token() }}">
                <input type="hidden" name="unit_type" value="{{ r.unit_type }}">
                <button type="submit" class="link-btn link-btn--danger">恢复默认</button>
              </form>
              {% endif %}
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
{% endblock %}
# ==== app/templates/ahp/matrix.html ====
{% extends "base.html" %}
{% block title %}判断矩阵录入{% endblock %}
{% block content %}
<div class="card">
  <h2>
    判断矩阵录入
    · {{ unit_type }}
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}
  </h2>
  <p class="muted">
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
    矩阵已按论文表 3 的权重预填。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <table style="width:auto;">
      <thead>
        <tr>
```

```
            <th></th>                                                    9151
            {% for name in names %}<th scope="col">{{ name               9152
    }}</th>{% endfor %}                                                  9153
          </tr>                                                          9154
        </thead>                                                         9155
        <tbody>                                                          9156
          {% for i in range(names|length) %}                             9157
          <tr>                                                           9158
            <th scope="row">{{ names[i] }}</th>                          9159
            {% for j in range(names|length) %}                           9160
              {% if i == j %}                                            9161
              <td class="muted num">1</td>                               9162
              {% elif i < j %}                                           9163
              <td>                                                       9164
                <label class="muted" style="position:absolute;           9165
    left:-9999px;"                                                       9166
                       for="m-{{ i }}-{{ j }}">{{ names[i] }}            9167
    相对 {{ names[j] }} 的重要性</label>                                 9168
                <input id="m-{{ i }}-{{ j }}" type="number"               9169
    step="0.01" min="0.01"                                               9170
                       name="m_{{ i }}_{{ j }}" value="{{                9171
    default[i][j] }}">                                                   9172
              </td>                                                      9173
              {% else %}                                                 9174
              <td class="muted num">{{ default[i][j] }}</td>             9175
              {% endif %}                                                9176
            {% endfor %}                                                 9177
          </tr>                                                          9178
          {% endfor %}                                                   9179
        </tbody>                                                         9180
    </table>                                                             9181
    <p class="actions">                                                  9182
      <button type="submit" class="btn" name="action"                    9183
    value="calc">计算权重</button>                                       9184
      {% if result and result.consistent %}                              9185
      <button type="submit" class="btn" name="action"                    9186
    value="save">保存并生效</button>                                     9187
      {% endif %}                                                        9188
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>                9189
    </p>                                                                 9190
  </form>                                                                9191
  <p class="muted" style="margin-top:10px;">                             9192
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label             9193
    }}</strong>。                                                        9194
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；  9195
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。       9196
  </p>                                                                   9197
  {% if result %}                                                        9198
  <hr style="margin:22px 0;border:none;border-top:1px solid              9199
    var(--line);">                                                       9200
```

```
<h3>计算结果</h3>
<table style="width:auto;">
  <thead>
    <tr>
      <th scope="col">指标</th>
      {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">权重</th>
      {% for w in result.weights %}<td class="mono">{{ w }}</td>{% endfor %}
    </tr>
  </tbody>
</table>
<div class="stat-row">
  <div>
    <div class="stat__label">最大特征值 λmax</div>
    <div class="stat__value mono">{{ result.lambda_max }}</div>
  </div>
  <div>
    <div class="stat__label">一致性指标 CI</div>
    <div class="stat__value mono">{{ "%.4f"|format(result.ci) }}</div>
  </div>
  <div>
    <div class="stat__label">一致性比例 CR</div>
    <div class="stat__value mono">{{ "%.4f"|format(result.cr) }}</div>
  </div>
  <div>
    <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>
    <div class="stat__value">
      {% if result.consistent %}<span class="ok">通过</span>
      {% else %}<span class="warn">不通过</span>{% endif %}
    </div>
  </div>
</div>
{% if not result.consistent and result.most_inconsistent %}
<p style="margin-top:14px;padding:12px;border-radius:6px;background:var(--danger-soft);
          border-left:4px solid var(--danger);">
  <strong class="warn">建议优先调整：</strong>
  第 {{ result.most_inconsistent.i + 1 }} 行、第 {{ result.most_inconsistent.j + 1 }} 列的比较
  （当前 <span class="mono num">{{ result.most_inconsistent.v
```

```
      alue|round(3) }}</span>,                                                    9251
      理想约 <span class="mono num">{{ result.most_inconsistent.e                 9252
      xpected|round(3) }}</span>)                                                 9253
   </p>                                                                           9254
   {% endif %}                                                                    9255
   {% endif %}                                                                    9256
</div>                                                                            9257
{% endblock %}                                                                    9258
# ==== app/templates/ahp/indicators.html ====                                     9259
{% extends "base.html" %}                                                         9260
{% block title %}指标定义维护{% endblock %}                                        9261
{% block content %}                                                               9262
<div class="card">                                                                9263
   <h2>指标体系维护</h2>                                                          9264
   <p class="actions" style="margin-bottom:14px;">                                9265
      <a class="btn btn--sm btn--ghost"                                           9266
      href="/ahp/">返回权重计算</a>                                               9267
      <a class="btn btn--sm btn--ghost"                                           9268
      href="/ahp/experts">多专家矩阵</a>                                          9269
      <form method="post" action="/ahp/indicators/reset"                          9270
      style="display:inline;"                                                     9271
            onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 9272
      确定继续？');">                                                             9273
         {{ form.hidden_tag() }}                                                  9274
         <button type="submit" class="btn btn--sm                                 9275
      btn--danger">按论文取值重置</button>                                        9276
      </form>                                                                     9277
   </p>                                                                           9278
   {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,                  9279
      "三级指标")] %}                                                             9280
   {% set level_items = all_ind|selectattr("level",                               9281
      "equalto", level)|list %}                                                   9282
   {% if level_items %}                                                           9283
   <h3>{{ title }}（{{ level_items|length }} 项）</h3>                            9284
   <table>                                                                        9285
      <thead>                                                                     9286
         <tr>                                                                     9287
            <th scope="col">指标</th>                                             9288
            <th scope="col">上级指标</th>                                         9289
            <th scope="col">单位</th>                                             9290
            <th scope="col">标准化类型</th>                                       9291
            <th scope="col">理想取值</th>                                         9292
            <th scope="col">启用单元</th>                                         9293
            <th scope="col">限制性</th>                                           9294
            <th scope="col">操作</th>                                             9295
         </tr>                                                                    9296
      </thead>                                                                    9297
      <tbody>                                                                     9298
         {% for i in level_items %}                                               9299
         <tr>                                                                     9300
```

```
          <td style="text-align:left;">                                    9301
            <span class="mono muted">{{ i.code                             9302
      }}</span><br>{{ i.name }}                                            9303
          </td>                                                            9304
          <td>{{ (all_ind|selectattr("code", "equalto",                   9305
      i.parent_code)|list|first).name if i.parent_code else                9306
      "—" }}</td>                                                          9307
          <td>{{ i.unit or "—" }}</td>                                     9308
          <td>{{ type_label(i.ideal_type) }}</td>                          9309
          <td class="mono">{{ i.ideal_value or "—" }}</td>                 9310
          <td style="text-align:left;">{{ i.enabled_units or               9311
      "—" }}</td>                                                          9312
          <td>                                                             9313
            {% if i.is_constraint %}<span class="tag                       9314
      tag--warn">是</span>                                                 9315
            {% else %}<span class="tag tag--muted">否</span>{%             9316
      endif %}                                                             9317
          </td>                                                            9318
          <td class="nowrap">                                              9319
            <span class="row-actions">                                     9320
              <a class="link-btn" href="/ahp/indicators/{{                 9321
      i.code }}/edit">编辑</a>                                             9322
            </span>                                                        9323
          </td>                                                            9324
        </tr>                                                              9325
        {% endfor %}                                                       9326
      </tbody>                                                             9327
    </table>                                                               9328
    {% endif %}                                                            9329
    {% endfor %}                                                           9330
</div>                                                                     9331
{% endblock %}                                                             9332
# ==== app/templates/ahp/indicator_edit.html ====                          9333
{% extends "base.html" %}                                                  9334
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}                   9335
{% block content %}                                                        9336
<div class="card" style="max-width:680px;">                                9337
  <h2>编辑指标 · {{ ind.name }}</h2>                                       9338
  <p class="muted">                                                        9339
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{             9340
    ind.level }} 级，                                                      9341
    由论文指标体系确定，不可修改。                                         9342
  </p>                                                                     9343
  <form method="post">                                                     9344
    {{ form.hidden_tag() }}                                                9345
    <div class="field">                                                    9346
      <label class="field__label" for="name">指标名称</label>              9347
      <input id="name" type="text" name="name"                             9348
    style="width:100%;" required                                           9349
           value="{{ ind.name }}">                                         9350
```

```
</div>
<div class="field">
  <label class="field__label" for="unit">计量单位</label>
  <input id="unit" type="text" name="unit"
style="width:180px;"
         value="{{ ind.unit or '' }}">
</div>
<div class="field">
  <label class="field__label" for="ideal-type">标准化类型</la
bel>
  <select id="ideal-type" name="ideal_type">
    {% for code, label in [("benefit","越大越优"),("cost","越
小越优"),("mid","适中型"),("discrete","离散赋值")] %}
    <option value="{{ code }}" {% if ind.ideal_type ==
code %}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="ideal-value">理想取值</la
bel>
  <input id="ideal-value" type="text"
name="ideal_value" style="width:100%;"
         value="{{ ind.ideal_value or '' }}">
  <p class="muted" style="margin:4px 0 0;">
    如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
  </p>
</div>
<fieldset style="border:1px solid
var(--line);border-radius:6px;padding:10px
14px;margin:0 0 12px;">
  <legend style="padding:0 6px;font-weight:600;">启用该指标的微
地质单元</legend>
  {% for t in unit_types %}
  <label class="checkbox" style="margin-right:18px;">
    <input type="checkbox" name="enabled_units"
value="{{ t }}"
           {% if t in ind.units_list() %}checked{%
endif %}>
    <span>{{ t }}</span>
  </label>
  {% endfor %}
  <p class="muted" style="margin:8px 0 0;">
    论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。
  </p>
</fieldset>
<div class="field">
  <label class="checkbox" for="is-constraint">
    <input id="is-constraint" type="checkbox"
name="is_constraint" value="1"
```

```
                    {% if ind.is_constraint %}checked{% endif %}>
          <span>同时作为限制性指标（参与一票否决判定）</span>
        </label>
      </div>
      <div class="field">
        <label class="field__label" for="description">取值依据说明</
      label>
        <textarea id="description" name="description"
      style="min-height:88px;">{{ ind.description or ''
      }}</textarea>
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列
      表</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/ahp/experts.html ====
{% extends "base.html" %}
{% block title %}多专家判断矩阵{% endblock %}
{% block content %}
<div class="card">
    <h2>多专家判断矩阵 · 录入与汇总</h2>
    <form method="get" class="actions" style="margin-bottom:14
      px;">
      <div>
        <label class="field__label" for="unit-type">微地质单元类型</l
      abel>
        <select id="unit-type" name="unit_type"
      onchange="this.form.submit()">
          {% for t in ["岩质边坡", "碎石边坡", "平台"] %}
          <option value="{{ t }}" {% if unit_type == t
      %}selected{% endif %}>{{ t }}</option>
          {% endfor %}
        </select>
      </div>
      <div>
        <label class="field__label" for="level">指标层级</label>
        <select id="level" name="level"
      onchange="this.form.submit()">
          <option value="1" {% if level == 1 %}selected{%
      endif %}>一级指标</option>
          <option value="2" {% if level == 2 %}selected{%
      endif %}>二级指标</option>
        </select>
      </div>
    </form>
    <p class="muted">
```

```
    当前范围：<strong>{{ unit_type }}</strong> ·
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif
    %}。
    若选二级指标，请先到
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。
  </p>
  {% if rows %}
  <table>
    <caption class="muted">已录入 {{ rows|length }}
    位专家的矩阵</caption>
    <thead>
      <tr>
        <th scope="col">专家</th>
        <th scope="col">矩阵规模</th>
        <th scope="col">该矩阵 CR</th>
        <th scope="col">一致性</th>
        <th scope="col">录入时间</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.expert_name }}</td>
        <td class="num">{{ n }} × {{ n }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
    not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span
    class="tag tag--ok">通过</span>
          {% else %}<span class="tag tag--warn">不通过</span>{%
    endif %}
        </td>
        <td class="mono nowrap">{{ r.created_at | localdt
    }}</td>
        <td>
          <form method="post" action="/ahp/experts/{{ r.id
    }}/delete"
                onsubmit="return confirm('确定删除专家「{{
    r.expert_name }}」的矩阵吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm
    btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="actions" style="margin-bottom:10px;">
```

```
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{ 9501
    unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% 9502
    endif %}"> 9503
      汇总计算（几何平均） 9504
    </a> 9505
  </p> 9506
  {% else %} 9507
  <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p> 9508
  {% endif %} 9509
  <h3>录入一位专家的判断矩阵</h3> 9510
  <form method="post"> 9511
    {{ form.hidden_tag() }} 9512
    <input type="hidden" name="action" value="add"> 9513
    <input type="hidden" name="unit_type" value="{{ 9514
    unit_type }}"> 9515
    <input type="hidden" name="level" value="{{ level }}"> 9516
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1 9517
    }}">{% endif %} 9518
    <div class="field"> 9519
      <label class="field__label" for="expert-name">专家姓名</la 9520
    bel> 9521
      <input id="expert-name" type="text" name="expert_name" 9522
             style="width:220px;" placeholder="如：张工"> 9523
    </div> 9524
    <table style="width:auto;"> 9525
      <thead> 9526
        <tr> 9527
          <th></th> 9528
          {% for name in names %}<th scope="col">{{ name 9529
    }}</th>{% endfor %} 9530
        </tr> 9531
      </thead> 9532
      <tbody> 9533
        {% for i in range(n) %} 9534
        <tr> 9535
          <th scope="row">{{ names[i] }}</th> 9536
          {% for j in range(n) %} 9537
            {% if i == j %} 9538
            <td class="muted num">1</td> 9539
            {% elif i < j %} 9540
            <td> 9541
              <label class="muted" style="position:absolute; 9542
    left:-9999px;" 9543
                     for="e-{{ i }}-{{ j }}">{{ names[i] }} 9544
    相对 {{ names[j] }} 的重要性</label> 9545
              <input id="e-{{ i }}-{{ j }}" type="number" 9546
    step="0.01" min="0.01" 9547
                     name="m_{{ i }}_{{ j }}" value="{{ 9548
    default[i][j] }}"> 9549
            </td> 9550
```

```
                {% else %}                                                  9551
                <td class="muted num">{{ default[i][j] }}</td>              9552
                {% endif %}                                                 9553
              {% endfor %}                                                  9554
            </tr>                                                           9555
            {% endfor %}                                                    9556
          </tbody>                                                          9557
        </table>                                                            9558
        <p class="actions" style="margin-top:12px;">                        9559
          <button type="submit" class="btn">录入该专家矩阵</button>          9560
          <a class="btn btn--ghost" href="/ahp/indicators">指标维护</ 9561
        a>                                                                  9562
          <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>           9563
        </p>                                                                9564
      </form>                                                               9565
</div>                                                                      9566
{% endblock %}                                                              9567
# ==== app/templates/ahp/experts_aggregate.html ====                        9568
{% extends "base.html" %}                                                   9569
{% block title %}多专家矩阵汇总{% endblock %}                               9570
{% block content %}                                                         9571
<div class="card">                                                          9572
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{% 9573
    else %}二级指标（{{ l1 }}）{% endif %}</h2>                             9574
  {% if not result %}                                                       9575
  <p class="muted">                                                         9576
    该范围内尚未录入专家矩阵。                                              9577
    <a href="/ahp/experts?unit_type={{ unit_type                            9578
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                     9579
    %}">去录入</a>                                                          9580
  </p>                                                                      9581
  {% else %}                                                                9582
  <div class="stat-row">                                                    9583
    <div>                                                                   9584
      <div class="stat__label">参与汇总的专家</div>                         9585
      <div class="stat__value">{{ result.count }} 位</div>                  9586
    </div>                                                                  9587
    <div>                                                                   9588
      <div class="stat__label">λmax</div>                                   9589
      <div class="stat__value mono">{{ result.lambda_max                    9590
    }}</div>                                                                9591
    </div>                                                                  9592
    <div>                                                                   9593
      <div class="stat__label">CI</div>                                     9594
      <div class="stat__value mono">{{                                      9595
    "%.4f"|format(result.ci) }}</div>                                       9596
    </div>                                                                  9597
    <div>                                                                   9598
      <div class="stat__label">CR（要求 &lt; 0.1）</div>                    9599
      <div class="stat__value mono">{{                                      9600
```

```
    "%.4f"|format(result.cr) }}</div>                                    9601
    </div>                                                               9602
    <div>                                                                9603
      <div class="stat__label">一致性检验</div>                          9604
      <div class="stat__value">                                          9605
        {% if result.consistent %}<span class="ok">通过</span>           9606
        {% else %}<span class="warn">不通过</span>{% endif %}            9607
      </div>                                                             9608
    </div>                                                               9609
  </div>                                                                 9610
  <h3>汇总后权重</h3>                                                    9611
  <table style="width:auto;">                                            9612
    <thead>                                                              9613
      <tr>                                                               9614
        <th scope="col">指标</th>                                        9615
        {% for name in names %}<th scope="col">{{ name                   9616
    }}</th>{% endfor %}                                                  9617
      </tr>                                                              9618
    </thead>                                                             9619
    <tbody>                                                              9620
      <tr>                                                               9621
        <th scope="row">权重</th>                                        9622
        {% for w in result.weights %}<td class="mono">{{ w               9623
    }}</td>{% endfor %}                                                  9624
      </tr>                                                              9625
    </tbody>                                                             9626
  </table>                                                               9627
  <h3>各专家矩阵的一致性</h3>                                            9628
  <table style="width:auto;">                                            9629
    <thead>                                                              9630
      <tr><th scope="col">专家</th><th scope="col">该矩阵                9631
    CR</th><th scope="col">判定</th></tr>                                9632
    </thead>                                                             9633
    <tbody>                                                              9634
      {% for r in rows %}                                                9635
      <tr>                                                               9636
        <td>{{ r.expert_name }}</td>                                     9637
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is               9638
    not none else "—" }}</td>                                            9639
        <td>                                                             9640
          {% if r.cr is not none and r.cr < 0.1 %}<span                  9641
    class="ok">通过</span>                                               9642
          {% else %}<span class="warn">不通过</span>{% endif %}          9643
        </td>                                                            9644
      </tr>                                                              9645
      {% endfor %}                                                       9646
    </tbody>                                                             9647
  </table>                                                               9648
  {% if not result.consistent and result.most_inconsistent               9649
    %}                                                                   9650
```

```
  <p style="padding:12px;border-radius:6px;background:var(--
    danger-soft);
            border-left:4px solid var(--danger);">
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{
    result.most_inconsistent.j + 1 }} 列的比较
    （当前 <span class="mono">{{ result.most_inconsistent.value
    |round(3) }}</span>，
    理想约 <span class="mono">{{ result.most_inconsistent.expec
    ted|round(3) }}</span>）。
    该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
  </p>
  {% endif %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <form method="post"
          action="/ahp/experts?unit_type={{ unit_type
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"
          style="display:inline;"
          onsubmit="return confirm('保存后，「{{ unit_type
    }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">
      {{ form.hidden_tag() }}
      <input type="hidden" name="action" value="save_agg">
      <button type="submit" class="btn" {% if not
    result.consistent %}disabled{% endif %}>
        保存为生效权重
      </button>
    </form>
    <a class="btn btn--ghost" href="/ahp/experts?unit_type={
    { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1
    }}{% endif %}">返回录入</a>
    <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/indicator/index.html ====
{% extends "base.html" %}
{% block title %}指标数据采集{% endblock %}
{% block content %}
<div class="card">
  <h2>指标数据采集与管理</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV
    / Excel）</a>
  </p>
  {% if not patches %}
  <p class="muted">
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
```

```
</p>
{% else %}
{% for p in patches %}
<div class="patch-panel">
  <div class="patch-panel__head">
    <h3 class="patch-panel__name">{{ p.name }}</h3>
    <span class="patch-panel__meta">
      所属项目：{{ p.project.name if p.project else "—" }}
    </span>
    <span class="patch-panel__meta">
      微地质单元：{{ p.units.count() }} 个
    </span>
    {% if p.units.count() %}
    <span class="patch-panel__meta patch-panel__tail">
      <a class="link-btn" href="/indicator/import">批量导入本图斑数据</a>
    </span>
    {% endif %}
  </div>
  {% if not p.units.count() %}
  <div class="notice">
    <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
    <p style="margin:0;">
      指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
      请先完成划分，再回到本页录入。
    </p>
    <p style="margin:10px 0 0;">
      <a class="btn btn--sm" href="/unit/{{ p.id }}">去划分微地质单元</a>
    </p>
  </div>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">数据状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in p.units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{ "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td>
```

```
                {% if u.indicator_data %}                                    9751
                <span class="tag tag--ok">已录入</span>                      9752
                {% else %}                                                   9753
                <span class="tag tag--muted">未录入</span>                   9754
                {% endif %}                                                  9755
              </td>                                                          9756
              <td class="nowrap">                                            9757
                <span class="row-actions">                                   9758
                  <a class="link-btn" href="/indicator/{{ u.id               9759
    }}">录入 / 编辑</a>                                                      9760
                </span>                                                      9761
              </td>                                                          9762
            </tr>                                                            9763
            {% endfor %}                                                     9764
          </tbody>                                                           9765
        </table>                                                             9766
        {% endif %}                                                          9767
      </div>                                                                 9768
      {% endfor %}                                                           9769
      {% endif %}                                                            9770
</div>                                                                       9771
{% endblock %}                                                               9772
# ==== app/templates/indicator/edit.html ====                                9773
{% extends "base.html" %}                                                    9774
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}           9775
{% block content %}                                                          9776
<div class="card" style="max-width:820px;">                                  9777
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>                               9778
  <p class="muted">                                                          9779
    除标注「可选」的项外，本页字段均为必填：填齐后评价才能执行，缺项会在评价页逐条列出。   9780
    <br>标注「未检测」的重金属可以留空，按论文注记以「未受污染」计入污染指标，           9781
    评价结果与导出报告里会如实写明是哪几项未检测 —— 它不等于实测达标。                 9782
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。           9783
  </p>                                                                       9784
  <form method="post">                                                       9785
    {{ form.hidden_tag() }}                                                  9786
    {% for group_name, fields, has_hm in groups %}                           9787
    <h3>{{ group_name }}</h3>                                                9788
    {% for fname, cname, unit_ in fields %}                                  9789
    <div class="field">                                                      9790
      <label class="field__label" for="f-{{ fname }}">                       9791
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}                  9792
      </label>                                                               9793
      {% if fname == "aspect" %}                                             9794
      <select id="f-{{ fname }}" name="aspect">                              9795
        <option value="">— 不适用 —</option>                                 9796
        {% for a in aspect_types %}                                          9797
        <option value="{{ a }}" {% if current.get("aspect")                  9798
    == a %}selected{% endif %}>{{ a }}</option>                              9799
        {% endfor %}                                                         9800
```

```
        </select>
        {% else %}
        <input id="f-{{ fname }}" type="number" step="any"
    name="{{ fname }}"
               value="{{ current.get(fname) if
    current.get(fname) is not none else '' }}">
        {% endif %}
    </div>
    {% endfor %}
    {% endfor %}
    {% for group_name, fields, has_hm in groups %}
        {% if has_hm %}
        <h3>重金属含量（mg/kg）<span class="muted">（可选，未检测可留空）</span></h3>
        <p class="muted">未检测的金属可留空：按论文注释以「未受污染」计入污染指标；
           评价结果页与导出报告中会写明未检测的是哪几项，避免被误读成实测达标。</p>
        <table style="width:auto;">
          <thead>
            <tr>
              {% for m in heavy_metals %}<th scope="col">{{ m
    }}</th>{% endfor %}
            </tr>
          </thead>
          <tbody>
            <tr>
              {% for m in heavy_metals %}
              <td>
                <label class="muted" style="position:absolute;
    left:-9999px;"
                       for="hm-{{ m }}">{{ m }} 含量</label>
                <input id="hm-{{ m }}" type="number"
    step="any" name="hm_{{ m }}"
                       value="{{ current.get('heavy',
    {}).get(m) if current.get('heavy', {}).get(m) is not
    none else '' }}">
              </td>
              {% endfor %}
            </tr>
          </tbody>
        </table>
        {% endif %}
    {% endfor %}
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/indicator/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
```

```
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
  <h2>批量导入指标数据</h2>
  <p class="muted">
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。
    先下载模板按格式填写，可减少出错。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/indicator/import/template">下载导入模板（CSV）</a>
    <a class="btn btn--sm btn--ghost"
    href="/indicator/">返回指标数据</a>
  </p>
  <form method="post" enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="patch-id">目标矿山图斑</lab
    el>
      <select id="patch-id" name="patch_id" required>
        <option value="">— 请选择 —</option>
        {% for p in patches %}
        <option value="{{ p.id }}">{{ p.name }}（{{
    p.units.count() }} 个单元）</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="datafile">数据文件</label
    >
      <input id="datafile" type="file" name="datafile"
             accept=".csv,.xlsx,.xls" required>
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">开始导入</button>
    </p>
  </form>
  <h3>模板列说明</h3>
  <p class="muted">
    共 {{ headers|length }} 列。留空表示该指标「不适用」。
    坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
    （按论文注释视为未受污染）。
  </p>
  <p class="mono muted" style="line-height:1.9;">{{
    headers|join(" , ") }}</p>
  <h3>导入规则</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
```

```
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
      <span class="mono">10cm</span>）。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
    </div>
    {% if report.errors %}
    <h3>未导入的行</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
    </ul>
    {% else %}
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
    </p>
</div>
{% endif %}
{% endblock %}
#  ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% block title %}适宜性定量评价{% endblock %}
{% block content %}
<div class="card">
    <h2>适宜性定量评价</h2>
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
```

```
<table>
  <thead>
    <tr>
      <th scope="col">所属项目</th>
      <th scope="col">矿山图斑</th>
      <th scope="col">前置条件</th>
      <th scope="col">最近评价结果</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in patches %}
    {% set pr = progress.get(p.id) %}
    <tr>
      <td>{{ p.project.name if p.project else "—" }}</td>
      <td style="text-align:left;">{{ p.name }}</td>
      <td style="text-align:left;">
        {% if pr and pr.ready %}
        <span class="tag tag--ok">已具备</span>
        {% elif pr %}
        <span class="tag tag--warn">缺：{{ pr.blocking[0].label }}</span>
        <span class="muted">{{ pr.blocking[0].reason }}</span>
        {% endif %}
      </td>
      <td>
        {% if latest.get(p.id) %}
        F = <span class="mono num">{{ latest[p.id].final_score }}</span>,
        <span class="tag tag--ok">{{ latest[p.id].grade }}</span>
        {% else %}
        <span class="tag tag--muted">未评价</span>
        {% endif %}
      </td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/evaluate/{{ p.id }}">执行评价</a>
          {% if latest.get(p.id) %}
          <a class="link-btn" href="/evaluate/result/{{ latest[p.id].id }}">查看结果</a>
          {% endif %}
        </span>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
```

```
  {% endif %}                                                                  10001
</div>                                                                         10002
{% endblock %}                                                                 10003
# ==== app/templates/evaluate/run.html ====                                    10004
{% extends "base.html" %}                                                      10005
{% import "_macros.html" as ui %}                                              10006
{% block title %}执行评价 · {{ patch.name }}{% endblock %}                     10007
{% block content %}                                                            10008
<div class="card" style="max-width:880px;">                                    10009
  <h2>执行适宜性评价 · {{ patch.name }}</h2>                                   10010
  {{ ui.workflow_steps(progress) }}                                            10011
  {{ ui.workflow_blocking(progress) }}                                         10012
  {% if progress.ready %}                                                      10013
  <form method="post">                                                         10014
    {{ form.hidden_tag() }}                                                    10015
    <p class="actions">                                                        10016
      <button type="submit" class="btn">开始评价</button>                      10017
      <a class="btn btn--ghost" href="/evaluate/">返回</a>                     10018
    </p>                                                                       10019
  </form>                                                                      10020
  {% else %}                                                                   10021
  <p class="actions">                                                          10022
    <button type="button" class="btn" disabled>开始评价</button>              10023
    <a class="btn btn--ghost" href="/evaluate/">返回</a>                       10024
  </p>                                                                         10025
  {% endif %}                                                                  10026
</div>                                                                         10027
<div class="card" style="max-width:880px;">                                    10028
  <h2>当前流程状态</h2>                                                       10029
  <table>                                                                      10030
    <thead>                                                                    10031
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th              10032
    scope="col">说明</th></tr>                                                 10033
    </thead>                                                                   10034
    <tbody>                                                                    10035
      <tr>                                                                     10036
        <td>限制性条件判定</td>                                               10037
        <td>                                                                   10038
          {% if progress.constraint.passed %}<span                             10039
    class="tag tag--ok">已通过</span>                                          10040
          {% elif progress.constraint.done %}<span                             10041
    class="tag tag--warn">未通过</span>                                        10042
          {% else %}<span class="tag tag--muted">待完成</span>{               10043
    % endif %}                                                                 10044
        </td>                                                                  10045
        <td style="text-align:left;">{{                                        10046
    progress.constraint.detail }}</td>                                         10047
      </tr>                                                                    10048
      <tr>                                                                     10049
        <td>划分微地质单元</td>                                               10050
```

```
            <td>                                                                    10051
              {% if progress.units.done %}<span class="tag                          10052
      tag--ok">已完成</span>                                                        10053
              {% else %}<span class="tag tag--muted">待完成</span>{                 10054
      % endif %}                                                                    10055
            </td>                                                                   10056
            <td style="text-align:left;">{{                                        10057
      progress.units.detail }}</td>                                                 10058
          </tr>                                                                     10059
          <tr>                                                                      10060
            <td>录入指标数据</td>                                                   10061
            <td>                                                                    10062
              {% if progress.data.ready %}<span class="tag                          10063
      tag--ok">已完成</span>                                                        10064
              {% else %}<span class="tag tag--muted">待完成</span>{                 10065
      % endif %}                                                                    10066
            </td>                                                                   10067
            <td style="text-align:left;">{{                                        10068
      progress.data.detail }}</td>                                                  10069
          </tr>                                                                     10070
      </tbody>                                                                      10071
    </table>                                                                        10072
</div>                                                                              10073
{% if weight_sources %}                                                             10074
<div class="card" style="max-width:880px;">                                         10075
    <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">                 10076
      <thead>                                                                       10077
        <tr><th scope="col">微地质单元</th><th                                      10078
      scope="col">权重来源</th></tr>                                                10079
      </thead>                                                                      10080
      <tbody>                                                                       10081
        {% for ut, label in weight_sources.items() %}                               10082
        <tr>                                                                        10083
          <td style="text-align:left;">{{ ut }}</td>                               10084
          <td style="text-align:left;">                                             10085
            {% if "自定义" in label %}                                              10086
            <span class="tag tag--warn">{{ label }}</span>                          10087
            {% else %}                                                              10088
            <span class="tag tag--muted">{{ label }}</span>                         10089
            {% endif %}                                                             10090
          </td>                                                                     10091
        </tr>                                                                       10092
        {% endfor %}                                                                10093
      </tbody>                                                                      10094
    </table>                                                                        10095
</div>                                                                              10096
{% endif %}                                                                         10097
{% endblock %}                                                                      10098
# ==== app/templates/evaluate/result.html ====                                      10099
{% extends "base.html" %}                                                           10100
```

```
{% block title %}评价结果{% endblock %}
{% block content %}
<div class="card">
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{ r.final_score
    }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ r.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
      <div style="font-size:15px;">{{ r.measures }}</div>
    </div>
  </div>
  {% if unit_scores %}
  <h3>各微地质单元得分</h3>
  <table style="max-width:860px;">
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">单元评价因子 F′</th>
        <th scope="col">加权贡献</th>
        <th scope="col">权重来源</th>
      </tr>
    </thead>
    <tbody>
      {% for u in unit_scores %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
        <td class="mono">{{ u.unit_score }}</td>
        <td class="mono">{{ "%.4f"|format(u.ratio *
    u.unit_score) }}</td>
        <td class="muted">{{ u.weight_source or "论文表 3
    默认权重" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  <p class="actions" style="margin-top:18px;">
    <a class="btn btn--ghost" href="/report/{{ r.patch_id
    }}">查看得分明细</a>
```

```
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/report/index.html ====
{% extends "base.html" %}
{% block title %}报告输出{% endblock %}
{% block content %}
<div class="card">
    <h2>报告与结果输出</h2>
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">矿山图斑</th>
          <th scope="col">可生成报告</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set pr = progress.get(p.id) %}
        <tr>
          <td>{{ p.project.name if p.project else "—" }}</td>
          <td style="text-align:left;">{{ p.name }}</td>
          <td style="text-align:left;">
            {% if pr and pr.ready %}
            <span class="tag tag--ok">可以</span>
            {% elif pr %}
            <span class="tag tag--warn">暂不可</span>
            <span class="muted">{{ pr.blocking[0].reason
}}</span>
            {% endif %}
          </td>
          <td class="nowrap">
            {% if pr and pr.ready %}
            <span class="row-actions">
              <a class="link-btn" href="/report/{{ p.id
}}">查看得分明细</a>
              <a class="link-btn" href="/report/{{ p.id
}}/pdf">导出 PDF</a>
              <a class="link-btn" href="/report/{{ p.id
}}/word">导出 Word</a>
            </span>
            {% else %}
            <span class="muted">—</span>
            {% endif %}
```

```
            </td>                                                    10201
          </tr>                                                      10202
          {% endfor %}                                               10203
        </tbody>                                                     10204
    </table>                                                         10205
    {% endif %}                                                      10206
</div>                                                               10207
{% endblock %}                                                       10208
# ==== app/templates/report/detail.html ====                         10209
{% extends "base.html" %}                                            10210
{% block title %}得分明细 · {{ patch.name }}{% endblock %}           10211
{% block content %}                                                  10212
<div class="card">                                                   10213
    <h2>得分明细 · {{ patch.name }}</h2>                             10214
    <div class="stat-row">                                           10215
      <div>                                                          10216
        <div class="stat__label">最终评价因子 F</div>                10217
        <div class="stat__value mono">{{                            10218
      "%.4f"|format(result.final_score) }}</div>                     10219
      </div>                                                         10220
      <div>                                                          10221
        <div class="stat__label">适宜性等级</div>                    10222
        <div class="stat__value ok">{{ result.grade }}</div>         10223
      </div>                                                         10224
      <div style="flex:1;min-width:240px;">                          10225
        <div class="stat__label">推荐措施</div>                      10226
        <div style="font-size:15px;">{{                              10227
      result.grade_info.measure }}</div>                             10228
      </div>                                                         10229
    </div>                                                           10230
    <p class="muted" style="margin-top:12px;">                       10231
      本页与导出报告均取自                                           10232
      <strong>{% if record %}{{ record.created_at |                  10233
      localdt("%Y-%m-%d %H:%M:%S") }}{% endif %}                     10234
      记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else  10235
      "—" }}），                                                     10236
      不会因事后修改指标数据而变化。                                 10237
    </p>                                                             10238
    {% if undetected %}                                              10239
    <p class="muted" style="margin-top:8px;">                        10240
      数据完整度说明：以下重金属项<strong>未检测</strong>，按论文注记以「未受污染」计入污染指标，  10241
      并非实测达标 ——                                               10242
      {% for unit_type, metals in undetected %}                      10243
        {{ unit_type }}：{{ metals|join("、") }}{% if not            10244
      loop.last %}；{% endif %}                                      10245
      {% endfor %}。                                                 10246
    </p>                                                             10247
    {% endif %}                                                      10248
    <p class="actions" style="margin-top:16px;">                     10249
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF      10250
```

```
      报告</a>                                                                    10251
      <a class="btn btn--ghost" href="/report/{{ patch.id                         10252
      }}/word">导出 Word 报告</a>                                                 10253
      <a class="btn btn--ghost" href="/report/">返回</a>                          10254
   </p>                                                                           10255
</div>                                                                            10256
<div class="card">                                                                10257
   <h2>结果图表</h2>                                                              10258
   <h3>各微地质单元得分对比</h3>                                                  10259
   <div>{{ chart_units|safe }}</div>                                              10260
   <h3>各一级指标平均得分</h3>                                                    10261
   <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p> 10262
   <div>{{ chart_weight|safe }}</div>                                             10263
</div>                                                                            10264
{% for u in result.unit_results %}                                                10265
<div class="card">                                                                10266
   <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *                      10267
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>                   10268
   {% if u.weight_source %}                                                       10269
   <p class="muted" style="margin-top:-6px;">权重来源：{{                         10270
      u.weight_source }}</p>                                                      10271
   {% endif %}                                                                    10272
   <table>                                                                        10273
      <thead>                                                                     10274
         <tr>                                                                     10275
            <th scope="col">一级指标</th>                                         10276
            <th scope="col">二级指标</th>                                         10277
            <th scope="col">实测值</th>                                           10278
            <th scope="col">标准化值</th>                                         10279
            <th scope="col">权重</th>                                             10280
            <th scope="col">贡献</th>                                             10281
         </tr>                                                                    10282
      </thead>                                                                    10283
      <tbody>                                                                     10284
         {% for d in u.details %}                                                 10285
         <tr>                                                                     10286
            <td>{{ d.l1 }}</td>                                                   10287
            <td style="text-align:left;">{{ d.l2 }}</td>                          10288
            <td class="mono" style="text-align:left;">{{ "—" if                   10289
      d.raw is none else d.raw }}</td>                                            10290
            <td class="mono">{{ "—" if d.std_value is none else                   10291
      "%.4f"|format(d.std_value) }}</td>                                          10292
            <td class="mono">{{ "%.4f"|format(d.weight) }}</td>                    10293
            <td class="mono">{{ "—" if d.contribution is none                     10294
      else "%.4f"|format(d.contribution) }}</td>                                  10295
         </tr>                                                                    10296
         {% endfor %}                                                             10297
      </tbody>                                                                    10298
   </table>                                                                       10299
   <p class="muted" style="margin:0;">                                            10300
```

```
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span
      class="num">{{ "%.4f"|format(v) }}</span>{% if not
      loop.last %};{% endif %}{% endfor %}
    </p>
</div>
{% endfor %}
{% endblock %}
# ==== app/templates/report/print.html ====
<!DOCTYPE html>
<html lang="zh-CN">
<head>
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
  @page { size: A4; margin: 18mm 16mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #23262b; background: #fff;
    font-family: "Microsoft YaHei", "SimSun", sans-serif;
    font-size: 10.5pt; line-height: 1.65;
  }
  h1 { font-size: 17pt; text-align: center; margin: 0 0
    6px; }
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
       border-bottom: 1px solid #dfe3e8; padding-bottom:
    4px; }
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0
    6px; }
  .sub { text-align: center; color: #6b7280; font-size:
    9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
          font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
    text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace; }
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
```

```
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 V1.0 ·
    生成时间 {{ generated_at | localdt }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</td><td>{{ patch.name }}</td></tr>
    <tr><th>所在行政区</td><td>{{ patch.admin_region or "—" }}</td></tr>
    <tr><th>图斑面积（公顷）</td><td>{{ patch.area_ha if patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</td><td>{{ patch.mineral_type or "—" }}</td></tr>
    <tr><th>开采方式</td><td>{{ patch.mining_method or "—" }}</td></tr>
    <tr><th>责任主体缺失</td><td>{{ "是" if patch.responsible_entity_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</td>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,
            多样性指数基准 {{ reference.shannon_benchmark }}</td></tr>
    {% endif %}
  </table>
  <h2>二、评价结论</h2>
  <table class="kv">
    <tr><th>最终评价因子 F</td><td class="mono"><strong>{{ "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</td><td><strong>{{ result.grade }}</strong></td></tr>
    <tr><th>判定依据</td><td>{{ result.grade_info.meaning }}</td></tr>
    <tr><th>推荐措施</td><td>{{ result.grade_info.measure }}</td></tr>
  </table>
  <h2>三、各微地质单元得分</h2>
  <table>
    <thead>
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子 F'</th><th>加权贡献</th></tr>
    </thead>
    <tbody>
      {% for u in result.unit_results %}
      <tr>
        <td>{{ u.unit_type }}</td>
```

```
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
  }}</td>
        <td class="mono">{{ "%.4f"|format(u.unit_score)
  }}</td>
        <td class="mono">{{ "%.4f"|format(u.weighted)
  }}</td>
      </tr>
      {% endfor %}
    </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>五、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
  100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
    <thead>
      <tr>
        <th>一级指标</th><th>二级指标</th><th>实测值</th>
        <th>标准化值</th><th>权重</th><th>贡献</th>
      </tr>
    </thead>
    <tbody>
      {% for d in u.details %}
      <tr>
        <td>{{ d.l1 }}</td>
        <td class="left">{{ d.l2 }}</td>
        <td class="mono left">{{ "—" if d.raw is none else
  d.raw }}</td>
        <td class="mono">{{ "—" if d.std_value is none else
  "%.4f"|format(d.std_value) }}</td>
        <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
        <td class="mono">{{ "—" if d.contribution is none
  else "%.4f"|format(d.contribution) }}</td>
      </tr>
      {% endfor %}
    </tbody>
</table>
<p class="note">
  一级指标得分：
  {% for k, v in u.l1_scores.items() %}{{ k }} = {{
  "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{%
  endfor %}
</p>
{% endfor %}
<h2>六、评价依据说明</h2>
<ul class="note">
  <li>指标标准化按论文式（8） 执行，按指标性质分为越大越优、越小越优、
```

```
            适中型与离散赋值四类。</li>
        <li>单元评价因子按论文式（9）  由二级指标权重加权求得；
            整矿评价因子由各微地质单元按面积占比加权合成。</li>
        <li>适宜性等级按论文表  4  划分：F&lt;0.3  不适宜、0.3≤F≤0.7  基本适宜、F≥0.7
        适宜。</li>
        <li>指标权重取自论文表  3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
    </ul>
    <p class="sig">
        评价人（签字）：_____    日期：_____
    </p>
</body>
</html>
# ==== app/templates/search.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="card">
    <h2>搜索结果</h2>
    {% if not q %}
    <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区 / 矿种、
        标准名称与文号。</p>
    {% else %}
    <p class="muted">
        关键词「<strong>{{ q }}</strong>」共找到
        <span class="num">{{ total }}</span> 条结果。
    </p>
    {% if not total %}
    <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
    {% endif %}
    {% endif %}
</div>
{% if projects %}
<div class="card">
    <h2>评价项目（{{ projects|length }}）</h2>
    <table>
        <thead>
            <tr>
                <th scope="col">项目名称</th>
                <th scope="col">编号</th>
                <th scope="col">状态</th>
                <th scope="col">图斑数</th>
                <th scope="col">操作</th>
            </tr>
        </thead>
        <tbody>
            {% for p in projects %}
            <tr>
                <td style="text-align:left;">{{ p.name }}</td>
                <td class="mono">{{ p.code or "—" }}</td>
```

```
            <td><span class="tag tag--muted">{{ p.status }}</span></td>
            <td class="num">{{ p.patches.count() }}</td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/project/{{ p.id }}/patches">矿山图斑</a>
                <a class="link-btn" href="/project/{{ p.id }}/edit">编辑</a>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
    {% endif %}
    {% if patches %}
    <div class="card">
      <h2>矿山图斑（{{ patches|length }}）</h2>
      <table>
        <thead>
          <tr>
            <th scope="col">所属项目</th>
            <th scope="col">图斑名称</th>
            <th scope="col">行政区</th>
            <th scope="col">矿种</th>
            <th scope="col">操作</th>
          </tr>
        </thead>
        <tbody>
          {% for m in patches %}
          <tr>
            <td>{{ m.project.name if m.project else "—" }}</td>
            <td style="text-align:left;">{{ m.name }}</td>
            <td>{{ m.admin_region or "—" }}</td>
            <td>{{ m.mineral_type or "—" }}</td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/constraint/{{ m.id }}">限制性判定</a>
                <a class="link-btn" href="/unit/{{ m.id }}">微地质单元</a>
                <a class="link-btn" href="/evaluate/{{ m.id }}">执行评价</a>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
```

```
    </table>
</div>
{% endif %}
{% if standards %}
<div class="card">
    <h2>标准与政策文件（{{ standards|length }}）</h2>
    <table>
      <thead>
        <tr>
          <th scope="col">类别</th>
          <th scope="col">省份</th>
          <th scope="col">名称</th>
          <th scope="col">文号</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for s in standards %}
        <tr>
          <td>{{ s.category }}</td>
          <td>{{ s.province }}</td>
          <td style="text-align:left;">{{ s.name }}</td>
          <td class="mono">{{ s.doc_number or "—" }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
{% endif %}
{% endblock %}
# ==== seed_demo.py ====
# -*- coding: utf-8 -*-
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。
运行：python seed_demo.py                    幂等，只补写缺失的演示数据，
                                                不会删除用户自建的项目与图斑
      python seed_demo.py --reset            连同标准库与指标定义一并重建
"""
import json
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
```

```
                                MicroUnit, IndicatorData, Standard,
    StandardRule,
                                ConstraintCheck, EvaluationResult,
    ScoreDetail)
app = create_app()
# 内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」
DEMO_CODE = "DEMO-001"
# 论文表 1：11 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="江西",
    name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
    publish_date="2026-02-02",
        implement_date="2026-08-01",
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",
        doc_number=None, publish_date="2026-06-18",
    implement_date=None,
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),
```

```
                ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="政策文件", province="安徽",
         name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",
         doc_number="皖自然资修函[2020]34号",
         publish_date="2020-03-03", implement_date=None,
         summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",
         rules=[("责任主体", "==", "政府", "必要条件", None),
                ("无地质灾害", "==", "是", "必要条件", None),
                ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="山东",
         name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",
         doc_number="鲁自然资字[2022]104号",
         publish_date="2022-08-02", implement_date=None,
         summary="条件与安徽省相同，另增加自然恢复认定标准",
         rules=[("无地质灾害", "==", "是", "必要条件", None),
                ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="湖南",
         name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",
         doc_number="湘自资办发[2023]58号",
         publish_date="2023-07-20", implement_date=None,
         summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
>75%、连续3年正向增长",
         rules=[("植被覆盖度", ">", "75", "必要条件", "%"),
                ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),
                ("无地质灾害", "==", "是", "必要条件", None),
                ("符合国土空间规划", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="新疆",
         name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
         doc_number="新自然资源办函〔2024〕219号",
         publish_date="2024-12-20", implement_date=None,
         summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
         rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),
                ("无地质灾害", "==", "是", "必要条件", None),
                ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="广西",
         name="广西壮族自治区历史遗留矿山自然恢复技术指南",
         doc_number="桂自然资办〔2024〕250号",
         publish_date="2024-12-30", implement_date=None,
         summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
         rules=[("无地质灾害", "==", "是", "必要条件", None),
                ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="西藏",
         name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
         doc_number="藏自然资发〔2025〕19号",
         publish_date="2025-09-24", implement_date=None,
         summary="含修复成本高、生态修复效益不显著等适用条件",
         rules=[("责任主体", "==", "政府", "必要条件", None),
                ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                ("无地质灾害", "==", "是", "必要条件", None)]),
```

```
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.2319,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
                     "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                      "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                     "coverage": 58.42, "shannon": 0.72},
        },
    ],
}
def clean_case(reset=False):
    """清理数据。
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
    以便把演示环境恢复到初始状态。
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
    会在每次启动时被删除。
    """
    from app.models import EvaluationResult, ScoreDetail,
    ConstraintCheck
    demo_projects = Project.query.filter_by(code=DEMO_CODE)
    all()
    demo_ids = [p.id for p in demo_projects]
```

```
    patch_ids = []
    if demo_ids:
        patch_ids = [p.id for p in MinePatch.query.filter(
            MinePatch.project_id.in_(demo_ids)).all()]
    if patch_ids:
        unit_ids = [u.id for u in MicroUnit.query.filter(
            MicroUnit.patch_id.in_(patch_ids)).all()]
        if unit_ids:
            IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).delete(
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
(
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids)).
delete(
            synchronize_session=False)
    if demo_ids:
        Project.query.filter(Project.id.in_(demo_ids)).delet
e(
            synchronize_session=False)
    if reset:
        StandardRule.query.delete()
        Standard.query.delete()
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，
    #  避免长期运行后 score_details 表堆积无效数据。
    _purge_orphan_details()
    db.session.commit()
    return len(demo_ids)
def demo_patch():
    """取内置演示案例的图斑。
```

```
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。
    没有演示案例时才退回第一个图斑。
    """
    proj = Project.query.filter_by(code=DEMO_CODE).first()
    if proj is not None:
        m = MinePatch.query.filter_by(project_id=proj.id).first()
        if m is not None:
            return m
    return MinePatch.query.first()
def _purge_orphan_details():
    """清理指向已不存在评价记录的孤儿得分明细。
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。
    """
    live_ids = [r.id for r in EvaluationResult.query.all()]
    if live_ids:
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(
    live_ids)).delete(
            synchronize_session=False)
    else:
        ScoreDetail.query.delete()
    db.session.commit()
def seed(reset=False):
    with app.app_context():
        db.create_all()          # 全新环境下先建表，保证本脚本可独立运行
        _purge_orphan_details()
        admin = User.query.filter_by(username="admin").first()
        if not admin:
            from werkzeug.security import generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_hash("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        # 幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
        existing = Project.query.filter_by(code=DEMO_CODE).first()
        if existing and not reset:
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")
            _report()
            _seed_reference_data(reset=False)
            return
        clean_case(reset=reset)
        project = Project(name=CASE["name"], code=DEMO_CODE,
                          description="论文第 5 节典型案例",
    created_by=admin.id)
```

```
        db.session.add(project)
        db.session.flush()
        patch = MinePatch(project_id=project.id,
name=CASE["patch_name"],
                          admin_region="河北省唐山市玉田县",
                          area_ha=CASE["area_ha"],
                          mineral_type=CASE["mineral_type"],
                          mining_method="露天开采",
stop_years=None,
                          responsible_entity_missing=True)
        db.session.add(patch)
        db.session.flush()
        ref = ReferenceEcosystem(patch_id=patch.id,
                                 coverage_benchmark=75.0,
shannon_benchmark=0.7)
        db.session.add(ref)
        for u in CASE["units"]:
            unit = MicroUnit(patch_id=patch.id,
unit_type=u["unit_type"],
                             ratio=u["ratio"], area=None)
            db.session.add(unit)
            db.session.flush()
            d = u["data"]
            ind = IndicatorData(
                unit_id=unit.id,
                soil_ratio=d.get("soil_ratio"),
                thickness=d.get("thickness"),
                ph=d.get("ph"),
                organic_matter=d.get("organic_matter"),
                heavy_metals=json.dumps(d["heavy_metals"],
ensure_ascii=False)
                             if d.get("heavy_metals") else
None,
                coverage=d.get("coverage"),
                shannon=d.get("shannon"),
                slope=d.get("slope"),
                aspect=d.get("aspect"),
            )
            db.session.add(ind)
        #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。
        db.session.add(ConstraintCheck(
            patch_id=patch.id,
            policy_pass=True, location_pass=True,
pollution_pass=True,
            geo_pass=True, farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        print("已写入案例：", CASE["name"])
        print("  图斑：", CASE["patch_name"])
```

```
        print("  微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台
    23.19%")
        print("  管理员账号：admin / admin123")
        _seed_reference_data(reset=reset)
        _report()
def _seed_reference_data(reset=False):
    """写入标准库与指标定义。
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护
    的标准条目与指标定义。
    """
    if reset or Standard.query.count() == 0:
        seed_standards()
    from app.services.indicator_service import
    seed_indicators
    n = seed_indicators()
    print("指标定义：%d 项" % n)
def _report():
    total_p = Project.query.count()
    total_patch = MinePatch.query.count()
    print()
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))
    if total_p > total_patch:
        print("  提示：项目是归档单位，评价针对的是图斑。")
        print("        新项目需先建立图斑，才会出现在后续各模块中。")
def seed_standards():
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""
    StandardRule.query.delete()
    Standard.query.delete()
    db.session.commit()
    for s in STANDARDS:
        std = Standard(category=s["category"],
    province=s["province"],
                       name=s["name"],
    doc_number=s["doc_number"],
                       publish_date=s["publish_date"],
                       implement_date=s["implement_date"],
                       summary=s["summary"])
        db.session.add(std)
        db.session.flush()
        for item, op, thr, ctype, note in s["rules"]:
            db.session.add(StandardRule(standard_id=std.id,
    item=item,
                                        operator=op,
    threshold=thr,
    condition_type=ctype, note=note))
    db.session.commit()
    print("已写入标准/政策文件：%d 项，规则 %d 条"
          % (len(STANDARDS), sum(len(s["rules"]) for s in
    STANDARDS)))
```

```
if __name__ == "__main__":                                                    10951
    #  --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。          10952
    #  无论是否 --reset，都不会删除用户自建的项目与图斑。                       10953
    seed(reset="--reset" in sys.argv[1:])                                     10954
#  ==== 测试/test_ahp.py ====                                                 10955
#  -*- coding: utf-8 -*-                                                      10956
"""地基自检：验证 AHP 引擎与应用可启动。"""                                    10957
import sys                                                                    10958
sys.stdout.reconfigure(encoding="utf-8")                                      10959
import numpy as np                                                            10960
import os                                                                     10961
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                   10962
#  使 `from app import ...` 与 app/app.db 的定位不受影响。                    10963
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil                  10964
    e__)))                                                                    10965
if ROOT not in sys.path:                                                      10966
    sys.path.insert(0, ROOT)                                                  10967
from app.services import ahp_service as ahp                                   10968
LINE = "=" * 66                                                               10969
#  ------------------------------------------------------------------         10970
    ------------------                                                        10971
#  1） 完全一致矩阵 → 权重应精确还原，CR=0                                    10972
#  ------------------------------------------------------------------         10973
    ------------------                                                        10974
print(LINE)                                                                   10975
print("测试 1  完全一致判断矩阵的权重还原与一致性")                           10976
print(LINE)                                                                   10977
#  由权重构造完全一致矩阵：a_ij = w_i / w_j                                   10978
w_true = np.array([0.44, 0.31, 0.25])                                         10979
A = np.outer(w_true, 1.0 / w_true)                                            10980
r = ahp.weights_and_consistency(A)                                            10981
print("  判断矩阵 A：")                                                      10982
for row in A:                                                                 10983
    print("    " + "  ".join(f"{v:8.4f}" for v in row))                      10984
print()                                                                       10985
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in                          10986
    r["weights"]))                                                            10987
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))                  10988
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"                      10989
      % (r["lambda_max"], r["ci"], r["cr"],                                   10990
    r["consistent"]))                                                         10991
assert np.allclose(r["weights"], w_true, atol=1e-6),                          10992
    "权重未精确还原"                                                          10993
assert r["consistent"], "完全一致矩阵应通过检验"                              10994
#  ------------------------------------------------------------------         10995
    ------------------                                                        10996
#  2） 上三角输入 → 完整矩阵 → 一致性定位                                     10997
#  ------------------------------------------------------------------         10998
    ------------------                                                        10999
print()                                                                       11000
```

```
print(LINE)
print("测试 2  上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("  权重：", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))
print("  最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
         bad["expected"]))
# ------------------------------------------------------------------
# 3) 多专家聚合（几何平均）
# ------------------------------------------------------------------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合（几何平均）")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")
print("  聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ------------------------------------------------------------------
# 4) 论文表 3 权重可还原性
# ------------------------------------------------------------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f  CR = %.3e  一致: %s" %
      (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------------
# 5) 应用可启动 + 建表
# ------------------------------------------------------------------
print()
```

```
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表：" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
#  ==== 测试/test_evaluate.py ====
#  -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
#  ------------------------------------------------------------------
#  论文第 5 节案例数据（表 5，逐字抄录）
#  ------------------------------------------------------------------
#  三类单元面积占比：岩质边坡 13.86%、碎石边坡 62.95%、平台 23.19%
#  注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
```

```
            "data": {
                "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
                "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                                 "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
                "coverage": 87.92, "shannon": 0.60,
                "slope": 28.75, "aspect": "阳坡",
            },
        },
        {
            "unit_type": "平台",
            "ratio": 0.2319,
            "data": {
                "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
                "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                                 "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
                "coverage": 58.42, "shannon": 0.72,
            },
        },
]
# ------------------------------------------------------------------
    ------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
```

```
        print("   %s   %.2f%% × %.4f = %.4f"
              % (ur["unit_type"], ur["ratio"] * 100,
        ur["unit_score"], ur["weighted"]))
print("   " + "-" * 60)
print("   最终评价因子 F = %.4f" % total)
print()
print("   适宜性等级：%s" % res["grade"])
print("   判定依据   ：%s" % res["grade_info"]["meaning"])
print("   推荐措施   ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("   - 评价引擎按论文式（8）（9） 与表 3 权重完整跑通")
print("   - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("   - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("   - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("     该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
# ==== 测试/test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
```

```
            db.session.add(User(username="admin",                    11201
                                password_hash=generate_password_     11202
        hash("admin123"),                                            11203
                                role="admin"))                       11204
            db.session.commit()                                      11205
client.post("/login", data={"username": "admin",                     11206
        "password": "admin123"})                                     11207
print(LINE)                                                          11208
print("测试 1  首页与 AHP 入口")                                      11209
print(LINE)                                                          11210
r = client.get("/")                                                  11211
print("  首页状态码:", r.status_code, "(应 200)")                     11212
assert r.status_code == 200                                          11213
r = client.get("/ahp/")                                              11214
print("  AHP 入口状态码:", r.status_code, "(应 200)")                 11215
assert r.status_code == 200                                          11216
assert "微地质单元类型" in r.get_data(as_text=True)                    11217
print()                                                              11218
print(LINE)                                                          11219
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")                     11220
print(LINE)                                                          11221
r = client.post("/ahp/", data={                                      11222
        "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",  11223
}, follow_redirects=True)                                            11224
body = r.get_data(as_text=True)                                      11225
print("  跳转后页面含「判断矩阵录入」:", "判断矩阵录入" in body)        11226
print("  含三个一级指标(土壤/植被/地形):", all(n in body for n in     11227
        ("土壤指标", "植被指标", "地形指标")))                         11228
assert "判断矩阵录入" in body                                          11229
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))    11230
print()                                                              11231
print(LINE)                                                          11232
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")                      11233
print(LINE)                                                          11234
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角:          11235
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =             11236
        0.31/0.25 = 1.24                                             11237
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={     11238
        "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",         11239
}, follow_redirects=True)                                            11240
body = r.get_data(as_text=True)                                      11241
print("  结果页含「计算结果」:", "计算结果" in body)                   11242
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in       11243
        body, " 含 0.25:", "0.25" in body)                           11244
print("  含一致性检验通过:", "通过" in body)                           11245
assert "计算结果" in body                                              11246
assert "0.44" in body and "0.31" in body and "0.25" in body          11247
print()                                                              11248
print(LINE)                                                          11249
print("测试 4  不一致矩阵应提示不通过并定位")                          11250
```

```
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
```

```
        "password": "admin123"},                                    11301
                        follow_redirects=True)                      11302
body = r.get_data(as_text=True)                                     11303
#  工作台：页首横幅 + 评价流程分区 + 数据概览                          11304
print("  登录后含「评价流程」：", "评价流程" in body)                  11305
print("  含账户 admin：", "admin" in body)                           11306
print("  含页首横幅：", 'class="hero"' in body)                      11307
print("  含数据概览：", 'class="stat-card"' in body)                 11308
assert "评价流程" in body and 'class="hero"' in body                 11309
print()                                                             11310
print(LINE)                                                         11311
print("测试 3  项目与矿山档案")                                      11312
print(LINE)                                                         11313
r = client.get("/project/")                                         11314
body = r.get_data(as_text=True)                                     11315
print("  项目列表状态码：", r.status_code)                           11316
print("  含案例项目：", "玉田县典型历史遗留废弃矿山案例" in body)      11317
assert "玉田县典型历史遗留废弃矿山案例" in body                       11318
#  新建项目                                                         11319
client.post("/project/add", data={"name": "测试项目A", "code":      11320
        "T-001"})                                                   11321
r = client.get("/project/")                                         11322
print("  新建项目后含「测试项目A」：", "测试项目A" in r.get_data(as_text=True11323
        ))                                                          11324
assert "测试项目A" in r.get_data(as_text=True)                       11325
#  清理测试数据，避免污染演示库                                      11326
with app.app_context():                                             11327
        from app.models import db, Project                          11328
        for p in Project.query.filter_by(name="测试项目A").all():    11329
                db.session.delete(p)                                11330
        db.session.commit()                                         11331
print("  已清理测试项目")                                            11332
print()                                                             11333
print(LINE)                                                         11334
print("测试 4  微地质单元（面积占比自动计算）")                      11335
print(LINE)                                                         11336
from app.models import MinePatch                                    11337
with app.app_context():                                             11338
        patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f11339
        irst()                                                      11340
        pid = patch.id                                              11341
r = client.get("/unit/%d" % pid)                                    11342
body = r.get_data(as_text=True)                                     11343
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡",    11344
        "碎石边坡", "平台")))                                        11345
print("  含占比合计：", "占比合计" in body)                          11346
assert "岩质边坡" in body and "占比合计" in body                     11347
print()                                                             11348
print(LINE)                                                         11349
print("测试 5  标准知识库与省际口径比对")                            11350
```

```
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准:", all(p in body for p in ("辽宁", "湖南",
    "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组:", "植被覆盖度" in body)
print("  含多省份并列:", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件:", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
```

```
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
```

```
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.id.
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_report.py ====
#  -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
```

```
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("        逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
```

```
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
print("  含得分明细章节：", "得分明细" in text)
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
```

```
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
    ----------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
```

```
                follow_redirects=False)                                  11651
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if                    11652
    h.startswith("remember_token=")]                                     11653
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2        11654
    else "[不符合]"))                                                    11655
assert not rt2                                                           11656
# 4) 携带 remember_token 的新会话应自动登录                              11657
c3 = app.test_client()                                                   11658
c3.set_cookie("remember_token",                                          11659
              rt[0].split(";")[0].split("=", 1)[1],                      11660
    domain="localhost")                                                  11661
r3 = c3.get("/", follow_redirects=False)                                 11662
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"        11663
      % ("[符合]" if r3.status_code == 200 else "[不符合]",              11664
    r3.status_code))                                                     11665
assert r3.status_code == 200                                             11666
# ------------------------------------------------------------------11667
    ----------------                                                     11668
print()                                                                  11669
print(LINE)                                                              11670
print("二、删除功能")                                                    11671
print(LINE)                                                              11672
client.post("/login", data={"username": "admin",                         11673
    "password": "admin123"})                                             11674
# 项目：列表页须有删除按钮                                               11675
html = client.get("/project/").get_data(as_text=True)                    11676
has_del = "/delete" in html and "btn--danger" in html                    11677
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))  11678
assert has_del, "项目列表缺少删除按钮"                                   11679
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else     11680
    "[不符合]"))                                                         11681
assert "confirm(" in html                                                11682
# 图斑列表：地址用的是项目主键，不是图斑主键                             11683
with app.app_context():                                                  11684
    from app.models import MinePatch, Project                            11685
    import seed_demo                                                     11686
    patch = seed_demo.demo_patch()                                       11687
    patch_id = patch.id                                                  11688
    project_id = patch.project_id                                        11689
html = client.get("/project/%d/patches" %                                11690
    project_id).get_data(as_text=True)                                   11691
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in     11692
    html and "btn--danger" in html else "[不符合]"))                     11693
assert "/project/patch/" in html and "btn--danger" in html               11694
# 微地质单元                                                             11695
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr11696
    ue)                                                                  11697
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in     11698
    html else "[不符合]"))                                               11699
assert "/unit/delete/" in html                                           11700
```

```
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
```

```
    MicroUnit, IndicatorData,                                          11751
                            Indicator, ExpertMatrix,                   11752
    OperationLog)                                                      11753
from app.utils.schema import ensure_schema                             11754
from app.utils import browser, charts                                  11755
LINE = "=" * 72                                                        11756
BASE = ROOT                                                            11757
class TestConfig:                                                      11758
    SECRET_KEY = "test"                                                11759
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                           11760
    os.path.join(BASE, "app", "app.db")                                11761
    SQLALCHEMY_TRACK_MODIFICATIONS = False                             11762
    WTF_CSRF_ENABLED = False                                           11763
    TESTING = True                                                     11764
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",                11765
    "uploads")                                                         11766
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024                              11767
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",                  11768
    "webp", "bmp", "tif", "tiff"}                                      11769
app = create_app(TestConfig)                                           11770
with app.app_context():                                                11771
    db.create_all()                                                    11772
    ensure_schema(db)                                                  11773
    ExpertMatrix.query.delete()                                        11774
    db.session.commit()                                                11775
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响          11776
    import seed_demo                                                   11777
    patch_id = seed_demo.demo_patch().id                               11778
c = app.test_client()                                                  11779
c.post("/login", data={"username": "admin", "password":                11780
    "admin123"})                                                       11781
def ok(flag, label, extra=""):                                         11782
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",          11783
    label, extra))                                                     11784
    assert flag, label                                                 11785
# ------------------------------------------------------------------11786
    ----------------                                                   11787
print(LINE)                                                            11788
print("操作日志与用户维护")                                                11789
print(LINE)                                                            11790
before = None                                                          11791
with app.app_context():                                                11792
    before = OperationLog.query.count()                                11793
c.post("/project/add", data={"name": "日志核对项目", "code":             11794
    "LOG-T"})                                                          11795
with app.app_context():                                                11796
    after = OperationLog.query.count()                                 11797
    last = OperationLog.query.order_by(OperationLog.id.desc(11798
    )).first()                                                         11799
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,    11800
```

```
    after))                                                              11801
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)          11802
ok("project" in (last.target_type or ""), "日志记录了对象类型",          11803
    last.target_type)                                                    11804
r = c.get("/profile?tab=logs")                                           11805
body = r.get_data(as_text=True)                                          11806
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")      11807
ok("新建项目" in body, "日志页显示操作记录")                               11808
# 账号编辑 / 删除（职务只有管理员与操作员两种）                           11809
with app.app_context():                                                  11810
    u = User(username="tmp_user", role="operator",                       11811
             password_hash=User.query.first().password_hash)             11812
    db.session.add(u); db.session.commit()                               11813
    uid = u.id                                                           11814
r = c.get("/users/%d/edit" % uid)                                        11815
ok(r.status_code == 200, "账号编辑页可访问")                               11816
# 保护校验要趁系统里只有一名管理员时做：                                  11817
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效       11818
with app.app_context():                                                  11819
    admin_id = User.query.filter_by(role="admin").first().id             11820
c.post("/users/%d/edit" % admin_id,                                      11821
       data={"role": "operator", "active": "1"})                         11822
with app.app_context():                                                  11823
    ok(db.session.get(User, admin_id).role == "admin",                   11824
    "拒绝把最后一个管理员降级")                                          11825
c.post("/users/%d/edit" % uid, data={"role": "admin",                    11826
    "active": "1"})                                                      11827
with app.app_context():                                                  11828
    ok(db.session.get(User, uid).role == "admin",                        11829
    "职务已修改并保存")                                                  11830
c.post("/users/%d/delete" % uid, follow_redirects=True)                  11831
with app.app_context():                                                  11832
    ok(db.session.get(User, uid) is None, "账号删除生效")                11833
# 清理日志核对用的项目                                                   11834
with app.app_context():                                                  11835
    for p in Project.query.filter_by(name="日志核对项目").all():          11836
        db.session.delete(p)                                             11837
    db.session.commit()                                                  11838
# ------------------------------------------------------------------11839
    ----------------                                                     11840
print()                                                                  11841
print(LINE)                                                              11842
print("遥感影像挂接")                                                    11843
print(LINE)                                                              11844
doc = fitz.open()                                                        11845
pg = doc.new_page(width=320, height=200)                                 11846
pg.insert_text((20, 60), "RS image", fontname="china-s",                 11847
    fontsize=16)                                                         11848
png = doc[0].get_pixmap(dpi=100).tobytes("png")                          11849
doc.close()                                                              11850
```

```
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
```

```
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="coverage").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type": "碎石边坡",
                              "level": "1", "expert_name": name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡", level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>", body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) < 0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
   vals[0] if vals else "")
# ------------------------------------------------------------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,58.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
```

```
                    "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
                content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
                data={"patch_id": str(patch_id),
                    "datafile": (io.BytesIO(b"x"), "d.pdf")},
                content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
    "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
# 报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
    from app.views.evaluate import load_patch_input,
    save_evaluation
    save_evaluation(patch_id, load_patch_input(patch_id))
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
```

```
        d = fitz.open(stream=data, filetype="pdf")                    12001
        txt = "".join(d[i].get_text() for i in                        12002
        range(d.page_count))                                          12003
        ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")                12004
        ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")      12005
        ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)  12006
        d.close()                                                     12007
print()                                                               12008
print(LINE)                                                           12009
print("补齐模块自检全部通过 √")                                          12010
print(LINE)                                                           12011
#  ==== 测试/test_workflow.py ====                                    12012
#  -*- coding: utf-8 -*-                                              12013
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。                          12014
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。            12015
本测试逐个构造「缺前置」的场景，验证系统不会放行。                             12016
"""                                                                   12017
import os                                                             12018
import sys                                                            12019
sys.stdout.reconfigure(encoding="utf-8")                              12020
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                    12021
#  使 `from app import ...` 与 app/app.db 的定位不受影响。              12022
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil          12023
    e__)))                                                            12024
if ROOT not in sys.path:                                              12025
    sys.path.insert(0, ROOT)                                          12026
from app import create_app                                            12027
from app.models import (db, Project, MinePatch, MicroUnit,            12028
    IndicatorData,                                                    12029
                        ConstraintCheck, EvaluationResult)            12030
from app.services import workflow as wf                               12031
LINE = "=" * 72                                                       12032
BASE = ROOT                                                           12033
class TestConfig:                                                     12034
    SECRET_KEY = "test"                                               12035
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                          12036
    os.path.join(BASE, "app", "app.db")                               12037
    SQLALCHEMY_TRACK_MODIFICATIONS = False                            12038
    WTF_CSRF_ENABLED = False                                          12039
    TESTING = True                                                    12040
app = create_app(TestConfig)                                          12041
c = app.test_client()                                                 12042
c.post("/login", data={"username": "admin", "password":               12043
    "admin123"})                                                      12044
_fails = []                                                           12045
def ok(flag, label, extra=""):                                        12046
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",         12047
    label, extra))                                                    12048
    if not flag:                                                      12049
        _fails.append(label)                                          12050
```

```
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
location_pass=passed,
            pollution_pass=True, geo_pass=True,
farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
"未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
soil_ratio=0.34, thickness=10,
                                     ph=9.24,
organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
()
```

```
        uids = [u.id for u in MicroUnit.query.filter_by(patch_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id.in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" + tag).first()
        if pr:
            db.session.delete(pr)
        db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
```

```
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
    "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
    "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
    "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
    "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
    "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
    "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
    "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
    "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
```

```
        soil_ratio=0.3, thickness=10,                                   12201
                                              ph=8.0,                   12202
        organic_matter=5.0, coverage=80.0,                              12203
                                              shannon=0.6, slope=20.0,  12204
        aspect="阳坡"))                                                 12205
        db.session.commit()                                             12206
r = post_eval(pid)                                                      12207
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")                12208
cleanup(pid, "5")                                                       12209
print()                                                                 12210
print(LINE)                                                             12211
print("二、前置条件齐备时才放行")                                        12212
print(LINE)                                                             12213
pid = new_patch("OK"); add_units(pid); add_constraint(pid,              12214
        passed=True); add_data(pid)                                     12215
with app.app_context():                                                 12216
        pr = wf.patch_progress(pid)                                     12217
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")              12218
r = post_eval(pid)                                                      12219
n = results_of(pid)                                                     12220
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)                   12221
print("   重定向到:", r.headers.get("Location"))                        12222
print()                                                                 12223
print(LINE)                                                             12224
print("三、报告不得先于评价产生")                                        12225
print(LINE)                                                             12226
pid2 = new_patch("R")     # 什么都不做                                  12227
r = get_report(pid2)                                                    12228
ok(r.status_code == 302 and "/report/" ==                               12229
        (r.headers.get("Location") or ""),                              12230
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)     12231
for suffix in ("/pdf", "/word"):                                        12232
        rr = c.get("/report/%d%s" % (pid2, suffix),                     12233
        follow_redirects=False)                                         12234
        ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP12235
        %d" % rr.status_code)                                           12236
cleanup(pid2, "R")                                                      12237
cleanup(pid, "OK")                                                      12238
print()                                                                 12239
print(LINE)                                                             12240
print("四、前置齐备但从未执行评价时，报告仍不得产生")                    12241
print(LINE)                                                             12242
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，  12243
#  也不应凭空推算出结论。                                                12244
pid3 = new_patch("N")                                                   12245
add_units(pid3); add_constraint(pid3, passed=True);                     12246
        add_data(pid3)                                                  12247
with app.app_context():                                                 12248
        pr = wf.report_progress(pid3)                                   12249
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")                  12250
```

```
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
    "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",
    "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
    "报告数值取自评价记录，而非重新计算",
    "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).fi
    irst()
```

```
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
           follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
       follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first()
    ).id
r = c.post("/unit/edit/%d" % uid,
           data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
```

```
    cleanup(pid4, "D")
    print()
    print(LINE)
    if _fails:
        print("存在未通过项：")
        for f in _fails:
            print("  -", f)
        sys.exit(1)
    print("流程门禁自检全部通过 √")
    print(LINE)
# ==== 测试/test_weights.py ====
# -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
```

```
def ok(flag, label, extra=""):                                          12401
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",          12402
    label, extra))                                                      12403
    if not flag:                                                        12404
        _fails.append(label)                                            12405
TAG = "W"                                                               12406
UNIT_TYPE = "碎石边坡"                                                  12407
def build_case():                                                       12408
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""  12409
    with app.app_context():                                             12410
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"          12411
    % TAG)                                                              12412
        db.session.add(p); db.session.flush()                           12413
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"       12414
    % TAG)                                                              12415
        db.session.add(patch); db.session.flush()                       12416
        db.session.add(MicroUnit(patch_id=patch.id,                     12417
    unit_type=UNIT_TYPE,                                                12418
                                 area=1.0, ratio=1.0))                  12419
        db.session.flush()                                              12420
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi             12421
    st()                                                                12422
        db.session.add(IndicatorData(                                   12423
            unit_id=u.id, soil_ratio=0.34, thickness=10,                12424
    ph=9.24,                                                            12425
            organic_matter=4.89, coverage=87.92,                        12426
    shannon=0.60,                                                       12427
            slope=28.75, aspect="阳坡"))                                12428
        db.session.add(ConstraintCheck(                                 12429
            patch_id=patch.id, policy_pass=True,                        12430
    location_pass=True,                                                 12431
            pollution_pass=True, geo_pass=True,                         12432
    farmland_pass=True,                                                 12433
            passed=True, reason="全部通过"))                            12434
        db.session.commit()                                             12435
        return patch.id                                                 12436
def cleanup(pid):                                                       12437
    with app.app_context():                                             12438
        from app.models import ScoreDetail                              12439
        rids = [r.id for r in EvaluationResult.query.filter_            12440
    by(patch_id=pid)]                                                   12441
        if rids:                                                        12442
            ScoreDetail.query.filter(ScoreDetail.result_id.i            12443
    n_(rids)).delete(                                                   12444
                synchronize_session=False)                              12445
        EvaluationResult.query.filter_by(patch_id=pid).delet            12446
    e()                                                                 12447
        ConstraintCheck.query.filter_by(patch_id=pid).delete            12448
    ()                                                                  12449
        uids = [u.id for u in MicroUnit.query.filter_by(patc            12450
```

```
    h_id=pid)]                                                          12451
        if uids:                                                        12452
            IndicatorData.query.filter(IndicatorData.unit_id        12453
    .in_(uids)).delete(                                                 12454
                synchronize_session=False)                              12455
        MicroUnit.query.filter_by(patch_id=pid).delete()                12456
        m = db.session.get(MinePatch, pid)                              12457
        if m:                                                           12458
            db.session.delete(m)                                        12459
        p = Project.query.filter_by(code="WEIGHT-%s" %                  12460
    TAG).first()                                                        12461
        if p:                                                           12462
            db.session.delete(p)                                        12463
        db.session.commit()                                             12464
def run_eval(pid):                                                      12465
    c.post("/evaluate/%d" % pid, data={},                               12466
    follow_redirects=False)                                             12467
    with app.app_context():                                             12468
        return EvaluationResult.query.filter_by(patch_id=pid        12469
    ) \                                                                 12470
            .order_by(EvaluationResult.id.desc()).first().fi        12471
    nal_score                                                           12472
print(LINE)                                                             12473
print("一、默认应使用论文表 3 的权重")                                  12474
print(LINE)                                                             12475
with app.app_context():                                                 12476
    ws.reset_weights(UNIT_TYPE)                                         12477
    w, label, scheme = ws.effective_weights(UNIT_TYPE)                  12478
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",  12479
    label)                                                              12480
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",      12481
   "土壤指标 %.2f" % w["l1"]["土壤指标"])                               12482
pid = build_case()                                                      12483
base_score = run_eval(pid)                                              12484
print("  默认权重下的评价值 F = %.4f" % base_score)                    12485
print()                                                                 12486
print(LINE)                                                             12487
print("二、保存 AHP 算出的权重后，评价应改用新权重")                    12488
print(LINE)                                                             12489
with app.app_context():                                                 12490
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）  12491
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",             12492
    0.20),                                                              12493
                                ("terrain", 0.10)])                     12494
    n_items = WeightItem.query.count()                                  12495
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)    12496
with app.app_context():                                                 12497
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)               12498
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",  12499
    label2)                                                             12500
```

```
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
    "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
    "未覆盖的二级权重仍保留论文取值",
    "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("   新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
    "权重变化确实改变了评价结果",
    "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
    "评价记录中标注了权重来源",
    stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
    label3)
back_score = run_eval(pid)
print("   恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
    base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
```

```
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/services/cleanup.py ====
# -*- coding: utf-8 -*-
"""级联清理：删图斑、删项目时，把挂在它下面的数据一并清掉。
为什么单独抽出来：`Project.patches` 与 `MinePatch.units` 上有 ORM 级联，
但图斑下面的**指标数据、限制性判定、评价结果、得分明细、参照生态系统**
都没有级联关系。原先只有「删除图斑」那条路径里手写了清理，
于是「删除项目」会留下一批孤儿行——自检实测：删一个项目后，
上面五张表各残留 1 行（项目与图斑本身都删掉了，数据却还在库里）。
两条删除路径都调用这里，才不会漏。
"""
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail,
    ConstraintCheck,
                        ReferenceEcosystem)
def delete_patch_children(patch_id):
    """清掉挂在某个图斑下的全部附属数据（不含图斑本身）。
    返回各表清理条数，便于调用方打印或断言。
    """
    counts = {}
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patch_id=patch_id).all()]
    if unit_ids:
```

```
        counts["indicator_data"] = IndicatorData.query.filte
    r(
            IndicatorData.unit_id.in_(unit_ids)).delete(synch
    ronize_session=False)
    result_ids = [r.id for r in
                  EvaluationResult.query.filter_by(patch_id=
    patch_id).all()]
    if result_ids:
        counts["score_details"] = ScoreDetail.query.filter(
            ScoreDetail.result_id.in_(result_ids)).delete(sy
    nchronize_session=False)
    counts["evaluation_results"] = EvaluationResult.query.fi
    lter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["constraint_checks"] = ConstraintCheck.query.filt
    er_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["micro_units"] = MicroUnit.query.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["reference_ecosystems"] = ReferenceEcosystem.quer
    y.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    return counts
def delete_patch(patch_id):
    """删除一个图斑及其全部附属数据。图斑不存在时返回 (None, {})。"""
    patch = db.session.get(MinePatch, patch_id)
    if patch is None:
        return None, {}
    counts = delete_patch_children(patch_id)
    db.session.delete(patch)
    db.session.commit()
    return patch, counts
def delete_project(project):
    """删除一个项目：先清掉每个图斑的附属数据，再删项目本身。
    项目下的图斑由 `Project.patches` 的 ORM 级联删除，但图斑下面那五张表
    必须先手工清，否则会留孤儿。
    """
    cleaned = 0
    for patch in project.patches.all():
        counts = delete_patch_children(patch.id)
        cleaned += sum(counts.values())
    name = project.name
    db.session.delete(project)
    db.session.commit()
    return name, cleaned
# ==== app/services/unit_config.py ====
# -*- coding: utf-8 -*-
"""标准化参数与必填项：让界面上能改的东西真正参与计算。
以前有个说不通的地方：图斑的「参照生态系统基准」、指标定义里的「理想取值」
和「启用单元」在界面上都能改、也能落库，**但计算链只读代码常量**
```

```
（`app/utils/constants.py`），从不查这两张表。于是报告里写着「基准　60%」，
实际按常量　75%　算，同一份报告两个口径。
现在统一从这里取：
- 基准值（植被覆盖度、Shannon）来自图斑的　`reference_ecosystems`；
- 各指标的理想取值来自　`indicators.ideal_value`（能解析成数字的才覆盖，
  像「5~9」「25°　分界」这种区间 / 规则型仍按代码里的算法处理）；
- 启用单元来自　`indicators.enabled_units`（岩质边坡无土壤、平台无地形）。
库里没有行、或值不是数字时一律回落到代码常量　——　所以历史数据与
`验证数据/`　里的　8　个算例结果不会因为这次改动而变化。
"""
from app.models import Indicator, ReferenceEcosystem
from app.utils.constants import (
    UNIT_TYPES, IDEAL_SOIL_RATIO, IDEAL_THICKNESS,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METALS,
)
#　二级指标编码　→　标准化参数名（同时也是　IndicatorData　的字段名）
NUMERIC_IDEALS = {
    "soil_ratio": ("soil_ratio", IDEAL_SOIL_RATIO),
    "thickness": ("thickness", IDEAL_THICKNESS),
    "organic_matter": ("organic_matter", IDEAL_ORGANIC),
    "coverage": ("coverage", IDEAL_COVERAGE),
    "shannon": ("shannon", IDEAL_SHANNON),
}
#　重金属编码　→　金属名
METAL_CODES = {"hm_" + m: m for m in HEAVY_METALS}
#　必填的实测字段（污染指标按论文注记允许「未检测」，故不在必填之列）
REQUIRED_CODES = ["soil_ratio", "thickness", "ph",
    "organic_matter",
                  "coverage", "shannon", "slope", "aspect"]
#　字段　→　界面上的中文名，用于提示「缺什么」
FIELD_LABELS = {
    "soil_ratio": "土石比", "thickness": "土层厚度", "ph": "pH",
    "organic_matter": "有机质", "coverage": "植被覆盖度",
    "shannon": "生物多样性（Shannon）", "slope": "坡度", "aspect":
    "坡向",
}
def _numeric(text):
    """把理想取值的文本解析成数字；区间、规则型返回　None。"""
    if text is None:
        return None
    try:
        return float(str(text).strip())
    except (TypeError, ValueError):
        return None
class UnitSettings:
    """一个图斑上下文的标准化参数。空对象表示「全部按代码常量」。"""
    def __init__(self, params=None, enabled=None):
        self.params = params or {}              #　覆盖用的标准值
```

```
        self.enabled = enabled or {}                # 单元类型 → 启用的二级指标编码集合
    def ideal(self, code, fallback):
        """取某指标的理想取值：库里有数字就用库里的，否则用常量。"""
        value = self.params.get(code)
        return fallback if value is None else value
    def group_enabled(self, unit_type, l1_name, l1_code):
        """该单元类型下某个一级指标是否启用（库里没配就按代码常量）。"""
        codes = self.enabled.get(unit_type)
        if codes is None:
            return None                             # 交给调用方按常量判断
        return l1_code in codes
    def enabled_codes(self, unit_type):
        """该单元类型启用的全部二级指标编码；库里没配返回 None。"""
        return self.enabled.get(unit_type)
    def required_codes(self, unit_type):
        """该单元类型必填的实测字段。
        按「启用的二级指标」推导：岩质边坡不启用土壤，平台不启用地形。
        库里没配该项时用代码常量里的结构性规则。
        """
        from app.utils.constants import UNIT_ENABLED_INDICATORS
        codes = self.enabled.get(unit_type)
        if codes is None:
            groups = UNIT_ENABLED_INDICATORS.get(unit_type, ())
            l2_by_group = {
                "土壤指标": ["soil_ratio", "thickness", "ph", "organic_matter"],
                "植被指标": ["coverage", "shannon"],
                "地形指标": ["slope", "aspect"],
            }
            codes = {c for g in groups for c in l2_by_group.get(g, [])}
        return [c for c in REQUIRED_CODES if c in codes]
    def missing_fields(self, unit_type, data):
        """返回该单元缺哪些必填项（中文名列表）。"""
        missing = []
        for code in self.required_codes(unit_type):
            value = data.get(code)
            if value is None or (isinstance(value, str) and value.strip() == ""):
                missing.append(FIELD_LABELS.get(code, code))
        return missing
    def undetected_metals(self, data):
        """按论文注记，未检测的重金属按「未受污染」计入；这里列出是哪些。"""
        measured = data.get("heavy_metals") or {}
        return [m for m in HEAVY_METALS if measured.get(m) is None]
def for_patch(patch_id):
```

```
    """读某个图斑的标准化参数；库里有什么就用什么，没有就用常量。
    优先级：图斑自己的参照生态系统基准  **高于**  指标定义里的全局理想取值  ——
    前者说的是「这个图斑的参照生态系统是什么样」，后者是「一般情况下的理想值」，
    越具体的越应该说了算。两者都优先于代码常量。
    """
    params = {}
    enabled = {}
    for ind in Indicator.query.all():
        code = ind.code
        if code in NUMERIC_IDEALS:
            value = _numeric(ind.ideal_value)
            if value is not None:
                params[code] = value
        elif code in METAL_CODES:
            #  重金属的「理想值」存的是限值，作为越小越优的极值使用
            value = _numeric(ind.ideal_value)
            if value is not None:
                params[code] = value
        units = (ind.enabled_units or "").strip()
        if units:
            for unit in UNIT_TYPES:
                if unit in units:
                    enabled.setdefault(unit,
    set()).add(code)
    #  再用该图斑的参照生态系统基准覆盖：它是「这个图斑的参照生态系统」，
    #  比指标定义里的全局理想取值更具体，应该它说了算。
    ref = ReferenceEcosystem.query.filter_by(patch_id=patch_
    id).first()
    if ref is not None:
        if ref.coverage_benchmark is not None:
            params["coverage"] = float(ref.coverage_benchmar
    k)
        if ref.shannon_benchmark is not None:
            params["shannon"] = float(ref.shannon_benchmark)
    return UnitSettings(params=params, enabled=enabled)
def metal_limits(settings):
    """重金属的极值表：库里有配置的用库里的，其余用常量。"""
    limits = dict(HEAVY_METAL_LIMIT)
    for code, metal in METAL_CODES.items():
        value = settings.params.get(code) if settings else
    None
        if value is not None:
            limits[metal] = value
    return limits
#  ==== app/utils/__init__.py ====
#  -*- coding: utf-8 -*-
#  ==== app/utils/parsing.py ====
#  -*- coding: utf-8 -*-
"""表单数值的安全解析。
背景：页面上的数字输入框原来写成 `float(request.form.get("area"))`，
```

```
用户填「约 3．5」或全角数字就会撞上 ＶａｌｕｅＥｒｒｏｒ —— 页面上是默认英文 ５００ 页。
真人演示时这是最容易踩到的一类。统一走这里：
没填返回 Ｎｏｎｅ，填了但不是数字抛 ＦｉｅｌｄＥｒｒｏｒ，让视图 ｆｌａｓｈ 一句中文再回退。
"""
import math
class FieldError(ValueError):
    """表单里某个字段填得不对，携带给用户看的中文提示。"""
def parse_float(raw, field, *, minimum=None, maximum=None):
    """把表单值解析成 float。
    raw       表单原始值（可能是 None / "" / "abc" / "1e400"）
    field     字段名，用于拼提示
    minimum / maximum   可选的取值范围，越界同样抛 FieldError
    """
    if raw is None:
        return None
    text = str(raw).strip()
    if text == "":
        return None
    try:
        value = float(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填数字" % field)
    # nan / inf 能通过 float()，但存进库里会把标准化算成满分，必须挡住
    if not math.isfinite(value):
        raise FieldError("「%s」请填正常的数字" % field)
    if minimum is not None and value < minimum:
        raise FieldError("「%s」不能小于 %g" % (field, minimum))
    if maximum is not None and value > maximum:
        raise FieldError("「%s」不能大于 %g" % (field, maximum))
    return value
def parse_int(raw, field, *, default=None, minimum=None,
              maximum=None):
    """解析整数，用于查询参数与计数类字段。"""
    if raw is None:
        return default
    text = str(raw).strip()
    if text == "":
        return default
    try:
        value = int(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填整数" % field)
    if minimum is not None and value < minimum:
        value = minimum
    if maximum is not None and value > maximum:
        value = maximum
    return value
# ==== app/utils/timefmt.py ====
# -*- coding: utf-8 -*-
"""时间显示：库里一律存 ＵＴＣ，展示时换成服务器本地时间。
```

为什么不在写库时直接用本地时间：一个系统里时间基准只能有一个。
`app/models.py` 里所有 `created_at` / `checked_at` 都用 `datetime.utcnow`，
存的是 ＵＴＣ；如果写库改成本地时间，已经存进库的几千条记录会新旧混着差 8 小时。
所以统一在「展示」这一层换算，老数据不用动。
用法（Ｊｉｎｊａ）：
    {{ row.created_at | localdt }}   →
    2026-09-26 23:33
    {{ row.created_at | localdt("%Y-%m-%d %H:%M:%S") }} →
    2026-09-26 23:33:28
"""
from datetime import datetime, timezone
DEFAULT_FORMAT = "%Y-%m-%d %H:%M"
def to_local(value, fmt=DEFAULT_FORMAT, empty="—"):
    """把 ＵＴＣ 时间换算成服务器本地时间并格式化。
    传进来的值按「无时区即 ＵＴＣ」处理；带时区的原样换算，便于以后改用
    带时区的时间戳。空值返回占位符，免得页面上出现 None。
    """
    if value is None:
        return empty
    if value.tzinfo is None:
        value = value.replace(tzinfo=timezone.utc)
    return value.astimezone().strftime(fmt)
def now_utc():
    """取当前 ＵＴＣ 时间（带时区），用于只作展示、不落库的时间戳。"""
    return datetime.now(timezone.utc)
==== app/views/__init__.py ====
-*- coding: utf-8 -*-